

A Full Conservative Derivation of the Two-Body 3PN Sector from the Unified 4D Toy Model

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Abstract

We present the full conservative two-body 3PN assembly for the 4D toy-model program within the same declared closure hierarchy used at 1PN and 2PN. Starting from the exact 4D reduction backbone, the frozen Newtonian/1PN/2PN ledger, and the 2.5PN narrowing to the grouped real $P_{20} \oplus P_{21} \oplus P_{22}$ quadrupole bundle, we close the complete conservative near-zone ledger through order c^{-6} in the fixed ADM chart. The strict Schwarzschild one-body gate fixes a minimal three-parameter repair ledger

$$(\mu_{\rho,3}, d_3, s_{24}) = \left(\frac{1}{4}, -\frac{45}{4}, -\frac{1}{16} \right).$$

The comparable-mass sector then splits exactly into three lanes: the ordinary-Lagrangian counterimage of the universal free two-body relativistic Hamiltonian, an exact grouped real P_2 middle-block closure, and a unique geometry-side static completion. Equivalently,

$$\Delta \hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}} + \Delta l_{15}^{(g)} U^4,$$

with $\Delta H_3 = -\Delta L_3$ in the fixed ADM chart. The apparent pure-kinetic remainder collapses to the compiler image of the free 3PN Hamiltonian, while the remaining static slot is uniquely geometry-side after sigma-collapse. Thus the conservative 3PN bottleneck is no longer algebraic: within the declared hierarchy, the two-body 3PN ledger is fully assigned, and the live unresolved problem is the deeper moving-throat derivation of the grouped- P_2 and geometry constitutive lanes together with the already-narrow outgoing quadrupole-normalization gap inherited from the 2.5PN program.

1 Introduction and scope

1.1 Motivation: from full conservative 2PN and the 2.5PN narrowing to the first conservative grouped- P_2 gate

The earlier papers in this program already established a strong conservative backbone. On the reduction side, the unified 4D framework supplies the exact projection identities, the projected continuity law with leakage, the controlled Poisson hook, and the worldtube/coherent-defect reduction used to define the brane-facing point-particle sector. On the post-Newtonian side, the carried lower-order ledger fixes the Newtonian and conservative 1PN and 2PN blocks within a declared closure hierarchy, with the surviving viability conditions

$$\kappa_{\rho} = 1, \quad n = 5, \quad \kappa_{\text{add}} = \frac{1}{2}, \quad \kappa_{\text{PV}} = \frac{3}{2}, \quad \beta_{1\text{PN}} = 3.$$

Just as important, the conservative 2PN assembly sharpened the ontology of the particle itself. The defect can no longer be treated as a pure scalar monopole with small corrections; the solved conservative cross sector already looks like a small throat-response theory with a carried odd dipole wake, a genuine $P_0 \oplus P_2$ mouth/support layer, and a separate geometry-closure channel.

The 2.5PN program then made the higher-order bridge much sharper. Its main structural lesson was that the surviving universal point-particle route is not a vague “do more of the whole theory” direction, but the orbital/worldtube STF quadrupole branch, packaged conservatively as the grouped real

$$P_{20} \oplus P_{21} \oplus P_{22}$$

bundle. That narrowing changes the meaning of the next conservative step. After 2PN, the right question is no longer whether one can add more local coefficients somehow. The right question is whether the first honest higher-order conservative load carried by the grouped real P_2 sector can be closed exactly inside the same hierarchy that already worked at 1PN and 2PN.

That is the purpose of the present paper. At 3PN the near-zone two-body ledger through order c^{-6} must close all of the following at once: free-particle octic kinematics v^8 , one-body/self terms through U^4 , the full comparable-mass middle interaction block, the static quartic term at order G^4/r^4 , and the generic-frame uniqueness problem in the fixed ADM chart. This is the first conservative grouped- P_2 gate of the program. It is also the first place where the 2.5PN narrowing becomes operational rather than only organizational, because the grouped real quadrupole bundle now has to carry a full middle-block theorem burden instead of merely serving as a suggestive future interface.

A first structural consequence appears already at the strict one-body level. The exact isotropic Schwarzschild gate through 3PN is not reproduced by a single cubic extension of the carried 2PN denominator-style self sector. The minimal one-body repair ledger is instead

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16},$$

so the 3PN gate already proves that this order is not a one-parameter continuation of the 2PN one-body closure. But the one-body sector is still only the front door. The deeper algebraic issue is whether the full comparable-mass c^{-6} residual can be assigned uniquely once the lower-order ledger is frozen, the exact COM target is imported, and the generic-frame chart is fixed.

The main positive result of the paper is that it can. In the fixed ADM chart, the conservative 3PN comparable-mass residual splits exactly as

$$\begin{aligned} \Delta \hat{L}_3^{\text{GR}} = & \underbrace{\Delta l_1 v^8}_{\text{compiler image of the free COM Hamiltonian}} + \underbrace{L_{P_2}^{\text{mid}}}_{\text{exact grouped real } P_2 \text{ closure}} \\ & + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}. \end{aligned}$$

The middle interaction block is carried by a richer grouped real P_2 family, the remaining static slot is uniquely geometry-side after sigma-collapse, and the apparent pure-kinetic leftover is only the ordinary-Lagrangian counterimage of the universal free relativistic two-body COM Hamiltonian. A second crucial structural lesson is that this statement has to be reached Hamiltonian-first. Naive ordinary-Lagrangian COM reduction of the imported generic-frame target does not reproduce the exact COM ordinary target, whereas the exact cubic Legendre lift followed by COM reduction does.

The scope of the paper is therefore deliberately strong but intermediate. It is stronger than a notebook-level target match because the theorem chain is already enough to show that the conservative 3PN bottleneck is no longer an algebraic coefficient search. It is weaker than a fully first-principles moving-throat PDE theorem because the grouped real P_2 constitutive law, the geometry-side completion, and the passive/outgoing quadrupole normalization on the true branch still require a deeper dynamical derivation. The reader should therefore view this paper in the same way as the earlier full conservative 1PN and 2PN assemblies: a full assembly theorem *within a declared closure hierarchy*, not an assumption-free theorem of the fully solved bulk problem.

A second purpose of the paper is bookkeeping clarity. At 3PN the argument mixes carried exact identities, reduced-regime small-body and COM statements, protocol-dependent constitutive closure, and final theorem consequences. The reader should be able to see, line by line, which ingredients are inherited exactly from the lower-order stack, which depend on the declared reduction and chart choices, which are fixed only within the response hierarchy, and which are final assembly statements. That separation is part of the result.

1.2 What this paper claims

The main claims of the paper are the following.

1. The exact 4D parent reduction backbone from the earlier papers can be carried forward unchanged as the reduction framework of the present 3PN analysis.
2. The strict isotropic Schwarzschild one-body gate fixes a minimal three-parameter 3PN repair ledger,

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16},$$

so the 3PN one-body sector is not a one-parameter continuation of the carried 2PN closure.

3. The exact isotropic COM target can be imported, solved, and reduced to a finite 15-slot comparable-mass residual vector, and the exact cubic ordinary/Hamiltonian compiler gives the authoritative bridge between the COM ordinary and Hamiltonian charts.
4. In the fixed ADM chart the exact generic-frame cubic compiler reduces to

$$\Delta H_3 = -\Delta L_3,$$

so the generic-frame ordinary representative is unique once the Hamiltonian target is imposed. In particular, the correct generic-frame lift is Hamiltonian-first rather than naive ordinary-Lagrangian COM reduction.

5. The minimal local grouped- P_2 front end is too small, but the richer grouped real P_2 family

$$(T_{20}, T_{21}, T_{22}, S_{20}, S_{21}, S_{22}, V_{20}, V_{21}, V_{22})$$

closes the full 9-slot 3PN middle block exactly.

6. The remaining static slot is uniquely geometry-side after the exact sigma-collapse removes the apparent P_0 /geometry ambiguity.
7. The apparent pure-kinetic leftover is not a new throat datum. It is only the ordinary-Lagrangian counterimage of the universal free relativistic two-body COM Hamiltonian.
8. Therefore, within the same declared hierarchy used successfully at 1PN and 2PN, the conservative 3PN comparable-mass ledger is fully assigned in the fixed ADM chart, with exact split

$$\Delta \hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}} + \Delta l_{15}^{(g)} U^4.$$

What remains open is no longer algebraic coefficient matching. What remains is the deeper moving-throat derivation of the grouped- P_2 constitutive law, the geometry completion, and then their insertion into the already narrowed passive/outgoing quadrupole bridge.

Stated differently, the point of the paper is not that the model can be tuned until a 3PN-looking ledger appears. The point is that, once the lower-order stack and the fixed-chart target are handled correctly, the full c^{-6} problem narrows to a short list of named theorem gates. By the end of the present paper those gates are passed, and the remaining unresolved questions are constitutive and dynamical rather than algebraic.

1.3 What this paper does not claim

The negative side of the scope is equally important.

First, this paper does *not* claim an assumption-free theorem that begins with a fully solved moving-throat PDE and ends automatically with the complete point-particle 3PN dynamics. As at 1PN and 2PN, the particle limit used here is a controlled reduction inside a declared hierarchy. The main text therefore presents a reduced-theory theorem chain, not a finished theorem of the full moving-throat bulk problem.

Second, this paper does *not* claim that every grouped- P_2 or geometry coefficient has already been derived microscopically from the completed throat PDE. The theorem proved here is that the conservative ledger is fully assigned once the declared hierarchy is accepted. The constitutive origin of the richer grouped- P_2 family and the unique geometry completion remains a deeper follow-on problem.

Third, this paper does *not* claim any dissipative, radiative, or final 2.5PN outgoing-normalization result. The point of the present paper is to close the conservative 3PN front end that the 2.5PN narrowing had already identified as the relevant grouped real quadrupole lane. The passive/outgoing quadrupole normalization on the actual moving-throat branch remains outside the scope of the present theorem.

Fourth, the paper is restricted to the nonspinning conservative two-body sector through order c^{-6} . It does not attempt a treatment of spin couplings, radiation reaction, strong-field nonperturbative completion, generic many-body response, or higher-PN sectors beyond the present conservative 3PN scope. Nor does it claim to replace the broader script-side moving-throat or grouped response record by a single closed PDE derivation. The main text is intentionally organized around the smallest theorem chain needed for the present 3PN statement.

These non-claims are not qualifications to hide. They are part of the correct interpretation of the result. The paper should be read as establishing the sharpest honest conservative 3PN statement currently supported by the program: a full algebraic assembly theorem within the declared hierarchy, with the remaining open work shifted from coefficient matching to microscopic response and normalization.

1.4 Claim taxonomy

To keep the argument referee-safe, we retain the same bookkeeping discipline used in the earlier full conservative papers.

Exact. Identities that follow directly from the declared lower-order ledger, the exact one-body target, the exact compiler algebra, symmetry bookkeeping, or exact source-map manipulations once the chart and basis choices have been frozen.

Controlled reduction. Steps that require an explicit reduced description, such as the carried 4D-to-brane and worldtube reductions, the reduced isotropic COM basis, the grouped real P_2 packaging, or the Hamiltonian-first COM reduction rule used to recover the exact fixed-chart representative.

Protocol closure. Statements that depend on the same declared response hierarchy used below 3PN rather than on a fully solved moving-throat PDE. In the present paper the most important examples are the one-body repair packaging, the richer local grouped- P_2 constitutive family, and the interpretation of the remaining static slot as a geometry-side completion.

Full assembly. The final theorem consequence after the previous ingredients are inserted. In the present paper this means that, once the lower-order ledger, the exact 3PN targets, the fixed ADM chart, and the declared closure hierarchy are frozen, the final equality is exact algebra.

This taxonomy is not cosmetic. It prevents four logically distinct moves from being blurred together: carrying forward exact lower-order structure, using the declared reductions and chart choices, invoking the constitutive closure needed to close the grouped- P_2 and geometry lanes, and stating the final theorem consequence that survives after those ingredients are frozen.

1.5 Roadmap

The next section gives an executive summary of the final theorem ledger. Section 3 then records the minimal carry-forward dictionary from the earlier 4D, 1PN, 2PN, and 2.5PN papers and states the exact 3PN target. Section 4 begins the proof chain with the strict one-body Schwarzschild gate, the minimal repair ledger, the natural self/static seed, and the exact cubic compiler.

Section 5 lifts that one-body input into the full comparable-mass problem: the reduced COM basis, the imported exact COM target, the generic-frame scaffold, the COM-null and contact/gauge structure, the failure of naive ordinary-Lagrangian COM reduction, the Hamiltonian-first lift, and the fixed-chart uniqueness theorem. Section 6 then isolates the first conservative grouped- P_2 gate, explains why the minimal front end fails, and proves that the richer grouped family closes the full middle block exactly.

Section 7 shows that the remaining static slot is uniquely geometry-side after sigma-collapse, while Section 8 proves that the apparent pure-kinetic leftover is only a compiler image of the free two-body Hamiltonian rather than a new throat datum. Section 9 then states the final exact split in COM and generic-frame form and records the equality to the standard 3PN ADM target within the declared hierarchy.

The last main-text blocks serve two carry-forward purposes. Section 10 explains exactly how the present conservative theorem interfaces with the already-narrowed 2.5PN quadrupole program, why 3PN is the first conservative grouped- P_2 gate, what the isotropy data mean, and what the full moving-throat PDE still has to supply. Section 11 separates what is now fixed from what remains symbolic or open, Section 12 records the role and scope of the referee suite, and Section 13 closes the paper. The appendices collect the detailed algebraic ledgers, compiler derivations, grouped- P_2 constructions, geometry completion, and verification scaffolding that would obscure the theorem chain if left in the body of the paper.

2 Executive summary of results

This paper closes the first honest conservative grouped- P_2 gate of the program. The earlier papers already froze the exact 4D reduction backbone, carried the conservative Newtonian+1PN+2PN ledger through its declared closure hierarchy, and narrowed the live higher-order universal point-particle route to the orbital/worldtube STF quadrupole branch packaged as the grouped real

$$P_{20} \oplus P_{21} \oplus P_{22}.$$

The present question is therefore not whether one can add more local coefficients somehow. It is whether the full nonspinning conservative two-body ledger at order c^{-6} can be assigned uniquely once the strict one-body gate, the exact reduced COM target, and the fixed ADM chart are handled correctly. The answer is now sharp: within the same declared closure hierarchy used successfully at 1PN and 2PN, the conservative 3PN two-body ledger is fully assigned in the fixed ADM chart.

Table 1: Load-bearing results of the present 3PN paper.

Item	Result	Status
One-body gate	The strict isotropic Schwarzschild target through c^{-6} is fixed exactly, and the minimal repair ledger is $\mu_{\rho,3} = 1/4$, $d_3 = -45/4$, $s_{24} = -1/16$.	Settled
Fixed-chart compiler	The imported exact GR COM target becomes a finite 15-slot residual, the Hamiltonian-first route is the authoritative lift, and in the fixed ADM chart the generic-frame compiler reduces to $\Delta H_3 = -\Delta L_3$.	Settled
Grouped- P_2 middle block	The minimal local grouped- P_2 front end fails, but the richer family $(T_{20}, T_{21}, T_{22}, S_{20}, S_{21}, S_{22}, V_{20}, V_{21}, V_{22})$ closes the full 9-slot middle block exactly with $\det M_{\text{mid}} = -4/27$.	Exact closure
Static scalar lane	The grouped module predicts a definite static companion, and the remaining scalar gap collapses algebraically to one unique geometry-side representative after the exact sigma identity $\nu(p^3 + q^3) = (1 - 3\nu)(p^2 q + p q^2)$.	Unique completion
Pure-kinetic slot	The apparent leftover COM v^8 term is not a new throat datum; it is the exact ordinary-Lagrangian compensation required to reproduce the universal free relativistic two-body COM Hamiltonian.	Resolved
Remaining gap	The moving-throat PDE must still supply the microscopic origin of the grouped- P_2 data, the isotropy verdict $a_2 = b_2 = 0$ on the true branch, and the final passive/outgoing quadrupole normalization.	Open

Headline result. Within the stated hierarchy, the exact comparable-mass 3PN residual splits as

$$\Delta \hat{L}_3^{\text{GR}} = \underbrace{\Delta l_1 v^8}_{\text{compiler image of the free COM Hamiltonian}} + \underbrace{L_{P_2}^{\text{mid}}}_{\text{exact grouped real } P_2 \text{ closure}} + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}.$$

The two scalar coefficients worth keeping visible already at the executive level are

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}, \quad \Delta l_{15}^{(g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}.$$

The middle interaction block is carried entirely by a richer grouped real P_2 family, the remaining static scalar slot is uniquely geometry-side after sigma-collapse, and the apparent pure-kinetic leftover is only the ordinary-Lagrangian counterimage of the universal free relativistic two-body COM Hamiltonian. What remains open is therefore no longer algebraic coefficient matching. What remains open is the deeper moving-throat derivation of the grouped- P_2 constitutive law and geometry completion, followed by their insertion into the already narrowed passive/outgoing quadrupole bridge.

Table 1 lists the load-bearing results established in the present paper and the role each plays in the theorem chain.

How the ledger closes. The proof chain has three main positive gates and one decisive bookkeeping lesson. First, the strict one-body Schwarzschild target shows that 3PN is not a one-parameter continuation of the carried 2PN self sector; the $U^3 v^2$ and $U^2 v^4$ channels together force the three-parameter repair ledger $(\mu_{\rho,3}, d_3, s_{24})$. Second, the exact reduced COM target and the exact cubic Legendre compiler turn the comparable-mass problem into a finite algebraic residual rather than a blind coefficient search. Third, the grouped real P_2 ontology survives its first full conservative stress test: the minimal constitutive front end is too rigid, but the richer grouped family closes the entire nine-slot middle block exactly. The bookkeeping lesson is that the correct quotient principle is Hamiltonian-first. Naive ordinary-Lagrangian COM reduction of

the imported generic-frame target does not reproduce the exact COM ordinary target, whereas the exact cubic Legendre lift followed by COM reduction does. Once this is done, the fixed-chart generic-frame compiler collapses to the slotwise sign flip

$$\Delta H_3 = -\Delta L_3,$$

so no true algebraic ambiguity survives after the Hamiltonian target is imposed.

One-body gate and self/static seed. The strict one-body target is the exact isotropic Schwarzschild expansion through 3PN,

$$\frac{L_{3\text{PN},1\text{body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4.$$

A single cubic extension of the carried 2PN denominator-style self sector is not enough to reproduce this answer. The exact repair ledger is instead

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16},$$

where $\mu_{\rho,3}$ is the cubic static gate, d_3 is the cubic denominator datum, and s_{24} is the genuinely new U^2v^4 self slot. Symmetric promotion of the exact one-body target then gives the natural 3PN self/static seed,

$$L_{3,\text{seed}} = \frac{5}{128}(m_A v_A^8 + m_B v_B^8) + \frac{11Gm_A m_B}{16r}(v_A^6 + v_B^6) + \cdots - \frac{G^4 m_A m_B}{8r^4}(m_B^3 + m_A^3),$$

which isolates all universal one-body/self/static content before the genuine comparable-mass lift begins.

Fixed-chart algebra and unique generic-frame representative. In the standard isotropic COM basis the exact cubic ordinary/Hamiltonian compiler is diagonal. Importing the exact GR 3PN COM target and subtracting the frozen self/static seed therefore yields an explicit 15-slot residual vector. At the generic-frame level the interaction scaffold has block sizes

$$24, \quad 17, \quad 8, \quad 1,$$

for the G/r , G^2/r^2 , G^3/r^3 , and G^4/r^4 blocks. COM projection of that scaffold reveals a small middle-block obstruction if one keeps only the naive seed placement, and it also shows that COM-blind directions are larger than the actual ordinary contact/gauge orbit. None of this is a failure of the scaffold itself. The decisive point is that once the Hamiltonian target is imposed in the fixed ADM chart, the generic-frame residual compiler becomes

$$\Delta H_3 = -\Delta L_3,$$

so the generic-frame ordinary representative is unique. At that stage the bottleneck is no longer algebraic span or chart ambiguity.

The grouped real P_2 middle block. At 3PN the conservative grouped data are the three $O(\omega^2)$ coefficients

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)},$$

or equivalently the grouped trace/anomaly variables

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \quad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \quad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2}.$$

The first isotropy test for the minimal branch is therefore

$$a_2 = 0, \quad b_2 = 0.$$

A direct time-local grouped front end built only from the leading three $O(\omega^2)$ couplings is too rigid: after the unique local demotion it lands in the correct nine interaction slots but has rank three and obeys six exact relations violated by the GR target. The grouped ontology itself is therefore not the problem; only the minimal constitutive choice is too small. Enlarging the local grouped family to

$$(T_{20}, T_{21}, T_{22}, S_{20}, S_{21}, S_{22}, V_{20}, V_{21}, V_{22})$$

gives an exact 9×9 middle-block compiler with constant determinant

$$\det M_{\text{mid}} = -\frac{4}{27}.$$

This richer grouped lift reproduces the full GR middle block $(\Delta l_6, \dots, \Delta l_{14})$ with no residual left.

Geometry completion and pure-kinetic collapse. Once the grouped real P_2 block is closed, the only remaining conservative scalar task is the one-dimensional static remainder. The grouped module itself predicts a definite static companion,

$$\Delta l_{15}^{(P_2 \text{ pred})} = \frac{\nu(293 - 308\nu - 102\nu^2)}{24},$$

so the actual scalar gap is explicit before any geometry interpretation is chosen. Allowing both body-local P_0 -type and pair/geometry representatives initially creates an apparent two-family ambiguity, but the exact mass identity

$$\nu(p^3 + q^3) = (1 - 3\nu)(p^2q + pq^2)$$

collapses that ambiguity algebraically. The resulting static counterterm is therefore unique and geometry-side,

$$\Delta L_{3,\text{ct}}^{\text{scalar}} = \frac{G^4 pq}{96r^4} (408\nu^2 + 1232\nu - 2080 + 63\pi^2)(p^2q + pq^2),$$

with Hamiltonian image $\Delta H_{3,\text{ct}}^{\text{scalar}} = -\Delta L_{3,\text{ct}}^{\text{scalar}}$. The final apparent pure-kinetic leftover is equally non-mysterious. At generic-frame level the free 3PN ordinary piece is simply

$$L_{3,\text{free}}^{\text{gen}} = \frac{5}{128}(m_A v_A^8 + m_B v_B^8),$$

while the exact free COM Hamiltonian yields

$$h_1^{\text{free}} = \frac{-5 + 35\nu - 70\nu^2 + 35\nu^3}{128}, \quad h_2 = h_3 = h_4 = h_5 = 0.$$

Applying the exact COM compiler then gives precisely the leftover

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16},$$

so the pure-kinetic slot is only the ordinary-Lagrangian compensation required to reproduce the universal free relativistic two-body COM Hamiltonian.

What the result fixes for the 2.5PN bridge. The main positive interface result is that the grouped real P_2 ontology survives its first honest conservative stress test. The richer local grouped compiler closes the full nine-slot middle block exactly, so the throat side does not need a qualitatively new non- P_2 conservative carrier for that sector. The remaining static slot is uniquely geometry-side, and the pure-kinetic slot is only a compiler image of the free COM Hamiltonian, so neither competes with the grouped quadrupole module for the middle-block burden. The remaining 2.5PN gap is therefore much narrower than “derive 3PN somehow.” The moving-throat PDE still has to supply the actual low-frequency constitutive origin of the grouped real P_2 coefficients, the isotropy verdict $a_2 = b_2 = 0$ on the true branch, and then the final passive/outgoing quadrupole normalization on that branch. The present theorem should therefore be read as removing the conservative algebraic uncertainty beneath that outgoing-normalization gap rather than as closing the gap itself.

3 Carry-forward dictionary and theorem target

This section is intentionally downstream of the parent 4D action paper, the full conservative 1PN and 2PN assembly papers, and the 2.5PN theorem-audit package. We do not repeat those longer derivations here. Instead, we record only the imported objects, frozen lower-order values, chart conventions, and exact target statements that the present 3PN proof chain actually uses. The purpose of the section is fourfold: to freeze notation, to separate exact imports from controlled reductions, to make clear what is being carried rather than rederived, and to state the theorem target that organizes the sections that follow.

The guiding principle is the same one used in the earlier conservative papers. What is exact in the parent reduction framework should remain visibly exact; what enters only after a declared near-zone, small-body, COM, or grouped-channel assumption should remain visibly reduced; and what is fixed only within the response hierarchy should remain visibly hierarchical rather than disguised as an assumption-free theorem of the fully solved moving-throat PDE. The present paper is therefore built on an imported dictionary rather than on a fresh rederivation of the full bulk problem.

3.1 Minimal imported 4D reduction dictionary

The parent theory lives on a $(4 + 1)$ -dimensional spacetime with coordinates

$$X^M = (t, x, y, z, w), \quad \mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \quad \mathbf{x} = (x, y, z) \in \mathbb{R}^3.$$

The fundamental bulk objects carried into the present paper are the matter field $\psi(\mathbf{X}, t)$, its density

$$\rho(\mathbf{X}, t) = |\psi(\mathbf{X}, t)|^2,$$

the optional localized gauge field $A_M(\mathbf{X}, t)$ with localization profile $Z(w)$, and the effective throat geometry variables

$$a(t), \quad L(t).$$

When electric charge is named explicitly in the inherited 4D notation, it is always the corrected branch quantity

$$q_* = \eta_Q e_*, \quad q_{\text{eff}} = \frac{q_*}{\sqrt{Z_{\text{int}}}}, \quad Z_{\text{int}} = \int_{-\infty}^{\infty} Z(w) dw,$$

not the gravity-side scalar coefficient κ_ρ that appears in the post-Newtonian ledger below.

A brane observer does not see the full bulk fields directly. Brane observables are introduced by a fixed normalized projection kernel $W(w)$,

$$\int_{-\infty}^{\infty} W(w) dw = 1,$$

through the projection map

$$\mathcal{P}_W[Q](t, \mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w) Q(t, \mathbf{x}, w) dw.$$

In particular,

$$\rho_{\text{br}} = \mathcal{P}_W[\rho], \quad \mathbf{J}_{\text{br}} = \mathcal{P}_W[(j^x, j^y, j^z)], \quad \mathbf{v}_{\text{br}} = \frac{\mathbf{J}_{\text{br}}}{\rho_{\text{br}}} \quad (\rho_{\text{br}} > 0).$$

This projection operation is distinct from *reduction*. Projection defines what a brane observer measures; reduction refers to the further controlled elimination of the extra coordinate under an explicit ansatz or scaling regime.

The exact projected continuity law carried forward from the parent theory is

$$\partial_t \rho_{\text{br}} + \nabla_3 \cdot \mathbf{J}_{\text{br}} = S_{\text{leak}},$$

with leakage source

$$S_{\text{leak}} = -[W(w)j^w]_{-\infty}^{+\infty} + \int_{-\infty}^{\infty} W'(w)j^w dw.$$

After Helmholtz decomposition of the brane velocity,

$$\mathbf{v}_{\text{br}} = \nabla_3 \phi_{\text{br}} + \mathbf{v}_T, \quad \nabla_3 \cdot \mathbf{v}_T = 0,$$

one obtains the exact longitudinal identity

$$\rho_{\text{br}} \nabla_3^2 \phi_{\text{br}} = S_{\text{leak}} - \partial_t \rho_{\text{br}} - (\nabla_3 \rho_{\text{br}}) \cdot (\nabla_3 \phi_{\text{br}} + \mathbf{v}_T).$$

Under the carried quasi-static near-zone assumptions this reduces to the Poisson hook

$$\nabla_3^2 \phi_{\text{br}} \simeq \frac{S_{\text{eff}}}{\rho_0}, \quad \Phi_N = \lambda_\Phi \phi_{\text{br}},$$

which is the origin of the Newtonian scalar sector used in the lower-order particle reduction.

The second imported reduction step is the controlled worldtube/coherent-defect limit. For a compact defect of size $a \ll r$ in a smooth external field, the center-of-mass reduction gives the usual point-particle force law

$$M_A \ddot{\mathbf{X}}_A = -M_A \nabla \Phi_{\text{ext}}(\mathbf{X}_A) + O\left(\frac{a^2}{r^2} \frac{\Phi_{\text{ext}}}{r}\right),$$

so that the universal point-particle source at leading order is controlled by the worldline/worldtube multipoles rather than by arbitrary internal details of the defect. That worldtube dictionary is what later allows the higher-order universal bridge to be packaged in terms of the orbital/worldtube STF quadrupole rather than a generic internal support excitation.

3.2 Frozen conservative carry-forward ledger

The present paper imports the conservative Newtonian+1PN+2PN ledger as frozen input, not as open fit parameters. The surviving lower-order viability conditions are

$$\kappa_\rho = 1, \quad n = 5, \quad \kappa_{\text{add}} = \frac{1}{2}, \quad \kappa_{\text{PV}} = \frac{3}{2}, \quad \beta_{\text{1PN}} = 3.$$

The first four values are the scalar mass-dressing, optical/barotropic, added-mass, and adiabatic breathing-response coefficients frozen by the earlier program. The fifth is the resulting carried

1PN ledger value. The present paper is not allowed to repair a 3PN mismatch by retuning any of these numbers.

When referenced, the parity-even wake data inherited from the full conservative 1PN assembly are also treated as frozen branch data,

$$\alpha^2 = \frac{3}{4}, \quad a_H = 0, \quad K_{\text{vec}} = \frac{2}{\pi^2},$$

not as live higher-order fit parameters.

The conservative 2PN assembly sharpened the ontology of the defect further. By the start of the present 3PN paper, the particle could no longer be treated as a pure scalar monopole with small corrections. The carried conservative structure already looked like a small throat-response theory with three distinct pieces:

1. a carried odd dipole wake sector,
2. a genuine even $P_0 \oplus P_2$ mouth/support layer,
3. and a separate geometry-closure channel.

This matters conceptually because the present 3PN work is *not* starting from a blank EFT. It is extending a structure that was already forced by the conservative 2PN match. In particular, the grouped real quadrupole bundle is not being introduced ad hoc at 3PN; it is the conservative reorganization of a support/geometry structure that was already visible at lower order.

What remains genuinely open from the lower-order side is not the existence of these channels but their full dynamical completion. In particular, the dynamic pole or compliance scales

$$\Omega_{1\perp}, \Omega_{10}, \Omega_0, \Omega_{20}, \Omega_{21}, \Omega_{22}, \Omega_g$$

are not claimed as already derived from the fully solved moving-throat PDE. This is one reason the present paper continues to speak in terms of a declared closure hierarchy rather than an assumption-free first-principles theorem.

Two structural consequences of this imported ledger should be stated explicitly. First, the present paper is not allowed to repair a higher-order mismatch by changing lower-order conservative constants that were already fixed below 3PN. Second, the defect already carries enough internal support structure that the 3PN analysis must distinguish carefully between universal point-particle content, self/static carry-forward content, grouped- P_2 middle-block content, and the separate static geometry lane.

3.3 Why the 2.5PN notes narrow the next conservative target to the grouped real P_2 bundle

The 2.5PN theorem audit is what makes the present 3PN problem sharply focused rather than diffuse. By the end of that audit, the live universal point-particle route had already narrowed to the orbital/worldtube STF quadrupole branch. In grouped real-channel language, that means the only higher-order conservative payload that still matters for the minimal branch is the grouped real

$$P_{20} \oplus P_{21} \oplus P_{22}$$

bundle.

More specifically, the 2.5PN notes reduce the future conservative task to the first even coefficients of that grouped real P_2 response. On the minimal isotropic branch, once the common conservative quadrupole data

$$(c_0, c_2, c_4) \quad \text{or equivalently} \quad (\overline{K}_0, \overline{K}_2, \overline{K}_4)$$

are known, the effective pole Ω_Q and the odd Burke–Thorne coefficient $\bar{\Gamma}_5$ are fixed automatically. The isotropy gate itself is also explicit in grouped language: the grouped triples must collapse to one isotropic trace, with the anisotropy defects vanishing.

That is why the present paper uses the grouped real P_2 bundle as its conservative front end. The question is no longer “derive all of higher PN” or “add more coefficients somehow.” The question is whether the conservative 3PN calculation lands on the same grouped- P_2 branch that the 2.5PN notes identified as the only surviving universal quadrupole route.

At 3PN the grouped conservative data are precisely the three $O(\omega^2)$ coefficients

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)},$$

or, equivalently, the grouped trace/anomaly variables

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \quad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \quad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2}.$$

The first isotropy test for the minimal branch is therefore

$$a_2 = 0, \quad b_2 = 0.$$

The present paper does not pretend to settle that isotropy verdict from a fully solved moving-throat PDE. What it does is close the conservative 3PN middle block in exactly the grouped language that the later bridge will need.

3.4 PN bookkeeping and invariant variables

The conservative expansion is organized exactly as in the earlier papers:

$$L_{\text{cons}} = L_0 + c^{-2}L_1 + c^{-4}L_2 + c^{-6}L_3 + O(c^{-8}).$$

So throughout this paper, “3PN” always means the coefficient of c^{-6} .

In the generic frame, the basic two-body invariants remain

$$\begin{aligned} \mathbf{r} &= \mathbf{x}_A - \mathbf{x}_B, & r &= |\mathbf{r}|, & \mathbf{n} &= \mathbf{r}/r, \\ v_A^2 &= \mathbf{v}_A^2, & v_B^2 &= \mathbf{v}_B^2, & v_{AB} &= \mathbf{v}_A \cdot \mathbf{v}_B, \\ v_{An} &= \mathbf{v}_A \cdot \mathbf{n}, & v_{Bn} &= \mathbf{v}_B \cdot \mathbf{n}. \end{aligned}$$

The usual local pair potentials are

$$U_A = \frac{Gm_B}{r}, \quad U_B = \frac{Gm_A}{r},$$

and in the strict one-body gate we write

$$U = \frac{GM}{r}.$$

In the COM sector, the reduced isotropic ordinary-Lagrangian basis uses the velocity invariants (v^2, d^2, u) with

$$d \equiv \dot{r}, \quad u \equiv \frac{GM}{rc^2}, \quad \nu \equiv \frac{m_A m_B}{(m_A + m_B)^2}.$$

The matching COM Hamiltonian basis uses the corresponding reduced momentum invariants (p^2, p_r^2, u) . In the present paper the ordinary COM coefficients are denoted l_i , the COM Hamiltonian coefficients h_i , and the residuals beyond the frozen self/static seed by Δl_i and Δh_i . This placement matters because one of the main 3PN lessons was that COM bookkeeping is a compiler convenience, not a license to treat naive ordinary COM substitution as a generic-frame theorem. For later compact formulas it is also convenient to write

$$v_\perp^2 \equiv v^2 - d^2,$$

but the invariant generic-frame quantities displayed above remain primary.

3.5 COM versus generic-frame conventions

The fixed convention carried forward from the present session is simple: use the COM basis whenever it simplifies the exact target or the diagonal compiler map, but treat the *generic-frame Hamiltonian compiler* as the authoritative uniqueness statement.

The reason for this convention is now settled. A naive COM reduction of the imported generic-frame ordinary 3PN target does *not* reproduce the correct COM ordinary target. By contrast, the Hamiltonian-first route does:

1. import the generic-frame ordinary target,
2. apply the exact cubic Legendre lift first,
3. then reduce the Hamiltonian to COM.

That sequence recovers the standard 3PN COM Hamiltonian exactly. The next sections therefore treat Hamiltonian-first lifting as the authoritative quotient principle and do not reopen the earlier naive-COM-reduction detour.

A second settled convention follows from the fixed ADM chart result. Once the lower-order ledger is frozen, the full generic-frame 3PN residual compiler is exact and trivial in that chart,

$$\Delta H_3 = -\Delta L_3.$$

So COM-blind families or ordinary contact freedoms that appear before the Hamiltonian chart is fixed are no longer true ambiguities afterward. In the fixed chart, the generic-frame Hamiltonian target determines the ordinary representative uniquely.

3.6 Notation firewall

The notation firewall from the corrected 4D / EM / 2PN stack must be kept intact.

First, electric-charge notation is completely separate from the conservative 3PN coefficient ledger. When charge appears at all, it uses the corrected variables

$$\eta_Q = \pm 1, \quad q_* = \eta_Q e_*, \quad q_{\text{eff}} = \frac{q_*}{\sqrt{Z_{\text{int}}}},$$

while the historical gravity-side bare $q = 1$ remains renamed as

$$\kappa_\rho = 1.$$

Quantized circulation belongs to the magnetic/vortical sector, not to the electric-charge dictionary.

Second, several overloaded symbols from the older literature must not be allowed to drift. The appendix-side source-weight vector

$$J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right)$$

is not an electromagnetic current. The pure-potential geometry-closure deficit Δ_{geom} and the monopole wall add-on ΔK_{00} are distinct objects. And in the present paper the grouped labels 20, 21, 22 always denote grouped real P_2 channels, not tensor indices.

Third, the grouped real P_2 variables are kept separate from the COM coefficient ledgers. The grouped conservative data are written as

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)},$$

or, after regrouping, as

$$\bar{u}_2, \quad a_2, \quad b_2,$$

and are never to be confused with the COM coefficients l_i and h_i . Similarly, geometric throat variables (a, L) , pole scales such as Ω_Q , and PN coefficients such as d_3 or $\mu_{\rho,3}$ should remain visibly distinct. The rest of the paper uses all three layers at once, and most avoidable confusion at this stage is not physics but notation drift.

Finally, in the static-completion subsection it will be convenient to write

$$p \equiv m_A, \quad q \equiv m_B,$$

but only there, and only as mass shorthands. They are not electric charges.

3.7 Exact theorem target

The theorem target of the paper is not an unconstrained coefficient fit. It has a strict one-body front door, an exact imported reduced target, and a final fixed-chart split.

Strict one-body gate. The exact isotropic Schwarzschild target through c^{-6} is

$$\frac{L_{3\text{PN},1\text{body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4.$$

Any acceptable 3PN one-body/self sector must reproduce this answer exactly. The first task of the paper is therefore to show that the carried denominator-style package can be repaired minimally by a finite three-parameter ledger rather than by an uncontrolled extension of the lower-order self sector.

Exact reduced COM target. Once the natural symmetric self/static seed is subtracted from the imported exact GR COM target, the remaining 15-slot residual vector is

$$\begin{aligned} \Delta l_1 &= \frac{3\nu(3\nu-1)(4\nu-1)}{16}, & \Delta l_2 &= \Delta l_3 = \Delta l_4 = \Delta l_5 = 0, \\ \Delta l_6 &= \frac{\nu(38-116\nu-57\nu^2)}{16}, & \Delta l_7 &= \frac{\nu^2(20-69\nu)}{16}, \\ \Delta l_8 &= \frac{3\nu^2(3-11\nu)}{16}, & \Delta l_9 &= \frac{5\nu^3}{16}, \\ \Delta l_{10} &= \frac{\nu(129-98\nu+52\nu^2)}{16}, & \Delta l_{11} &= \frac{\nu(-3+52\nu+124\nu^2)}{16}, \\ \Delta l_{12} &= \frac{\nu(-5+11\nu+48\nu^2)}{12}, \\ \Delta l_{13} &= -\frac{\nu(244+3\pi^2+1272\nu+96\nu^2)}{192}, & \Delta l_{14} &= \frac{\nu(452+3\pi^2-384\nu-224\nu^2)}{64}, \\ \Delta l_{15} &= \frac{\nu(-908+63\pi^2)}{96}. \end{aligned}$$

This vector is the exact reduced target data against which every later compiler, grouped- P_2 , and geometry statement is checked.

Fixed-chart theorem target. Within the same declared closure hierarchy used successfully at 1PN and 2PN, the goal is to show all of the following.

1. The exact one-body repair ledger closes the strict Schwarzschild gate.

2. The fixed ADM chart generic-frame compiler reduces to

$$\Delta H_3 = -\Delta L_3,$$

so the correct generic-frame representative is unique once the Hamiltonian target is imposed.

3. The full nine-slot middle interaction block is carried exactly by the richer grouped real P_2 family

$$(T_{20}, T_{21}, T_{22}, S_{20}, S_{21}, S_{22}, V_{20}, V_{21}, V_{22}).$$

4. The remaining static scalar slot is uniquely geometry-side after the exact sigma-collapse removes the apparent P_0 /geometry ambiguity.
5. The apparent pure-kinetic leftover is only the ordinary-Lagrangian counterimage of the universal free relativistic two-body COM Hamiltonian, rather than a new throat datum.

Equivalently, the exact comparable-mass 3PN residual to be proved in the rest of the paper is

$$\Delta \hat{L}_3^{\text{GR}} = \underbrace{\Delta l_1 v^8}_{\text{compiler image of the free COM Hamiltonian}} + \underbrace{L_{P_2}^{\text{mid}}}_{\text{exact grouped real } P_2 \text{ closure}} + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}.$$

This is the most compact statement of the theorem target. It says that once the strict one-body gate, the exact reduced COM target, the Hamiltonian-first lift, the grouped real P_2 module, and the static sigma-collapse are all handled correctly, the conservative 3PN bottleneck is no longer algebraic.

The correct theorem-status reading is the same as in the earlier full conservative papers. The statement above is a *full assembly target within a declared closure hierarchy*, not an assumption-free theorem of the fully solved moving-throat PDE. What remains open after the present paper is therefore not “more 3PN coefficients.” What remains open is the deeper moving-throat derivation of the grouped- P_2 constitutive law and unique geometry lane, and then the insertion of that data into the already narrowed passive/outgoing quadrupole bridge.

4 One-body 3PN gate, self/static seed, and exact cubic compiler

The first genuine theorem gate of the 3PN program is the strict test-mass sector. Before any comparable-mass lift is attempted, one must know exactly what the carried one-body/self-sector is allowed to reduce to, and one must know whether the lower-order denominator/static packaging survives unchanged. At 3PN it does not. The strict one-body audit shows that the order- c^{-6} sector is already richer than a one-parameter continuation of the 2PN closure: there is an exact Schwarzschild target, a minimal three-parameter repair ledger, and then a unique natural symmetric self/static promotion that isolates all bodywise content before the genuine comparable-mass residual is solved. The second ingredient needed before that lift can become finite is the exact cubic-order perturbative Legendre compiler. Once the carried 1PN/2PN ledger is frozen, that compiler makes the 3PN Hamiltonian content a rigid algebraic image of the new ordinary-Lagrangian block rather than a blind full recomputation.

4.1 Exact Schwarzschild test-mass target through 3PN

The strict one-body gate is the isotropic Schwarzschild worldline action. Using

$$u \equiv \frac{U}{c^2} = \frac{GM}{rc^2},$$

the exact test-mass Lagrangian in isotropic Schwarzschild coordinates is

$$\frac{L_{\text{Schw}}}{m} = -c^2 \sqrt{\left(\frac{1-u/2}{1+u/2}\right)^2 - (1+u/2)^4 \frac{v^2}{c^2}}.$$

Expanding through total order c^{-6} gives

$$\begin{aligned} \frac{L_{\text{Schw}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left(\frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2 \right) \\ & + \frac{1}{c^4} \left(\frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 + \frac{1}{4}U^3 \right) \\ & + \frac{1}{c^6} \left(\frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4 \right) + O(c^{-8}). \end{aligned}$$

So the exact 3PN one-body destination is

$$\boxed{\frac{L_{\text{3PN,1body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4.}$$

This plays exactly the same role here that the strict Schwarzschild test-mass check played at lower order: it is the unique one-body answer that every later comparable-mass construction must reduce to.

Two features of the target are worth recording immediately. First, the universal v^8 and Uv^6 coefficients continue the lower-order relativistic pattern exactly. Second, the remaining three slots

$$U^2v^4, \quad U^3v^2, \quad U^4,$$

already show that the full 3PN self sector cannot be inferred by naive pattern continuation alone. The rest of the subsection fixes precisely how much new one-body information is needed.

4.2 Minimal one-body repair ledger

The first algebraic test is whether the carried denominator-style self sector can simply be pushed one order higher. Write the minimal cubic extension as

$$L_{\text{red}} = -mc^2(1-u)\sqrt{1 - \frac{v^2/c^2}{D(u)}}, \quad D(u) = 1 - 4u + 8u^2 + d_3u^3 + O(u^4).$$

Expanding through c^{-6} yields

$$\begin{aligned} \frac{L_{\text{red}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left(\frac{1}{8}v^4 + \frac{3}{2}Uv^2 \right) \\ & + \frac{1}{c^4} \left(\frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 \right) \\ & + \frac{1}{c^6} \left(\frac{5}{128}v^8 + \frac{11}{16}Uv^6 + 3U^2v^4 + \left(-4 - \frac{d_3}{2}\right)U^3v^2 \right) + O(c^{-8}). \end{aligned}$$

So the carried denominator logic already reproduces the universal v^8 and Uv^6 coefficients automatically, but it leaves the U^3v^2 and U^2v^4 slots unresolved. Matching the exact Schwarzschild coefficient of U^3v^2 forces

$$-4 - \frac{d_3}{2} = \frac{13}{8} \quad \implies \quad \boxed{d_3 = -\frac{45}{4}.}$$

It is useful to note that the unextended 2PN invariant-denominator philosophy would have predicted instead

$$d_3^{\text{carry}} = \frac{21783}{2432},$$

which is nowhere near the exact target. So already at the U^3v^2 level, 3PN requires a genuinely new cubic invariant correction if one wants to retain that denominator-style packaging.

The pure static slot gives an independent gate. Extend the local mass-scaling sector one order further by a cubic static coefficient $\mu_{\rho,3}$:

$$\frac{L_{\text{static,1body}}}{m} = U - \frac{1}{2} \frac{U^2}{c^2} + \frac{1}{4} \frac{U^3}{c^4} - \frac{\mu_{\rho,3}}{2} \frac{U^4}{c^6} + O(c^{-8}).$$

Matching the exact static Schwarzschild term

$$-\frac{1}{8} \frac{U^4}{c^6}$$

then gives

$$-\frac{\mu_{\rho,3}}{2} = -\frac{1}{8} \quad \Longrightarrow \quad \boxed{\mu_{\rho,3} = \frac{1}{4}}.$$

So the denominator datum and the static gate are both fixed exactly and independently.

At this point the one-body problem is still not closed. After imposing $d_3 = -45/4$ and $\mu_{\rho,3} = 1/4$, the denominator-style self sector still predicts

$$3U^2v^4 = \frac{48}{16}U^2v^4,$$

whereas the exact Schwarzschild target demands

$$\frac{47}{16}U^2v^4.$$

So a residual one-body discrepancy of

$$-\frac{1}{16}U^2v^4$$

remains. This cannot be absorbed by d_3 , which controls only the U^3v^2 slot, and it cannot be absorbed by $\mu_{\rho,3}$, which controls only the pure static U^4 slot. The minimal repair is therefore to admit one explicit new self invariant,

$$s_{24} \frac{U^2v^4}{c^6}, \quad \boxed{s_{24} = -\frac{1}{16}}.$$

Collecting everything, the exact one-body repair ledger is

$$\boxed{\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16}}.$$

This is the cleanest strict one-body lesson of the 3PN analysis. The 3PN gate does *not* say “continue the 2PN denominator idea by one more coefficient.” It says that three distinct data are already needed at the strict test-mass level: a cubic static gate, a cubic denominator datum, and one genuinely new self-sector slot.

4.3 Natural 3PN self/static seed

Once the exact one-body target and its minimal repair ledger are frozen, the natural next step is the same one that worked at 2PN: promote the strict one-body structure symmetrically into a two-body self/static seed while keeping all genuinely comparable-mass interaction data outside the seed. The rule is to remain strictly bodywise and symmetric so that the exact test-mass reduction is automatic.

Start from the exact one-body 3PN target

$$\frac{L_{3\text{PN},1\text{body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4.$$

For body A in the field of body B , use

$$U_A = \frac{Gm_B}{r}, \quad m \rightarrow m_A, \quad v \rightarrow v_A,$$

and for body B in the field of body A , use

$$U_B = \frac{Gm_A}{r}, \quad m \rightarrow m_B, \quad v \rightarrow v_B.$$

Adding the two bodywise copies gives the natural symmetric self/static 3PN seed,

$$L_{3,\text{seed}} = \frac{5}{128}(m_A v_A^8 + m_B v_B^8) + \frac{11Gm_A m_B}{16r}(v_A^6 + v_B^6)$$

$$+ \frac{47G^2 m_A m_B}{16r^2}(m_B v_A^4 + m_A v_B^4) + \frac{13G^3 m_A m_B}{8r^3}(m_B^2 v_A^2 + m_A^2 v_B^2) - \frac{G^4 m_A m_B}{8r^4}(m_B^3 + m_A^3),$$

with the understanding that this entire block is the coefficient of c^{-6} .

This seed has three properties that make it the correct starting object for the 3PN lift.

First, it is manifestly symmetric under $A \leftrightarrow B$.

Second, it preserves the strict one-body gate exactly. In the test-mass limit, the light-body sector reduces to the exact Schwarzschild target, while the heavy-body copy collapses to a self piece that does not contaminate the test-mass dynamics.

Third, it isolates the universal bodywise content before the genuine comparable-mass residual is solved. In the later COM chart this means the seed is sparse: before lower-order compiler feedback, only the self/static slots are switched on directly, and the remaining 3PN structure can be interpreted cleanly as residual beyond this frozen bodywise core.

That is why this seed is the right starting point for the full 3PN lift. It is neither a guess about the complete comparable-mass answer nor a vague “start somewhere” ansatz. It is the unique symmetric promotion of the exact one-body 3PN data into the same self/static lane that already organized the 2PN construction successfully.

4.4 Exact cubic Legendre-transform identity

The second indispensable ingredient is the exact cubic-order perturbative Legendre compiler. The 2PN assembly already relied on the perturbative Legendre transform through quadratic order. At 3PN the new content enters at cubic order, and once the carried 1PN/2PN ledger is frozen, the resulting Hamiltonian feedback is rigid.

Write the conservative ordinary Lagrangian as

$$L(v) = L_0(v) + \varepsilon L_1(v) + \varepsilon^2 L_2(v) + \varepsilon^3 L_3(v), \quad \varepsilon = c^{-2},$$

with quadratic zeroth-order block

$$L_0(v) = \frac{1}{2}v^T M v,$$

where the mass matrix M is constant at the order of the perturbative inversion. Define the Newtonian momentum-space velocity

$$v_0 = M^{-1}p,$$

and the lower-order derivative data at v_0 ,

$$A_0 \equiv \left. \frac{\partial L_1}{\partial v} \right|_{v_0}, \quad B_0 \equiv \left. \frac{\partial L_2}{\partial v} \right|_{v_0}, \quad C_0 \equiv \left. \frac{\partial^2 L_1}{\partial v^2} \right|_{v_0}.$$

Then the perturbative momentum inversion

$$v = v_0 + \varepsilon v_1 + \varepsilon^2 v_2 + O(\varepsilon^3)$$

solves exactly to

$$v_1 = -M^{-1}A_0, \quad v_2 = M^{-1}C_0M^{-1}A_0 - M^{-1}B_0.$$

Substituting into

$$H(p) = p \cdot v - L(v)$$

gives the cubic-order Hamiltonian expansion

$$H = H_0 + \varepsilon H_1 + \varepsilon^2 H_2 + \varepsilon^3 H_3 + O(\varepsilon^4),$$

with

$$H_1 = -L_1(v_0),$$

$$H_2 = -L_2(v_0) + \frac{1}{2}A_0^T M^{-1}A_0,$$

and the key 3PN identity

$$H_3 = -L_3(v_0) + A_0^T M^{-1}B_0 - \frac{1}{2}A_0^T M^{-1}C_0M^{-1}A_0.$$

This is the exact 3PN analogue of the quadratic compiler identity used at 2PN. Its practical meaning is that once the lower-order 1PN/2PN blocks are frozen, the genuinely new 3PN Hamiltonian content enters only through the explicit term $-L_3(v_0)$, while the feedback pieces are already determined by the carried lower-order ledger. That is precisely what turns the later comparable-mass lift into a finite algebraic residual problem rather than a blind variational search.

The next section will apply this compiler first in the reduced COM chart and then, more importantly, in the fixed generic-frame Hamiltonian chart. The key lesson to keep in view is already visible here: at 3PN the right bridge is not “reduce first and hope.” It is exact compiler algebra first, followed only then by the reduced comparison and fixed-chart uniqueness analysis.

5 Comparable-mass scaffold, COM target, and fixed-chart uniqueness

The strict one-body audit carried out immediately before this section fixes everything that is universal and bodywise at 3PN, but it does not yet solve the genuinely comparable-mass problem. The next task is to convert the full nonspinning order- c^{-6} ledger into a finite algebraic residual, to determine exactly what the center-of-mass reduction does and does not fix, and then to prove that the remaining apparent generic-frame ambiguity disappears once the fixed ADM Hamiltonian chart is imposed. This is therefore the section in which the 3PN program stops being “more coefficients somewhere” and becomes a rigid scaffold–compiler–quotient chain.

The logic is simple but unforgiving. First, one freezes a reduced 15-slot COM basis and the exact cubic ordinary/Hamiltonian compiler in that chart. Second, one imports the exact GR COM target and isolates the full residual beyond the frozen self/static seed. Third, one lifts that residual to a finite exchange-symmetric generic-frame scaffold and checks what COM projection still fails to determine. Fourth, one diagnoses the middle-block obstruction and repairs it by a tiny ν -dressed seed sector that vanishes in the strict one-body limit. Finally, one proves that the decisive quotient principle is Hamiltonian-first rather than naive ordinary-Lagrangian COM reduction, and that in the fixed ADM chart the full generic-frame residual compiler is just the slotwise sign flip

$$\Delta H_3 = -\Delta L_3.$$

At that point the comparable-mass 3PN bottleneck is no longer algebraic. What remains is the physical interpretation of the exact residual already isolated in this section.

5.1 Reduced COM 15-slot basis and diagonal ordinary/Hamiltonian compiler

In the standard isotropic center-of-mass chart, and suppressing the overall reduced-mass normalization already fixed by the lower-order papers, write the conservative reduced ordinary Lagrangian as

$$\hat{L} = \hat{L}_0 + c^{-2}\hat{L}_1 + c^{-4}\hat{L}_2 + c^{-6}\hat{L}_3 + O(c^{-8}),$$

with $\hat{L}_0$, $\hat{L}_1$, and $\hat{L}_2$ already frozen. The new 3PN coefficient is expanded in the natural 15-slot ordinary basis

$$\begin{aligned} \hat{L}_3 = & l_1 v^8 + l_2 v^6 \dot{r}^2 + l_3 v^4 \dot{r}^4 + l_4 v^2 \dot{r}^6 + l_5 \dot{r}^8 \\ & + \frac{1}{r} (l_6 v^6 + l_7 v^4 \dot{r}^2 + l_8 v^2 \dot{r}^4 + l_9 \dot{r}^6) \\ & + \frac{1}{r^2} (l_{10} v^4 + l_{11} v^2 \dot{r}^2 + l_{12} \dot{r}^4) + \frac{1}{r^3} (l_{13} v^2 + l_{14} \dot{r}^2) + \frac{l_{15}}{r^4}. \end{aligned}$$

The matching isotropic COM Hamiltonian basis is

$$\begin{aligned} \hat{H}_3 = & h_1 p^8 + h_2 p^6 p_r^2 + h_3 p^4 p_r^4 + h_4 p^2 p_r^6 + h_5 p_r^8 \\ & + \frac{1}{r} (h_6 p^6 + h_7 p^4 p_r^2 + h_8 p^2 p_r^4 + h_9 p_r^6) \\ & + \frac{1}{r^2} (h_{10} p^4 + h_{11} p^2 p_r^2 + h_{12} p_r^4) + \frac{1}{r^3} (h_{13} p^2 + h_{14} p_r^2) + \frac{h_{15}}{r^4}, \end{aligned}$$

where p^2 and p_r are the reduced isotropic COM momentum invariants.

The importance of this chart is that the exact cubic Legendre transform carried from the preceding one-body section diagonalizes slot by slot. The full forward map is

$$\begin{aligned} & \boxed{h_1 = -l_1 + \frac{3}{16}\nu - \frac{21}{16}\nu^2 + \frac{9}{4}\nu^3} \\ & \boxed{h_2 = -l_2, \quad h_3 = -l_3, \quad h_4 = -l_4, \quad h_5 = -l_5} \\ & \boxed{h_6 = -l_6 + \frac{1}{4} + \frac{7}{8}\nu - \frac{35}{8}\nu^2 - \frac{21}{4}\nu^3} \\ & \boxed{h_7 = -l_7 + \frac{11}{8}\nu^2 - \frac{9}{2}\nu^3, \quad h_8 = -l_8 + \frac{3}{4}\nu^2 - \frac{9}{4}\nu^3, \quad h_9 = -l_9} \\ & \boxed{h_{10} = -l_{10} + \frac{5}{4} + \frac{15}{8}\nu + \frac{123}{8}\nu^2 + \frac{13}{4}\nu^3} \end{aligned}$$

$$h_{11} = -l_{11} + \frac{7}{8}\nu + \frac{41}{8}\nu^2 + \frac{31}{4}\nu^3$$

$$h_{12} = -l_{12} + \frac{9}{2}\nu^2 + 4\nu^3$$

$$h_{13} = -l_{13} - \frac{3}{2} - \frac{59}{4}\nu - \frac{25}{4}\nu^2 - \frac{1}{2}\nu^3$$

$$h_{14} = -l_{14} + \frac{7}{4}\nu - \frac{31}{4}\nu^2 - \frac{7}{2}\nu^3, \quad h_{15} = -l_{15}.$$

The inverse map is immediate: one simply moves the known lower-order feedback polynomials to the opposite side. This is the decisive reduced-chart fact that makes the 3PN problem finite. Once the lower-order ledger is frozen, the imported GR COM target can be solved slot by slot, and the entire comparable-mass question reduces to an explicit residual rather than a blind variational search.

5.2 Imported exact GR COM target and residual extraction

With the diagonal compiler in hand, import the exact GR 3PN reduced COM target in the standard ADM chart and subtract the natural one-body/self-static seed constructed in the preceding one-body section. The result is the exact comparable-mass residual vector

$$\begin{aligned} \Delta l_1 &= \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}, & \Delta l_2 &= \Delta l_3 = \Delta l_4 = \Delta l_5 = 0, \\ \Delta l_6 &= \frac{\nu(38 - 116\nu - 57\nu^2)}{16}, & \Delta l_7 &= \frac{\nu^2(20 - 69\nu)}{16}, \\ \Delta l_8 &= \frac{3\nu^2(3 - 11\nu)}{16}, & \Delta l_9 &= \frac{5\nu^3}{16}, \\ \Delta l_{10} &= \frac{\nu(129 - 98\nu + 52\nu^2)}{16}, & \Delta l_{11} &= \frac{\nu(-3 + 52\nu + 124\nu^2)}{16}, \\ \Delta l_{12} &= \frac{\nu(-5 + 11\nu + 48\nu^2)}{12}, \\ \Delta l_{13} &= -\frac{\nu(244 + 3\pi^2 + 1272\nu + 96\nu^2)}{192}, & \Delta l_{14} &= \frac{\nu(452 + 3\pi^2 - 384\nu - 224\nu^2)}{64}, \\ \Delta l_{15} &= \frac{\nu(-908 + 63\pi^2)}{96}. \end{aligned}$$

This is the exact reduced 3PN target data set against which every later scaffold, quotient, grouped- P_2 , and geometry statement is checked.

The pattern of the residual is already highly informative. The first five slots show that the only free-particle mismatch beyond the self/static seed is the single ν^8 coefficient Δl_1 ; there is no comparable-mass residual in any of the other pure-kinetic monomials. The nine slots

$$(\Delta l_6, \dots, \Delta l_{14})$$

constitute the full middle interaction block, while the single static coefficient Δl_{15} is the entire quartic potential remainder. So before any generic-frame lifting is attempted, the COM data already exhibit the exact split that will survive to the end of the paper:

1. a one-dimensional free-kinetic correction,
2. a nine-dimensional middle interaction block,

3. and a one-dimensional static completion.

The rest of this section explains how these three pieces appear in the generic frame, why COM data alone do not yet fix the answer uniquely, and why the fixed Hamiltonian chart eventually removes that ambiguity completely.

5.3 Generic-frame residual scaffold

To lift the COM data back to the full two-body chart, use the exchange-symmetric generic-frame invariants

$$a = v_A^2, \quad b = v_B^2, \quad c = \mathbf{v}_A \cdot \mathbf{v}_B, \quad d = v_{An}, \quad e = v_{Bn},$$

with

$$p = m_A, \quad q = m_B.$$

After subtracting the frozen one-body/self-static seed, the honest generic-frame comparable-mass scaffold is

$$L_{3,\text{full}} = L_{3,\text{seed}} + L_{3,\text{res}},$$

with

$$L_{3,\text{res}} = \frac{Gm_A m_B}{r} \sum_{i=1}^{24} q_i Q_i + \frac{G^2 m_A m_B}{r^2} \sum_{j=1}^{17} t_j T_j + \frac{G^3 m_A m_B}{r^3} \sum_{k=1}^8 s_k S_k + \frac{G^4 m_A m_B}{r^4} u_1 U_{01},$$

where the unique exchange-symmetric static cross polynomial is

$$U_{01} = p^2 q + p q^2.$$

The important point is not the eventual sparsity of the solved answer but the size and symmetry of the residual search space before any contact/gauge quotient is imposed. The full generic-frame interaction problem is finite from the start: the block sizes are

$$24, \quad 17, \quad 8, \quad 1,$$

for the G/r , G^2/r^2 , G^3/r^3 , and G^4/r^4 blocks, for a total of 50 interaction directions.

Projecting this scaffold to COM gives the exact blockwise image dimensions

$$\boxed{12, \quad 6, \quad 4, \quad 1,}$$

corresponding to the four G/r slots (l_6, l_7, l_8, l_9) each supporting ν , ν^2 , and ν^3 , the three G^2/r^2 slots (l_{10}, l_{11}, l_{12}) supporting only ν and ν^2 , the two G^3/r^3 slots (l_{13}, l_{14}) again supporting only ν and ν^2 , and the single static slot l_{15} supporting only a coefficient proportional to ν . So the blockwise nullities are

$$\boxed{12, \quad 11, \quad 4, \quad 0,}$$

and the total COM-blind interaction freedom is

$$\boxed{27.}$$

This is the first exact statement of what the solved COM target can and cannot determine by itself. The scaffold is large enough; the question is how to place the seed correctly and how to quotient the COM-blind family once the Hamiltonian chart is imposed.

5.4 COM projection, middle-block obstruction, and seed repair

The first lift test is whether the natural one-body/self-static seed already lands in the correct COM image block by block. The answer is almost yes, but not quite. The G/r sextic block and the G^4/r^4 static slot are already compatible with the exact COM target. The only obstruction sits in the middle G^2/r^2 and G^3/r^3 blocks.

The reason is simple. In the constant-coefficient generic-frame image, the G^2/r^2 and G^3/r^3 blocks carry only ν and ν^2 support under COM projection. But after subtracting the natural seed, the exact COM residual still contains small ν^3 tails in precisely those blocks. The obstruction pieces are

$$\delta l_{10}^{\text{obs}} = \frac{13}{4}\nu^3, \quad \delta l_{11}^{\text{obs}} = \frac{31}{4}\nu^3, \quad \delta l_{12}^{\text{obs}} = 4\nu^3,$$

for the G^2/r^2 block, and

$$\delta l_{13}^{\text{obs}} = -\frac{1}{2}\nu^3, \quad \delta l_{14}^{\text{obs}} = -\frac{7}{2}\nu^3,$$

for the G^3/r^3 block. This is not a failure of the scaffold. It is a seed-placement problem: the natural self/static seed is exact in the strict one-body sector, but it is not yet aligned with the mass-degree structure of the generic-frame middle blocks.

The repair is remarkably small. Define the refined seed correction sector

$$\Delta T_\nu = \frac{pq}{4(p+q)} (13ab + 31cde + 16d^2e^2),$$

$$\Delta S_\nu = \frac{p^2q^2}{2(p+q)^2} (c + 7de).$$

Then the refined seed is

$$L_{3,\text{seed}}^{\text{ref}} = L_{3,\text{seed}} + \frac{G^2pq}{r^2} \Delta T_\nu + \frac{G^3pq}{r^3} \Delta S_\nu.$$

These extra pieces do exactly what is needed:

1. they vanish in the strict one-body limit because they are proportional to ν ,
2. they leave the G/r and G^4/r^4 blocks untouched,
3. and under COM reduction they reproduce precisely the missing ν^3 obstruction tails.

So the middle-block obstruction is bookkeeping rather than ontology. After this repair, each interaction block admits an exact COM-compatible generic-frame representative, and what remains is no longer a compatibility failure but a quotient problem.

5.5 The COM-null ideal and the canonical quotient slice

Once the refined seed is fixed, the remaining generic-frame freedom is exactly the family invisible in the COM quotient algebra. In the COM frame the body velocities satisfy the three linear-momentum identities

$$\mathcal{C}_1 \equiv pa + qc = 0, \quad \mathcal{C}_2 \equiv pc + qb = 0, \quad \mathcal{C}_3 \equiv pd + qe = 0,$$

and the three collinearity identities

$$\mathcal{C}_4 \equiv ab - c^2 = 0, \quad \mathcal{C}_5 \equiv ae - cd = 0, \quad \mathcal{C}_6 \equiv bd - ce = 0.$$

These generate the COM-null ideal

$$\mathcal{I}_{\text{COM}} = \langle \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4, \mathcal{C}_5, \mathcal{C}_6 \rangle.$$

The full 27-dimensional nullspace of the 24/17/8/1 COM projection matrix is exactly the generic-frame freedom invisible in the quotient algebra

$$\mathbb{Q}[a, b, c, d, e, p, q] / \mathcal{I}_{\text{COM}}.$$

A useful refinement is that the T - and S -block null directions already lie in the smaller linear-momentum ideal

$$\langle \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3 \rangle,$$

whereas the Q -block null directions need the full ideal $\mathcal{I}_{\text{COM}}$.

This makes it natural to define a canonical sparse algebraic representative by quotienting by the COM-null ideal and setting those 27 null coefficients to zero. In that canonical slice one obtains an explicit generic-frame answer that reproduces the full solved COM target exactly. The details are lengthy and are best recorded appendix-side, but the conceptual point belongs in the main text: this canonical quotient slice is an exact algebraic representative, not a claim that every COM-blind direction is ordinary gauge. That distinction becomes crucial in the next subsection.

5.6 Contact generator, gauge orbit, and the 11-versus-27 distinction

A 3PN ordinary contact transformation is generated by total derivatives of odd scalars. With the correct exchange rule

$$A \leftrightarrow B, \quad c \mapsto c, \quad d \mapsto -e, \quad e \mapsto -d, \quad p \leftrightarrow q,$$

the raw odd family contains

$$\begin{array}{ll} 31 & (r \times \text{degree-7}), \quad 12 \quad (G \times \text{degree-5}), \\ 8 & \left(\frac{G^2}{r} \times \text{degree-3} \right), \quad 2 \quad \left(\frac{G^3}{r^2} \times \text{degree-1} \right), \end{array}$$

for a total of

$$\boxed{53}$$

candidates. Imposing the scaffold-preserving condition cuts this to

$$\boxed{22}$$

dimensions, and imposing zero COM image cuts it once more. The actual scaffold-preserving and COM-preserving ordinary contact/gauge orbit is therefore

$$\boxed{11\text{-dimensional.}}$$

So the exact reduction chain is

$$53 \longrightarrow 22 \longrightarrow 11.$$

A sparse basis for this true ordinary contact orbit factors only through the smaller subideal generated by

$$\mathcal{C}_3, \quad \mathcal{C}_4, \quad \mathcal{C}_5, \quad \mathcal{C}_6,$$

not through all of $\mathcal{I}_{\text{COM}}$. In particular, no generator uses $\mathcal{C}_1$ or $\mathcal{C}_2$, and the simple ordinary contact family leaves the static block untouched. Its image inside the compact generic-frame scaffold has blockwise ranks

$$\boxed{\text{rank}_Q = 6, \quad \text{rank}_T = 9, \quad \text{rank}_S = 4, \quad \text{rank}_U = 0.}$$

So the actual ordinary contact orbit is only a strict subspace of the larger 27-dimensional COM-blind family. This is the exact content of the 11-versus-27 distinction:

COM blindness and ordinary contact/gauge freedom are not identical.

The canonical quotient slice is therefore the right sparse algebraic reference chart, but it is not yet the final physical uniqueness theorem.

5.7 Why naive ordinary COM reduction fails

The decisive subtlety appears once the full generic-frame ordinary target is imported from the standard 3PN literature, with ambiguity fixed by

$$\lambda = -\frac{3}{11}\omega_s - \frac{1987}{3080}, \quad \omega_s = 0, \quad \Rightarrow \quad \lambda = -\frac{1987}{3080}.$$

After subtraction of the frozen natural one-body/self-static seed, the exact ordinary generic-frame residual lands entirely inside the compact 24/17/8/1 scaffold and still vanishes in the strict test-mass branch, so the imported target itself is perfectly consistent with the one-body gate.

The problem is not the target. The problem is the order of reduction. Projecting the imported generic-frame *ordinary* target directly to COM does not reproduce the exact COM ordinary target. The mismatches are

$$\begin{aligned} \delta l_6 &= \frac{\nu(80\nu^2 + 48\nu - 17)}{16}, & \delta l_7 &= \frac{3\nu(24\nu^2 - 10\nu + 1)}{16}, & \delta l_8 &= \frac{3\nu^2(4\nu - 1)}{8}, & \delta l_9 &= 0, \\ \delta l_{10} &= \frac{\nu(-52\nu^2 - 123\nu + 34)}{16}, & \delta l_{11} &= \frac{\nu(-124\nu^2 - 97\nu + 32)}{16}, & \delta l_{12} &= \nu^2(1 - 4\nu), \\ \delta l_{13} &= \frac{\nu(4\nu^2 + 27\nu - 7)}{8}, & \delta l_{14} &= \frac{\nu(28\nu^2 + 69\nu - 19)}{8}, & \delta l_{15} &= 0. \end{aligned}$$

So naive ordinary COM reduction agrees only in the l_9 and l_{15} slots. This is not a contradiction. It is exactly the same subtlety already known from standard 3PN dynamics: the reduced relative ordinary Lagrangian does not follow from the general-frame ordinary representative by simple substitution. Naive ordinary COM reduction is therefore not the final quotient principle.

5.8 Hamiltonian-first lift and exact COM recovery

The cure is to apply the exact cubic Legendre map in the generic frame first, and only then reduce to COM. Carrying forward the exact identity from the preceding one-body section,

$$H_3 = -L_3(v_0) + A_0^T M^{-1} B_0 - \frac{1}{2} A_0^T M^{-1} C_0 M^{-1} A_0,$$

with

$$v_0 = M^{-1}P, \quad A_0 = \left. \frac{\partial L_1}{\partial v} \right|_{v_0}, \quad B_0 = \left. \frac{\partial L_2}{\partial v} \right|_{v_0}, \quad C_0 = \left. \frac{\partial^2 L_1}{\partial v^2} \right|_{v_0},$$

one computes the generic-frame 3PN Hamiltonian first and substitutes the COM relations only afterward. That procedure gives the exact theorem

$$H_{3,\text{COM}}^{(\text{lift})} = H_{3,\text{COM}}^{\text{target}}.$$

Equivalently,

$$\text{generic-frame Legendre lift first} \implies H_{3,\text{COM}}^{\text{lift}} = H_{3,\text{COM}}^{\text{GR}}.$$

So the earlier mismatch is diagnosed completely:

naive COM reduction of ordinary $L_3 \neq$ correct COM ordinary target,

but the imported generic-frame ordinary target itself is fine. The order of operations was wrong, not the target. This is the point where the 3PN lift stops being a suspicion about the generic frame and becomes a clean chart issue.

5.9 Fixed-chart generic-frame uniqueness theorem

Once the Hamiltonian-first route is accepted, the remaining ambiguity question is no longer “is the target right?” but “which ordinary representative survives in the fixed Hamiltonian chart?” The answer is now exact.

Use the same formal invariant symbols as in the ordinary scaffold, but reinterpret them in the generic-frame Hamiltonian chart as Newtonian-order momentum invariants:

$$a = \frac{P_A^2}{m_A^2}, \quad b = \frac{P_B^2}{m_B^2}, \quad c = \frac{P_A \cdot P_B}{m_A m_B}, \quad d = \frac{n \cdot P_A}{m_A}, \quad e = \frac{n \cdot P_B}{m_B},$$

with again

$$p = m_A, \quad q = m_B.$$

In this chart the generic-frame Hamiltonian interaction scaffold has exactly the same 24/17/8/1 block structure as the ordinary one. Write

$$L_3 = L_{3,\text{seed}} + \Delta L_3,$$

where $L_{3,\text{seed}}$ is the frozen one-body/self-static seed and ΔL_3 is the pure interaction residual on the 50-slot scaffold. Because the lower-order feedback terms in the cubic Legendre identity depend only on the already frozen 1PN/2PN ledger, they are insensitive to the new 3PN residual. Therefore the full generic-frame residual compiler is exact and trivial:

$$\boxed{\Delta H_3 = -\Delta L_3(v_0)}.$$

In the fixed generic-frame momentum basis, this is literally the slotwise statement

$$\boxed{\Delta h = -\Delta l},$$

so the full 50×50 compiler matrix is simply

$$\boxed{C_{\text{gen}} = -I_{50}}.$$

That is the uniqueness theorem. None of the 27 COM-blind directions is Hamiltonian-blind in the fixed ADM chart. Once the full generic-frame Hamiltonian target is imposed, the entire COM-null family disappears at once. Equivalently,

$$\boxed{\text{COM blindness is not Hamiltonian blindness.}}$$

The actual 11-generator ordinary contact family is therefore not invisible in the fixed Hamiltonian chart; it corresponds to moving to a different canonical chart. After the ADM chart is frozen, the Hamiltonian target selects a single ordinary representative.

At that point the comparable-mass 3PN lift is finished as an algebraic problem. What remains is entirely physical: the nine-slot middle interaction block must be carried by the grouped real P_2 conservative module, the last static slot must be assigned to the scalar/geometry lane, and the isolated v^8 remainder must be recognized as the ordinary-Lagrangian counterimage of the universal free relativistic two-body COM Hamiltonian. Those are precisely the tasks of the next sections.

6 The grouped real P_2 conservative module

The previous section reduced the full comparable-mass 3PN problem to three COM pieces: one pure-kinetic slot, a nine-slot middle interaction block, and a one-dimensional static remainder. The present section addresses the middle block. It is the first place where the 2.5PN narrowing

to the orbital/worldtube STF quadrupole branch becomes a hard conservative theorem gate rather than a later radiative aspiration. The question is not whether a quadrupole-flavored ontology looks plausible in the abstract; the question is whether the grouped real

$$P_{20} \oplus P_{21} \oplus P_{22}$$

bundle can actually reproduce the exact solved 3PN middle block in the fixed ADM chart.

The answer is two-part. The smallest local grouped- P_2 demotion is too rigid: it produces only a rank-three image and obeys six exact slot relations violated by the GR target. But the grouped ontology itself is not too small. Once the next natural local grouped families are added, the full nine-slot middle block is reached exactly, with no middle-block residual left. This is the first fully successful conservative grouped- P_2 stress test in the series.

6.1 Why grouped real $P_{20} \oplus P_{21} \oplus P_{22}$ is the right 3PN front end

The 2.5PN audit already reduced the surviving universal point-particle route to the orbital/worldtube STF quadrupole branch. In conservative language, that means the next higher-order front end is not an unfocused collection of generic 3PN coefficients, but the grouped real quadrupole bundle

$$P_{20} \oplus P_{21} \oplus P_{22}.$$

The exact 3PN target inherited from the comparable-mass analysis just completed makes that statement quantitative. After the frozen self/static seed is subtracted, the full COM residual splits into

$$\Delta l_1 v^8, \quad (\Delta l_6, \dots, \Delta l_{14}), \quad \Delta l_{15} U^4.$$

The grouped real P_2 lane can only hope to carry the middle block $(\Delta l_6, \dots, \Delta l_{14})$, so it faces a solved nine-slot target from the beginning.

At the same time, the grouped conservative front end through order $O(\omega^2)$ contains only three real data,

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)},$$

or equivalently one grouped trace plus two anisotropy defects. So even before any microscopic constitutive law is chosen, the conservative grouped problem is already overdetermined. On the isotropic branch it becomes still sharper, because only one real datum survives. This is why 3PN is the right first conservative grouped- P_2 gate: it is the earliest order at which the surviving quadrupole route is forced to confront a nontrivial exact target, but it is still rigid enough that failure or success is unambiguous.

6.2 Grouped conservative data and isotropy variables

At 3PN the grouped conservative payload is carried by exactly three real $O(\omega^2)$ moments,

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)}.$$

It is usually cleaner to rewrite them in terms of the grouped trace and two anisotropy defects,

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \quad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \quad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2}, \quad (1)$$

with inverse map

$$u_2^{(20)} = \bar{u}_2 + 4a_2, \quad u_2^{(21)} = \bar{u}_2 - a_2 + b_2, \quad u_2^{(22)} = \bar{u}_2 - a_2 - b_2. \quad (2)$$

The exact isotropy gate for the minimal branch is therefore

$$\boxed{a_2 = 0, \quad b_2 = 0.} \quad (3)$$

If that gate passes, then all three grouped moments coincide and one writes

$$u_2^{(20)} = u_2^{(21)} = u_2^{(22)} \equiv u_2.$$

The exact co-rotating pair-frame source map makes the front end completely explicit. Write

$$d \equiv \dot{r}, \quad u_\perp^2 \equiv v^2 - d^2, \quad U \equiv \frac{GM}{r},$$

so that the pair velocity decomposes as $\mathbf{v} = \mathbf{u} + d \mathbf{n}$ with $\mathbf{u} \cdot \mathbf{n} = 0$. Then the real-STF orbital source coefficients of $\tilde{I}_{ij}^{\text{orb}}/(2\mu)$ imply the exact grouped derivative norms

$$\dot{C}_{20}^2 = \frac{2d^2}{3r^2}(3u_\perp^2 - U)^2, \quad \dot{C}_{21c}^2 + \dot{C}_{21s}^2 = \frac{2u_\perp^2}{r^2}(u_\perp^2 - d^2 - U)^2, \quad \dot{C}_{22c}^2 + \dot{C}_{22s}^2 = \frac{2d^2 u_\perp^4}{r^2}. \quad (4)$$

No constitutive assumption has entered yet. Equation (4) is a kinematic identity of the grouped source map itself, and it is the unique starting point for any honest local 3PN grouped lift.

6.3 Minimal local grouped- P_2 front end and why it fails

The smallest time-local grouped conservative ansatz built from the exact source norms is

$$L_{P_2, \omega^2}^{\text{front}} = \frac{c_2^{(20)}}{3r^2} d^2 (3u_\perp^2 - U)^2 + \frac{c_2^{(21)}}{r^2} u_\perp^2 (u_\perp^2 - d^2 - U)^2 + \frac{c_2^{(22)}}{r^2} d^2 u_\perp^4. \quad (5)$$

Under the standard virial bookkeeping $U \sim v^2 \sim d^2$, every monomial in (5) has uniform weight five. So the direct front end is formally too high to be a 3PN ordinary-Lagrangian block. If one insists on the smallest possible local isotropic demotion, the unique one-step choice is multiplication by one power of $U^{-1} \sim r$.

That test fails exactly. After the one-step demotion, the grouped image lands only in the nine interaction slots $(l_6, \dots, l_{14})$, has rank three, and obeys six exact relations,

$$2l_6 + 2l_7 + l_8 = 0, \quad l_9 = l_6 + l_7, \quad l_{10} = -2l_6, \quad (6)$$

$$l_{11} + l_{12} = 2l_6, \quad l_{13} = l_6, \quad l_{14} = -\frac{1}{6}l_{11}. \quad (7)$$

Every minimally demoted local grouped- P_2 image must satisfy all six simultaneously. The exact GR middle block does not. The referee audit checks that each obstruction polynomial is generically nonzero once the solved target $(\Delta l_6, \dots, \Delta l_{14})$ is substituted.

The isotropic specialization is even more rigid. Setting $a_2 = b_2 = 0$ gives

$$\begin{aligned} l_6 &= u_2, & l_7 &= -u_2, & l_8 &= l_9 = 0, & l_{10} &= -2u_2, \\ l_{11} &= 4u_2, & l_{12} &= -2u_2, & l_{13} &= u_2, & l_{14} &= -\frac{2}{3}u_2. \end{aligned} \quad (8)$$

So the first moral of the section is negative but precise: the grouped real P_2 ontology is not ruled out by 3PN, but its smallest local constitutive lift is too rigid to carry the exact middle block.

6.4 Richer local grouped- P_2 family

The failure of the minimal lift does not mean that the grouped ontology is too small. It means only that the first constitutive guess was too poor. The next natural local grouped families are obtained by introducing the source-square blocks

$$C_{20}^2 = \frac{1}{6}(3d^2 - v^2 - 2U)^2, \quad C_{21}^2 = 2d^2 u_\perp^2, \quad C_{22}^2 = \frac{1}{2}u_\perp^4, \quad (9)$$

and then dressing them in the three simplest local ways compatible with the 3PN middle block,

$$T_{20} = \frac{1}{3}Ud^2(3u_\perp^2 - U)^2, \quad T_{21} = Uu_\perp^2(u_\perp^2 - d^2 - U)^2, \quad T_{22} = Ud^2u_\perp^4, \quad (10)$$

$$S_A = U^2C_A^2, \quad V_A = \frac{v^2}{U}S_A = Uv^2C_A^2, \quad A \in \{20, 21, 22\}. \quad (11)$$

The three minimally demoted dynamic norms (T_A) still span only rank three, and appending the static-support family (S_A) raises the rank only to six. But the full nine-element family

$$(T_{20}, T_{21}, T_{22}, S_{20}, S_{21}, S_{22}, V_{20}, V_{21}, V_{22})$$

acts on the nine-slot middle block by an exact 9×9 compiler matrix with constant determinant

$$\boxed{\det M_{\text{mid}} = -\frac{4}{27} \neq 0.} \quad (12)$$

So the richer grouped family is exactly invertible on the whole middle block. This is the decisive positive grouped theorem of the 3PN conservative audit: the grouped real P_2 ontology itself is already large enough to carry the full nine-slot middle block once the constitutive lift is enlarged in the smallest natural way.

It is therefore natural to define the richer grouped module by

$$L_{P_2}^{\text{mid}} = \sum_{A \in \{20, 21, 22\}} (\lambda_A T_A + \sigma_A S_A + \tau_A V_A). \quad (13)$$

Because of (12), every exact middle-block target has one and only one coefficient triple $(\lambda_A, \sigma_A, \tau_A)$.

6.5 Exact GR grouped coefficients

Substituting the solved GR middle block $(L_6, \dots, L_{14}) = (\Delta l_6, \dots, \Delta l_{14})$ into the exact inverse compiler yields the unique grouped coefficient set

$$\begin{aligned} \lambda_{20}^{\text{GR}} &= \frac{\nu(-276\nu^2 - 3154\nu + 3\pi^2 + 2672)}{32}, \\ \lambda_{21}^{\text{GR}} &= \frac{\nu(2568\nu^2 + 2236\nu - 3044 - 3\pi^2)}{192}, \\ \lambda_{22}^{\text{GR}} &= \frac{\nu(7650\nu^2 + 32858\nu - 30136 - 33\pi^2)}{96}, \end{aligned} \quad (14)$$

$$\begin{aligned} \sigma_{20}^{\text{GR}} &= \frac{\nu(-102\nu^2 - 308\nu + 293)}{16}, \\ \sigma_{21}^{\text{GR}} &= \frac{\nu(-4188\nu^2 - 23778\nu + 21\pi^2 + 22006)}{192}, \\ \sigma_{22}^{\text{GR}} &= \frac{\nu(3906\nu^2 + 2478\nu - 2791 - 3\pi^2)}{48}, \end{aligned} \quad (15)$$

$$\begin{aligned} \tau_{20}^{\text{GR}} &= \frac{3\nu(-154\nu^2 - 87\nu + 38)}{32}, \\ \tau_{21}^{\text{GR}} &= \frac{\nu(-6672\nu^2 - 5824\nu + 3\pi^2 + 4412)}{384}, \\ \tau_{22}^{\text{GR}} &= \frac{\nu(-2790\nu^2 - 3367\nu + 3\pi^2 + 3386)}{96}. \end{aligned} \quad (16)$$

With exactly these coefficients, the richer grouped lift (13) reproduces the entire solved GR middle block identically:

$$(l_6, \dots, l_{14}) = (\Delta l_6, \dots, \Delta l_{14}). \quad (17)$$

So the conservative 3PN answer to the grouped front-end question is now sharp:

$$\boxed{\text{grouped real } P_2 \text{ can close the full 3PN middle block exactly.}} \quad (18)$$

This is the first place in the program where the conservative quadrupole bundle has survived a fully overdetermined exact target rather than a qualitative consistency test.

6.6 Structural corollaries

Two corollaries are immediate and both matter for the rest of the paper.

First, every basis element in (T_A, S_A, V_A) carries at least one power of U . Therefore the richer grouped module forces

$$l_1 = l_2 = l_3 = l_4 = l_5 = 0 \quad (19)$$

identically. So the grouped real P_2 closure does *not* compete with the pure-kinetic sector. The free-particle residual remains outside the grouped module and will later be shown to be only the ordinary-Lagrangian counterimage of the universal free COM Hamiltonian.

Second, the richer grouped compiler predicts a definite static companion rather than leaving the quartic potential slot arbitrary. One finds

$$l_{15}^{(P_2 \text{ pred})} = \frac{2}{3}\sigma_{20} = l_{10} + l_{11} + l_{12} + 2(l_6 + l_7 + l_8 + l_9). \quad (20)$$

For the exact GR middle block this becomes

$$\Delta l_{15}^{(P_2 \text{ pred})} = \frac{\nu(293 - 308\nu - 102\nu^2)}{24}. \quad (21)$$

Comparing with the true static residual isolated in the exact COM target gives the leftover scalar gap

$$\Delta l_{15}^{(0/g)} \equiv \Delta l_{15}^{\text{GR}} - \Delta l_{15}^{(P_2 \text{ pred})} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}. \quad (22)$$

So once the richer grouped real P_2 compiler closes the nine middle slots exactly, the full solved 3PN COM residual already splits as

$$\Delta \hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}}[\lambda_A^{\text{GR}}, \sigma_A^{\text{GR}}, \tau_A^{\text{GR}}] + \Delta l_{15}^{(0/g)} U^4. \quad (23)$$

That is the exact point where the old scalar/geometry lane re-enters. The richer grouped module has done everything it can do: it has carried the whole middle block and fixed its unavoidable static shadow. The only remaining conservative scalar task is the one-dimensional static remainder, which is the subject of the next section.

7 Static scalar/geometry completion

The preceding section closed the entire nine-slot middle interaction block by the richer grouped real P_2 compiler. What remains at conservative 3PN is therefore not a diffuse family of leftover coefficients but a sharply isolated one-dimensional static remainder. The main point of the present section is that this scalar remainder is more rigid than it first appears. Inside the old constant-coefficient generic-frame scaffold, the static block is too small to carry the exact gap. If one temporarily enlarges the scalar lane so as to allow both a body-local P_0 carrier and a pair/geometry carrier, the center-of-mass projection seems to leave a one-parameter

family. But that family is algebraically redundant: an exact mass identity collapses it to a unique pair/geometry representative. In the fixed ADM chart, the Hamiltonian image of that completion is then obtained by the same exact sign flip that already governs the rest of the residual. After this section, the static conservative 3PN lane is completely fixed; only its microscopic moving-throat realization remains open.

7.1 Residual static slot after grouped closure

Once the richer grouped real P_2 compiler has reproduced the exact nine-slot middle block, the solved COM residual splits as

$$\Delta \hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}}[\lambda_A^{\text{GR}}, \sigma_A^{\text{GR}}, \tau_A^{\text{GR}}] + \Delta l_{15}^{(0/g)} U^4, \quad U \equiv \frac{GM}{r}. \quad (24)$$

So the grouped module has already done all it can do. The only remaining scalar job is to identify the carrier of the one-dimensional static remainder.

A useful way to display that remainder is to compare the static companion forced by the grouped middle-block fit with the exact GR target. The grouped compiler predicts

$$l_{15}^{(P_2 \text{ pred})} = \frac{\nu(-102\nu^2 - 308\nu + 293)}{24}, \quad (25)$$

whereas the exact GR coefficient is

$$l_{15}^{\text{GR}} = \frac{\nu(-908 + 63\pi^2)}{96}. \quad (26)$$

The static gap is therefore

$$\Delta l_{15}^{(0/g)} = l_{15}^{\text{GR}} - l_{15}^{(P_2 \text{ pred})} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}. \quad (27)$$

Equation (27) is the full conservative scalar datum left after the grouped real P_2 module succeeds.

7.2 U -block no-go inside the old constant-coefficient generic scaffold

The first exact result is negative. In the old constant-coefficient generic-frame interaction scaffold, the only static basis polynomial is

$$\mathcal{U}_1 = p^2 q + p q^2, \quad p \equiv m_A, \quad q \equiv m_B, \quad (28)$$

inside the common prefactor $G^4 p q / r^4$. Writing

$$M = p + q, \quad \mu = \frac{pq}{M}, \quad \nu = \frac{pq}{M^2}, \quad U = \frac{GM}{r}, \quad (29)$$

one finds the exact COM image

$$\frac{1}{\mu} \frac{G^4 p q}{r^4} \mathcal{U}_1 = \nu U^4. \quad (30)$$

So every constant-coefficient static interaction term in the old scaffold has a COM coefficient of the form

$$l_{15} = \nu u_1, \quad (31)$$

linear in ν .

But the exact remainder (27) still contains ν^2 and ν^3 after the overall factor of ν is extracted. Therefore no new constant coordinate inside the original 24/17/8/1 generic-frame scaffold can carry the actual 3PN static gap. The residual scalar lane must instead be represented as a ν -dressed counterterm family.

7.3 Apparent P_0 versus geometry ambiguity

Once one allows such ν -dressed scalar carriers, two natural static mass families appear. The first is a body-local scalar family,

$$\mathcal{U}_0 = p^3 + q^3, \quad (32)$$

which has the right bookkeeping character for a body-local P_0 -type wall or support dressing. The second is the intrinsic pair family

$$\mathcal{U}_g = p^2q + pq^2, \quad (33)$$

which is the natural candidate for a geometry/compressibility completion.

At COM level these two families span only one scalar observable. Indeed,

$$\frac{1}{\mu} \frac{G^4 pq}{r^4} \nu \mathcal{U}_0 = \nu(1 - 3\nu)U^4, \quad \frac{1}{\mu} \frac{G^4 pq}{r^4} \mathcal{U}_g = \nu U^4. \quad (34)$$

So before the exact mass algebra is used, the most natural generic-frame static counterterm looks like a regular one-parameter $P_0 \oplus g$ family,

$$\Delta L_{3,\text{ct}}^{(0)} = \frac{G^4 pq}{r^4} \sigma \nu \mathcal{U}_0, \quad (35)$$

$$\Delta L_{3,\text{ct}}^{(g)} = \frac{G^4 pq}{r^4} \left[\frac{408\nu^2 + 1232\nu - 2080 + 63\pi^2}{96} - \sigma(1 - 3\nu) \right] \mathcal{U}_g. \quad (36)$$

Here σ labels how much of the COM scalar lane is assigned to a body-local P_0 carrier rather than to the pair/geometry carrier. At this stage the split looks real, because COM fixes only the sum.

7.4 Sigma-collapse and unique geometry completion

The apparent σ -freedom is illusory. The exact mass identity

$$\boxed{\nu(p^3 + q^3) = (1 - 3\nu)(p^2q + pq^2)} \quad (37)$$

collapses the full family (35)–(36) identically. Equivalently,

$$\frac{d}{d\sigma} (\Delta L_{3,\text{ct}}^{(0)} + \Delta L_{3,\text{ct}}^{(g)}) = 0. \quad (38)$$

So the full static scalar counterterm reduces to one unique pair/geometry representative,

$$\boxed{\Delta L_{3,\text{ct}}^{\text{scalar}} = \frac{G^4 pq}{96r^4} (408\nu^2 + 1232\nu - 2080 + 63\pi^2) (p^2q + pq^2)}. \quad (39)$$

Its COM image is precisely the residual coefficient (27):

$$\frac{1}{\mu} \Delta L_{3,\text{ct}}^{\text{scalar}} = \Delta l_{15}^{(g)} U^4, \quad \Delta l_{15}^{(g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}. \quad (40)$$

It is also convenient to phrase the same result directly at the reduced-coefficient level:

$$\begin{aligned} c_{\text{target}} &= -\frac{227}{24} + \frac{21\pi^2}{32}, & c_{\text{pred}}^{(P_2)} &= \frac{293}{24} - \frac{77}{6}\nu - \frac{17}{4}\nu^2, \\ c_{\text{geom}} &= \frac{408\nu^2 + 1232\nu - 2080 + 63\pi^2}{96}, \end{aligned} \quad (41)$$

with

$$c_{\text{target}} = c_{\text{pred}}^{(P_2)} + c_{\text{geom}}. \quad (42)$$

So once the exact mass algebra is respected, there is no independent body-local P_0 dressing left. The leftover 3PN static scalar sector belongs uniquely to the geometry side.

7.5 Hamiltonian image of the static completion

The fixed-chart generic-frame compiler established earlier acts on the residual by exact sign flip,

$$\Delta H_3 = -\Delta L_3. \quad (43)$$

So the Hamiltonian image of the unique static completion follows immediately:

$$\boxed{\Delta H_{3,\text{ct}}^{\text{scalar}} = -\Delta L_{3,\text{ct}}^{\text{scalar}}.} \quad (44)$$

No further algebraic resolution is needed. The ordinary and Hamiltonian static remainders are already the same unique polynomial with opposite sign, and the only remaining question is microscopic rather than algebraic: what scalar-side wall or geometry dynamics on the moving-throat branch realizes the coefficient in (39)?

At this stage, therefore, the full conservative 3PN static lane is fixed. The only untreated COM slot is the apparent pure-kinetic v^8 remainder, and that slot is shown in the next section to be nothing more than the ordinary-Lagrangian counterimage of the universal free two-body relativistic Hamiltonian.

8 Pure-kinetic sector and the free two-body Hamiltonian

The preceding section showed that the last genuine scalar ambiguity collapses to a unique geometry-side completion. So after the grouped real P_2 middle block and the static scalar lane are both fixed, only one apparently separate center-of-mass pure-kinetic slot remains. The point of the present section is that this leftover is not a new throat-response datum. At generic-frame level, the exact free relativistic two-body Lagrangian contains no mixed comparable-mass 3PN pure-kinetic interaction monomial at all. The apparent center-of-mass residual appears only because the exact ordinary/Hamiltonian compiler maps the universal free relativistic two-body Hamiltonian to a shifted ordinary coefficient. Once that compiler compensation is taken into account, the entire pure-kinetic 3PN sector closes identically. So the conservative 3PN split really does stop with three lanes only: the compiler image of free kinematics, the exact grouped real P_2 middle block, and the unique geometry-side static completion.

8.1 Why the apparent pure-kinetic residual is not a new throat datum

After the exact grouped middle-block closure and the static scalar/geometry completion, the only seemingly independent center-of-mass kinetic remainder is

$$\boxed{\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}, \quad \Delta l_2 = \Delta l_3 = \Delta l_4 = \Delta l_5 = 0.} \quad (45)$$

In the standard isotropic 15-slot center-of-mass basis, this is the entire leftover pure-kinetic block. Taken in isolation, (45) looks as though one more independent 3PN kinetic throat coefficient must still be supplied.

That interpretation is too quick. A genuinely new kinetic datum would have to appear first as a new generic-frame interaction family. But the exact free-body audit shows that no such family exists. The whole residual arises instead from the exact ordinary/Hamiltonian compiler applied to the already-fixed universal free relativistic two-body Hamiltonian. So the right question is not “what new kinetic interaction must the throat supply?” but rather “what ordinary-side coefficient is required so that the fixed free Hamiltonian target is reproduced exactly in center-of-mass variables?”

8.2 Generic-frame free kinematics

At generic-frame level, the exact free ordinary relativistic two-body Lagrangian is simply the sum of the two separate free-body square roots,

$$L_{\text{free}}^{\text{gen}} = -m_A c^2 \sqrt{1 - \frac{v_A^2}{c^2}} - m_B c^2 \sqrt{1 - \frac{v_B^2}{c^2}}. \quad (46)$$

Expanding through 3PN gives

$$\begin{aligned} L_{\text{free}}^{\text{gen}} = & - (m_A + m_B) c^2 + \frac{1}{2} (m_A v_A^2 + m_B v_B^2) + \frac{1}{8 c^2} (m_A v_A^4 + m_B v_B^4) \\ & + \frac{1}{16 c^4} (m_A v_A^6 + m_B v_B^6) + \frac{5}{128 c^6} (m_A v_A^8 + m_B v_B^8) + O(c^{-8}). \end{aligned} \quad (47)$$

So the 3PN pure-kinetic block is already exactly

$$\boxed{L_{3,\text{free}}^{\text{gen}} = \frac{5}{128} (m_A v_A^8 + m_B v_B^8)}. \quad (48)$$

There is no mixed comparable-mass pure-kinetic interaction monomial here. In particular, the generic-frame free block contains no independent $m_A m_B$ -type *octic* interaction that could serve as a new 3PN constitutive datum.

If one now performs the naive center-of-mass reduction

$$\mathbf{v}_A = \frac{m_B}{M} \mathbf{v}, \quad \mathbf{v}_B = -\frac{m_A}{M} \mathbf{v}, \quad M \equiv m_A + m_B, \quad (49)$$

with symmetric mass ratio

$$\nu \equiv \frac{m_A m_B}{M^2}, \quad (50)$$

then the carried free ordinary seed coefficient becomes

$$\boxed{l_1^{\text{seed}} = \frac{5}{128} - \frac{35}{128} \nu + \frac{35}{64} \nu^2 - \frac{35}{128} \nu^3}. \quad (51)$$

This is exactly the coefficient that one obtains by reducing the separate-body free kinematics before the exact center-of-mass compiler is imposed.

8.3 Exact free COM Hamiltonian target

The correct reference object is instead the exact free relativistic two-body center-of-mass Hamiltonian,

$$H_{\text{free}}^{\text{COM}} = \sqrt{m_A^2 c^4 + P^2 c^2} + \sqrt{m_B^2 c^4 + P^2 c^2}, \quad (52)$$

with reduced momentum variable

$$P = \mu p, \quad \mu \equiv \frac{m_A m_B}{M}. \quad (53)$$

Subtracting the rest energy and dividing by μ gives the reduced expansion

$$\frac{H_{\text{free}}^{\text{COM}} - M c^2}{\mu} = \frac{1}{2} p^2 - \frac{1 - 3\nu}{8 c^2} p^4 + \frac{1 - 5\nu + 5\nu^2}{16 c^4} p^6 + \frac{h_1^{\text{free}}}{c^6} p^8 + O(c^{-8}), \quad (54)$$

where the exact 3PN free coefficient is

$$\boxed{h_1^{\text{free}} = \frac{-5 + 35\nu - 70\nu^2 + 35\nu^3}{128}, \quad h_2 = h_3 = h_4 = h_5 = 0}. \quad (55)$$

This coefficient is not a new observable of the throat. It is simply the universal free relativistic two-body Hamiltonian written in center-of-mass variables. Accordingly, the remaining kinetic question is entirely a compiler question: what ordinary coefficient corresponds to the already-fixed Hamiltonian slot (55)?

8.4 Collapse of the ordinary pure-kinetic residual

The exact center-of-mass compiler established earlier acts on the first kinetic slot by

$$l_1 = F_1(\nu) - h_1, \quad F_1(\nu) = \frac{3}{16}\nu - \frac{21}{16}\nu^2 + \frac{9}{4}\nu^3. \quad (56)$$

Applying (56) to the naive free ordinary seed (51) gives the Hamiltonian image of that seed,

$$h_1^{\text{seed}} = F_1(\nu) - l_1^{\text{seed}} = -\frac{5}{128} + \frac{59}{128}\nu - \frac{119}{64}\nu^2 + \frac{323}{128}\nu^3. \quad (57)$$

The entire pure-kinetic remainder is therefore fixed by the difference between this seed Hamiltonian image and the exact free Hamiltonian target:

$$\begin{aligned} \Delta l_1 &= h_1^{\text{seed}} - h_1^{\text{free}} \\ &= \frac{3}{16}\nu - \frac{21}{16}\nu^2 + \frac{9}{4}\nu^3 = \boxed{\frac{3\nu(3\nu-1)(4\nu-1)}{16}}. \end{aligned} \quad (58)$$

This is exactly the leftover coefficient displayed in (45). Equivalently, the full ordinary target is

$$l_1^{\text{GR}} = F_1(\nu) - h_1^{\text{free}} = \frac{5 - 11\nu - 98\nu^2 + 253\nu^3}{128}, \quad (59)$$

with the shift from (51) to (59) given entirely by the compiler compensation (58).

So the apparent pure-kinetic leftover is not a new constitutive 3PN throat coefficient. It is only the ordinary-Lagrangian counterimage of the universal free relativistic two-body center-of-mass Hamiltonian. Since the free target also has

$$h_2 = h_3 = h_4 = h_5 = 0, \quad (60)$$

no extra generic-frame pure-kinetic interaction module should be added in the fixed ADM chart. The grouped real P_2 module already closes the whole middle block, the previous section already fixes the static scalar lane uniquely on the geometry side, and the present computation shows that the only remaining octic center-of-mass term is generated automatically by the exact compiler. After this collapse, no separate algebraic conservative 3PN bottleneck remains.

9 Final conservative 3PN theorem

The preceding three sections have removed all remaining algebraic uncertainty from the conservative 3PN problem. The nine-slot middle interaction block is carried exactly by the richer grouped real P_2 module; the apparent static P_0/g ambiguity collapses to a unique geometry-side completion; and the remaining center-of-mass pure-kinetic slot is only the ordinary-Lagrangian image of the universal free relativistic two-body Hamiltonian. The final theorem is therefore much shorter than the derivation chain that leads to it. In the fixed ADM chart, the full comparable-mass c^{-6} coefficient is completely assigned, and what remains open is no longer algebraic coefficient matching but the deeper moving-throat derivation of the grouped real P_2 constitutive law, the geometry completion, and then the passive/outgoing quadrupole normalization that enters the later 2.5PN bridge.

9.1 Final exact split in COM form

Within the declared hierarchy, the exact comparable-mass 3PN coefficient in the isotropic COM chart is

$$\Delta \hat{L}_3^{\text{GR}} = \underbrace{\Delta l_1 v^8}_{\text{free-COM compiler image}} + \underbrace{L_{P_2}^{\text{mid}}}_{\text{exact grouped real } P_2 \text{ closure}} + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}. \quad (61)$$

Here $U = GM/r$ is the usual COM Newtonian potential. The two scalar coefficients worth keeping explicit are

$$\Delta l_1 = \frac{3\nu(3\nu-1)(4\nu-1)}{16}, \quad \Delta l_{15}^{(g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}. \quad (62)$$

The meaning of (61) is now completely sharp. The term $\Delta l_1 v^8$ is not a new throat-response coefficient; it is exactly the ordinary-side compensation required by the free relativistic two-body COM Hamiltonian. The grouped module

$$L_{P_2}^{\text{mid}} = \sum_{A \in \{20, 21, 22\}} (\lambda_A^{\text{GR}} T_A + \sigma_A^{\text{GR}} S_A + \tau_A^{\text{GR}} V_A) \quad (63)$$

reproduces the full solved nine-slot middle block

$$(\Delta l_6, \dots, \Delta l_{14})$$

exactly, with the coefficient triples $(\lambda_A^{\text{GR}}, \sigma_A^{\text{GR}}, \tau_A^{\text{GR}})$ fixed uniquely by the GR target. Finally, the static term $\Delta l_{15}^{(g)} U^4$ is not a free P_0/g admixture. After sigma-collapse it is the unique geometry-side completion of the remaining scalar slot.

So the COM theorem really does stop with three and only three lanes:

1. a compiler image of universal free kinematics;
2. an exact grouped real P_2 middle-block closure;
3. a unique geometry-side static completion.

There is no fourth independent conservative 3PN carrier hiding behind the center-of-mass bookkeeping. Every solved coefficient belongs to one of the three structures displayed in (61).

9.2 Final exact split in generic-frame form

The generic-frame statement is not a second independent theorem; it is the unique Hamiltonian-first lift of the same three-lane solution. Let

$$p \equiv m_A, \quad q \equiv m_B, \quad \nu \equiv \frac{pq}{(p+q)^2}.$$

Then the full generic-frame ordinary 3PN coefficient can be written as

$$L_3^{\text{gen}} = L_{3,\text{seed}} + \sum_{A \in \{20, 21, 22\}} (\lambda_A^{\text{GR}} T_A + \sigma_A^{\text{GR}} S_A + \tau_A^{\text{GR}} V_A) + \Delta L_{3,\text{ct}}^{\text{scalar}}, \quad (64)$$

where the natural self/static seed is the symmetric promotion of the exact Schwarzschild one-body target, the grouped real P_2 families (T_A, S_A, V_A) carry the full middle interaction block, and the unique static counterterm is

$$\Delta L_{3,\text{ct}}^{\text{scalar}} = \frac{G^4 pq}{96r^4} (408\nu^2 + 1232\nu - 2080 + 63\pi^2) (p^2 q + pq^2). \quad (65)$$

Two features of (64) should be emphasized. First, the free-body part of the seed remains exactly

$$L_{3,\text{free}}^{\text{gen}} = \frac{5}{128} (m_A v_A^8 + m_B v_B^8),$$

so there is no separate mixed comparable-mass generic-frame pure-kinetic interaction family to be added by hand. The COM term $\Delta l_1 v^8$ appears only after the exact compiler maps the universal free Hamiltonian back to the ordinary chart. Second, once the lower-order ledger is frozen, the fixed ADM chart obeys the exact generic-frame relation

$$\boxed{\Delta H_3 = -\Delta L_3.} \quad (66)$$

So the earlier COM-blind null directions and the smaller ordinary contact/gauge orbit are intermediate bookkeeping structures only. They do not survive as true freedom after the full generic-frame Hamiltonian target is imposed.

In this sense the generic-frame theorem is stronger than a COM target match. There is a unique ordinary representative in the fixed chart, and it is exactly the Hamiltonian-first lift of the three solved pieces already displayed in the COM decomposition.

9.3 Equality to the standard 3PN ADM target

Because the generic-frame Hamiltonian compiler has zero kernel in the fixed ADM chart, the assembled ordinary representative Legendre-transforms to the standard 3PN ADM-type Hamiltonian target in exactly the same sense that the full conservative 1PN paper closes to EIH and the full conservative 2PN paper closes to ADM. After the one-body repair ledger, the grouped middle-block closure, the unique geometry completion, and the pure-kinetic compiler image are all inserted, no residual Hamiltonian discrepancy remains:

$$\boxed{H_3^{\text{derived}} = H_3^{\text{ADM}}, \quad \Delta H_3 \equiv H_3^{\text{derived}} - H_3^{\text{ADM}} = 0.} \quad (67)$$

The role of the Hamiltonian-first route should be stated explicitly. A naive ordinary-Lagrangian COM reduction of the imported generic-frame target does not reproduce the exact COM ordinary target. By contrast, the sequence

$$\text{generic-frame ordinary target} \longrightarrow \text{exact cubic Legendre lift} \longrightarrow \text{COM reduction}$$

does reproduce the exact COM Hamiltonian and therefore fixes the correct generic representative. Once that route is used, the apparent 27-dimensional COM-blind family and the smaller 11-generator ordinary contact orbit are seen for what they are: bookkeeping artifacts of an incomplete chart choice rather than surviving physical or algebraic freedom.

So the conservative 3PN result should not be read as a target match obtained after a sequence of special chart adjustments. It is an exact final equality in the fixed ADM chart once the correct compiler route is respected.

9.4 Main theorem statement

Main theorem. Fix the carried Newtonian, 1PN, and 2PN conservative ledger and work within the same declared closure hierarchy used in the earlier full conservative papers. Let the exact one-body 3PN Schwarzschild gate be repaired by

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16}, \quad (68)$$

let the grouped real P_2 middle-block compiler be the richer (T_A, S_A, V_A) family with

$$\det M_{\text{mid}} = -\frac{4}{27}, \quad (69)$$

and let the fixed ADM chart be imposed by the exact generic-frame sign-flip identity (66). Then the full conservative two-body 3PN sector is uniquely fixed. Equivalently,

$$\boxed{L_3^{\text{gen}} = L_{3,\text{seed}} + L_{P_2}^{\text{mid}}[\lambda_A^{\text{GR}}, \sigma_A^{\text{GR}}, \tau_A^{\text{GR}}] + \Delta L_{3,\text{ct}}^{\text{scalar}},} \quad (70)$$

with COM residual

$$\Delta\hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}} + \Delta l_{15}^{(g)} U^4, \quad (71)$$

and Hamiltonian equality

$$H_3^{\text{derived}} = H_3^{\text{ADM}}. \quad (72)$$

The point of the theorem is not merely that the model can reproduce the 3PN target after enough algebra. The stronger conclusion is structural: the conservative 3PN bottleneck is no longer an algebraic coefficient search. The higher-order conservative burden has now been compressed to an exact grouped real P_2 module plus a unique geometry-side completion, while the apparent pure-kinetic remainder is absorbed completely by the exact free-Hamiltonian compiler image.

This should also be read with the same status discipline as the earlier full conservative papers. The theorem is exact *within the declared hierarchy*; it is not being claimed as an assumption-free theorem of a fully solved moving-throat PDE. What remains open is therefore not another conservative 3PN algebra problem. What remains open is the deeper moving-throat derivation of the grouped real P_2 constitutive law and the geometry completion, followed by their insertion into the already narrowed passive/outgoing quadrupole normalization bridge. That interface is the subject of the next section.

10 Interface to the 2.5PN program

The conservative 3PN theorem does not replace the 2.5PN program; it tells us exactly how that program should now be read. The 2.5PN audit had already reduced the universal higher-order route to the orbital/worldtube STF quadrupole, packaged conservatively as the grouped real

$$P_{20} \oplus P_{21} \oplus P_{22}$$

bundle. What remained unclear at that stage was whether the same grouped ontology could survive its first honest conservative stress test, or whether the future outgoing-normalization problem would have to compete with some different hidden conservative carrier. The present 3PN result answers that question. The full nine-slot middle interaction block is carried exactly by the richer local grouped- P_2 module, the residual static slot is uniquely geometry-side, and the apparent COM pure-kinetic leftover is only the ordinary-Lagrangian image of the universal free relativistic two-body Hamiltonian. So the 2.5PN bridge should now be read against a much sharper conservative background: the remaining gap is no longer a vague higher-order bookkeeping problem, but the passive/ outgoing normalization of one already identified grouped-quadrupole branch.

10.1 Why 3PN is the first conservative grouped- P_2 gate

The 2.5PN notes already showed that the surviving universal point-particle route is the orbital/worldtube STF quadrupole branch. In conservative language, that means the next higher-order front end is not an unfocused list of new PN coefficients, but the grouped real quadrupole bundle

$$P_{20} \oplus P_{21} \oplus P_{22}.$$

The exact 3PN comparable-mass residual makes that statement quantitative. After the carried self/static seed is subtracted, the solved COM target splits into three pieces,

$$\Delta\hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + (\Delta l_6, \dots, \Delta l_{14}) + \Delta l_{15} U^4, \quad (73)$$

so the grouped real P_2 lane can only hope to carry the nine-slot middle block $(\Delta l_6, \dots, \Delta l_{14})$. That is exactly what makes 3PN the first hard conservative grouped- P_2 gate. It is the earliest

order at which the grouped branch is forced to confront a nontrivial exact target, but it is still low enough that the pass/fail test is sharp rather than diffuse.

This point should be emphasized because it changes the interpretation of the later radiative program. If the grouped real P_2 ontology had failed at 3PN, then the 2.5PN narrowing would have been only suggestive packaging. Because it succeeds, the grouped quadrupole route becomes the unique surviving conservative carrier of the middle interaction block in the same fixed chart in which the later passive/outgoing normalization question is posed.

10.2 The $a_2 = b_2 = 0$ isotropy test and grouped low-frequency data

At the first conservative grouped stage, the low-frequency data are still only three real $O(\omega^2)$ moments,

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)}.$$

It is cleaner to rewrite them as one grouped trace plus two anisotropy defects,

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \quad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \quad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2}, \quad (74)$$

with inverse map

$$u_2^{(20)} = \bar{u}_2 + 4a_2, \quad u_2^{(21)} = \bar{u}_2 - a_2 + b_2, \quad u_2^{(22)} = \bar{u}_2 - a_2 - b_2. \quad (75)$$

So the first grouped branch test is simply

$$\boxed{a_2 = 0, \quad b_2 = 0.} \quad (76)$$

If (76) fails, then the minimal isotropic quadrupole branch is already dead before any outgoing-normalization calculation is attempted. If it passes, then the three grouped moments collapse to one trace datum $\bar{u}_2$, and the future conservative problem is narrowed again.

That is precisely where the 3PN and 2.5PN programs meet. The present paper fixes the first grouped data that a future moving-throat calculation must reproduce; the 2.5PN package then explains what comes next on the minimal isotropic branch. At the next even order one must determine $u_4^{(20)}, u_4^{(21)}, u_4^{(22)}$, test isotropy again, and then check the single-pole branch identity

$$\boxed{a_4 = b_4 = 0, \quad \bar{u}_4 = 4\bar{u}_2^2, \quad \Omega_Q^2 = \frac{1}{4\bar{u}_2}.} \quad (77)$$

Equivalently, once the isotropic conservative trace data are known through $O(\omega^4)$, the minimal outgoing quadrupole branch is parametrized by a pole scale Ω_Q and one static normalization. In the grouped-response language used by the 2.5PN package, the target normalization may be written as

$$\boxed{\bar{K}_0^{\text{target}} = \frac{2G}{45c^5\bar{u}_2^{5/2}} = \frac{64G}{45c^5}\Omega_Q^5.} \quad (78)$$

So the practical theorem test has become very concrete: 3PN supplies the first half of the grouped conservative verdict, and the later outgoing bridge asks whether the same branch continues through (77) and (78).

10.3 What the present 3PN result fixes for the minimal outgoing $\ell = 2$ branch

The main positive conclusion is that the grouped real P_2 ontology survives its first exact conservative stress test. The richer local grouped family closes the entire middle block,

$$L_{P_2}^{\text{mid}} = \sum_{A \in \{20, 21, 22\}} (\lambda_A^{\text{GR}} T_A + \sigma_A^{\text{GR}} S_A + \tau_A^{\text{GR}} V_A), \quad (79)$$

with no middle-block residual left. This is the crucial conservative interface statement. The throat side does *not* need a qualitatively new non- P_2 carrier to reproduce the nine-slot 3PN interaction burden. The remaining static scalar slot is uniquely geometry-side, and the apparent pure-kinetic COM residual is only the ordinary-chart counterimage of the exact free relativistic two-body Hamiltonian. Neither of those lanes competes with the grouped quadrupole module for the middle-block role.

This matters directly for the outgoing $\ell = 2$ branch isolated at 2.5PN. Once the grouped real P_2 sector is accepted as the correct conservative front end, the later passive/outgoing problem no longer has to search over unrelated higher-order carriers. It only has to determine the low-frequency constitutive data of the same grouped branch on the true moving-throat solution. On the minimal isotropic branch the payoff is large: the conservative and dissipative coefficients are no longer independent. If one writes the isotropic conservative module as

$$\bar{K}_Q^{\text{cons}}(\omega) = \bar{K}_0 \left[\frac{3}{4} + \frac{1}{4} \frac{1}{1 - \omega^2/\Omega_Q^2} \right], \quad (80)$$

then the higher even moment and the odd Burke–Thorne coefficient follow automatically,

$$\boxed{\bar{K}_2 = \frac{\bar{K}_0}{4\Omega_Q^2}, \quad \bar{K}_4 = \frac{\bar{K}_0}{4\Omega_Q^4}, \quad \bar{\Gamma}_5 = \frac{9\bar{K}_0}{32\Omega_Q^5}.} \quad (81)$$

So the present 3PN theorem does not yet determine the outgoing coefficient numerically, but it does determine something almost as important: where that coefficient must come from. It must arise from the grouped real P_2 lane, with the geometry lane supplying only the separate static completion and the free-kinetic lane supplying only the universal compiler image.

10.4 What the full moving-throat PDE still has to supply

The remaining theorem gap is therefore much narrower than “derive higher PN somehow.” What the full moving-throat PDE still has to supply is now specific. First, it must produce the actual low-frequency constitutive origin of the conservative grouped data on the true branch, rather than only their reduced bookkeeping image. Second, it must decide whether that branch is genuinely isotropic, first at $O(\omega^2)$ through (76) and then at $O(\omega^4)$ through (77). Third, it must realize the final passive/outgoing normalization of the surviving quadrupole route.

The last point is the one surviving universal 2.5PN theorem condition. In the canonically normalized STF language of the theorem audit, it may be written as

$$\boxed{\hat{m}_0^2 \Gamma_5 = \frac{2G}{5c^5}.} \quad (82)$$

On the minimal isotropic outgoing branch, (82) is equivalent to the static scaling law already displayed in (78). So even here the gap is no longer diffuse: the missing PDE work must compute one isotropic normalization product on the actual moving-throat branch, not reopen the whole coefficient problem.

That is the correct way to read the present 3PN theorem. It removes the conservative algebraic uncertainty beneath the 2.5PN bridge, but it does not pretend to close the remaining passive/outgoing normalization problem by itself. The future moving-throat calculation still has to show that the true branch realizes the same grouped- P_2 constitutive law, the same isotropy verdict, and the same normalized outgoing quadrupole coefficient that the reduced 2.5PN package has already isolated as the final serious theorem gate.

11 Fixed versus open quantities

The final theorem closes the conservative 3PN algebraic ledger, but it does not erase the status boundary that runs through the whole series. As in the full 1PN, 2PN, and 2.5PN papers, one must keep separate what is exact in the carried reduction backbone, what is frozen as lower-order input, what is fixed by the present 3PN audit, and what still belongs to the unsolved moving-throat completion. This section records that boundary in one place. Its purpose is not to weaken the theorem, but to state its true strength: the conservative 3PN coefficient problem is now closed in the fixed ADM chart, while the remaining open work has been narrowed to a much smaller dynamical and constitutive problem.

11.1 Frozen carry-forward inputs

The present paper does not begin from a blank effective theory. It imports the exact 4D parent reduction backbone already frozen by the earlier papers: the projection map, projected continuity with leakage, the exact longitudinal identity, the quasi-static Poisson hook, and the coherent-defect / worldtube particle reduction. These formulas remain the structural dictionary that turns the bulk ontology into the near-zone two-body language used throughout the post-Newtonian series.

The lower-order conservative ledger is imported in the same way. The carried Newtonian and 1PN constants are

$$\kappa_\rho = 1, \quad n = 5, \quad \kappa_{\text{add}} = \frac{1}{2}, \quad \kappa_{\text{PV}} = \frac{3}{2}, \quad \beta_{1\text{PN}} = 3, \quad (83)$$

and the full conservative Newtonian+1PN and 2PN assemblies are already frozen in the same declared closure hierarchy. In particular, the 1PN EIH match and the 2PN ADM match are inherited here as fixed lower-order output, not reopened as free data.

The 2.5PN narrowing is also a frozen input. The surviving universal higher-order route had already been reduced to the orbital/worldtube STF quadrupole, packaged conservatively as the grouped real bundle

$$P_{20} \oplus P_{21} \oplus P_{22}. \quad (84)$$

That means the present 3PN paper is not searching over arbitrary new higher-PN carriers. It is testing whether the same grouped real P_2 ontology survives its first exact conservative stress test.

11.2 Quantities fixed by the present 3PN audit

The first new fixed data are the strict one-body 3PN repair parameters. The exact Schwarzschild gate forces

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16}. \quad (85)$$

So the carried 2PN denominator-style self sector does *not* extend to 3PN with one new scalar knob. The one-body gate already fixes a cubic static datum, a cubic denominator datum, and one genuinely new self-sector slot.

The second new fixed object is the comparable-mass middle-block compiler. The richer grouped real P_2 family closes the full nine-slot middle interaction block exactly,

$$L_{P_2}^{\text{mid}} = \sum_{A \in \{20,21,22\}} (\lambda_A^{\text{GR}} T_A + \sigma_A^{\text{GR}} S_A + \tau_A^{\text{GR}} V_A), \quad (86)$$

with a nondegenerate exact compiler matrix obeying

$$\det M_{\text{mid}} = -\frac{4}{27}. \quad (87)$$

This is the precise sense in which the algebraic 3PN bottleneck is gone: the full middle block is no longer a symbolic comparable-mass residual waiting for a future solve.

The third new fixed result is the chart-level uniqueness statement. Once the lower-order ledger is frozen and the fixed ADM Hamiltonian target is imposed, the generic-frame cubic compiler obeys the exact sign-flip rule

$$\Delta H_3 = -\Delta L_3. \quad (88)$$

So the earlier COM-blind null directions and the smaller ordinary contact/gauge orbit are intermediate bookkeeping structures only. They do not survive as true freedom after the full generic-frame Hamiltonian target is imposed.

Finally, the scalar pieces of the exact 3PN split are fixed. The unique geometry-side completion is

$$\Delta l_{15}^{(g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}, \quad (89)$$

while the apparent pure-kinetic COM remainder is

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}. \quad (90)$$

The meaning of (90) is important: it is not a new throat-response datum. It is only the ordinary-Lagrangian counterimage of the universal free relativistic two-body COM Hamiltonian.

Taken together, the fixed 3PN theorem package may be summarized as

$$\Delta \hat{L}_3^{\text{GR}} = \underbrace{\Delta l_1 v^8}_{\text{free-Hamiltonian compiler image}} + \underbrace{L_{P_2}^{\text{mid}}}_{\text{exact grouped real } P_2 \text{ closure}} + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}. \quad (91)$$

So the quantities fixed by the present audit are not just isolated coefficients; they are the complete three-lane conservative 3PN decomposition in the chosen chart.

Object	Status	Role at 3PN
Parent reduction backbone	frozen	Exact/reduced dictionary inherited from the 4D parent theory
Lower-order Newtonian+1PN+2PN ledger	frozen	Fixed conservative input and chart conventions
$\mu_{\rho,3}, d_3, s_{24}$	fixed here	Exact one-body 3PN repair ledger
$\lambda_A^{\text{GR}}, \sigma_A^{\text{GR}}, \tau_A^{\text{GR}}$	fixed here	Exact grouped real P_2 closure of the nine-slot middle block
$\Delta l_{15}^{(g)}$	fixed here	Unique geometry-side static completion
Δl_1	fixed here	Free-COM compiler image, not a new throat datum

11.3 Quantities that remain symbolic or open

What remains open is no longer algebraic coefficient matching. The open problem is the deeper moving-throat realization of already identified lanes. First, the present paper does not derive the grouped real P_2 constitutive map from a completed moving-throat PDE. It proves that the grouped family is the correct conservative carrier of the middle block, but it does not yet compute the true branch-side low-frequency origin of the coefficients in (86). Likewise, the paper fixes the unique geometry-side completion coefficient (89), but it does not yet derive that completion from a finished distributed geometry problem on the moving throat.

Second, the dynamic throat-response observables carried forward from the earlier papers remain genuinely open. A representative list is

$$\Omega_{1\perp}, \quad \Omega_{10}, \quad \Omega_0, \quad \Omega_{20}, \quad \Omega_{21}, \quad \Omega_{22}, \quad \Omega_g. \quad (92)$$

These are not missing algebraic residues in the conservative 3PN theorem. They are the dynamical pole/compliance scales of the underlying mouth / support / geometry response problem. The present paper is allowed to use the already solved zero-frequency conservative ledger, but it is not allowed to pretend that (92) have already been derived from the completed moving-throat PDE.

Third, the narrowed 2.5PN quadrupole bridge remains open in exactly the sense already isolated by the theorem audit. The future moving-throat calculation still has to determine the actual low-frequency grouped real P_2 data on the true branch, decide the isotropy verdict

$$a_2 = 0, \quad b_2 = 0, \quad (93)$$

and then realize the final passive/outgoing quadrupole normalization on that branch. The 3PN theorem removes the conservative algebraic uncertainty *beneath* this bridge; it does not close the outgoing normalization problem itself.

Finally, several sectors remain outside the honest scope of the present paper: dissipative leakage and radiation reaction beyond the already narrowed quadrupole route, spin couplings, strong-field completion, and higher-PN work beyond the solved conservative 3PN scope. So the correct negative statement is the same one already used in the lower-order series: the present paper is a full conservative derivation within a declared hierarchy, not an assumption-free theorem of the fully solved moving-throat bulk dynamics.

11.4 Interpretation

The point of this section is therefore simple. After the present audit, the 3PN problem no longer waits on another coefficient search, a hidden contact-transformation freedom, or an unexplained comparable-mass residual. All of that is fixed. What remains is the dynamical throat explanation of data that are already algebraically solved: the grouped real P_2 constitutive law, the geometry-side completion, the dynamic pole structure, and the passive/outgoing quadrupole normalization on the true branch.

That is the sharpest way to state the current roadmap. The conservative 3PN ledger is now part of the fixed carry-forward backbone. The open work begins only when one asks the completed moving-throat PDE to realize the same three lanes microscopically and then feed them into the already narrowed 2.5PN quadrupole bridge.

12 Verification and reproducibility ledger

The derivation in this paper is analytic rather than computational. The accompanying symbolic archive is therefore not presented as a black-box generator of the theorem. Its role is narrower and more auditable: it serves as a referee ledger for the exact symbolic identities, compiler maps, residual extractions, grouped- P_2 closure statements, sigma-collapse formulas, and fixed-chart assembly relations used in the main text. This is the same stance already adopted elsewhere in the series. The archive verifies the paper as written *within the declared closure hierarchy*; it does not replace any analytic step, and it does not upgrade the present paper into an assumption-free theorem of a fully solved moving-throat bulk PDE.

At 3PN the verification story is naturally split into two layers. The first layer is the inherited lower-order archive carried forward from the 4D/1PN /2PN papers, which freezes the reduction backbone, conventions, and already solved coefficient ledger used unchanged here. The second

layer is the dedicated 3PN referee suite, whose job is to check the new theorem chain specific to the present paper: the strict one-body gate, the cubic ordinary/Hamiltonian compiler, the exact COM target extraction, the Hamiltonian-first generic-frame lift, the richer grouped real P_2 middle-block closure, the sigma-collapse and geometry completion, the pure-kinetic collapse, and the final conservative 3PN theorem package. In the published bundle this second layer is dual-CAS: the stagewise SymPy master runner is paired with a bundle-local Mathematica manifest whose standalone `.wl` audits mirror the same theorem chain and carry embedded **Output:** transcripts.

12.1 Role of the 3PN referee suite

The present referee suite should be read as a symbolic reproducibility archive, not as an independent source of physical assumptions. Its function is to make each load-bearing algebraic step re-runnable in a transparent way. In particular, the suite is designed to verify the statements that are most likely to matter to a referee reading the paper as a theorem chain:

- whether the exact Schwarzschild one-body gate really forces the repair ledger $\mu_{\rho,3} = 1/4$, $d_3 = -45/4$, and $s_{24} = -1/16$;
- whether the cubic Legendre-transform identity used throughout the comparable-mass lift is exact;
- whether the imported GR COM target and the reduced self/static seed really produce the quoted residual vector;
- whether the grouped real P_2 middle block closes exactly only in the richer local family, with the stated determinant and without hidden leftover middle-block freedom;
- and whether the apparent static and pure-kinetic leftovers really do collapse into the unique geometry-side completion and the free-Hamiltonian compiler image claimed in the main text.

So the archive is not being asked to prove a different theorem from the one stated in the paper. It is being asked to verify that the paper’s own exact, reduced, and closure-level statements close where they are claimed to close.

12.2 Master archives

The main SymPy verification artifact for the present paper is `3pn_referee_master_sympy_audit.py`. It is a standalone SymPy referee script whose explicit purpose is to replay the full symbolic derivation chain corresponding to the current 3PN writeup. The master runner replays a bundle-local stage archive, executes the stages in isolated namespaces, and stops immediately on the first failed identity. This design keeps the executable theorem chain visible to a referee: there is no hidden notebook state, no dependence on local project imports, and no ambiguity about which exact stage source is being executed.

Two implementation details are especially important for reproducibility. First, each extracted stage file carries a pinned SHA256 digest inside the master runner, so the suite is not merely a loose script collection but a self-checking archive with explicit integrity tags. Second, the master script uses only the Python standard library plus SymPy. That makes the audit easy to rerun on a clean symbolic environment and aligns with the paper’s stated goal: to provide an auditable backstop for the algebra, not a fragile local build system.

The parallel Mathematica entry point is the manifest file `mathematica/wl_notes.txt`. It lists the referee-facing `3pn*_mathematica_audit.wl` scripts that mirror the same theorem chain in grouped tranches: one-body and grouped kickoff, comparable-mass compiler foundations,

imported GR COM target, generic-frame projection/contact, Hamiltonian-first lift and fixed-chart compiler, grouped- P_2 closure, and the scalar/kinetic/final-theorem closeout. Each of those scripts runs under `math -script` and ends with an embedded `Output:` block so that the archived transcript can be diffed directly against live stdout.

12.3 Stage audits replayed by the master runner

The SymPy stages are organized in the same order as the theorem chain in the paper, and the Mathematica manifest mirrors the same decomposition at a coarser but still referee-readable tranche level.

1. **Stages 1–4: one-body gate, grouped kickoff, and exact COM compiler.** The first block consists of `3pn_onebody_audit.py`, `3pn_grouped_p2_audit.py`, `3pn_comparable_mass_audit.py`, and `3pn_com_linear_map_audit.py`. These stages expand the exact Schwarzschild target, solve the minimal one-body repair ledger, freeze the natural self/static seed, set up the grouped real P_2 bookkeeping, and derive the exact diagonal COM ordinary/Hamiltonian compiler.
2. **Stages 5–11: imported GR target, generic-frame projection, and Hamiltonian-first fixed-chart lift.** The second block consists of `3pn_com_gr_target_audit.py`, `3pn_generic_frame_com_projection_audit.py`, `3pn_seed_repair_and_com_null_ideal_audit.py`, `3pn_contact_generator_and_gauge_orbit_audit.py`, `3pn_generic_frame_target_import_audit.py`, `3pn_hamiltonian_level_lift_audit.py`, and `3pn_generic_frame_hamiltonian_compiler_audit.py`. This block checks the exact GR COM target and residual extraction, the COM projection and seed-repair story, the COM-null and contact/gauge structures, the Hamiltonian-first lift, and the fixed-chart uniqueness theorem.
3. **Stages 12–18: grouped real P_2 closure, static geometry completion, and final theorem ledger.** The last block consists of `3pn_grouped_p2_target_pack_audit.py`, `3pn_grouped_p2_middle_block_test_audit.py`, `3pn_grouped_p2_richer_lift_audit.py`, `3pn_static_p0_geometry_counterterm_audit.py`, `3pn_sigma_collapse_and_unique_geometry_completion_audit.py`, `3pn_pure_kinetic_collapse_audit.py`, and `3pn_conservative_theorem_audit.py`. These stages check the grouped target pack, the failure of the minimal grouped lift, the success of the richer grouped family, the sigma-collapse, the unique geometry-side static completion, the pure-kinetic collapse to the free COM Hamiltonian, and the final conservative 3PN split.

This modular staging is intentional. It keeps the one-body gate, the comparable-mass compiler, the grouped-middle-block closure, and the final static and kinetic collapses separate, so that failure in one sector cannot be hidden by compensating changes in another.

12.4 What the suite verifies

The suite verifies the load-bearing symbolic statements of the paper sector by sector.

First, the early stages verify the exact one-body and compiler backbone. They check the strict Schwarzschild 3PN target, confirm that the carried 2PN one-body package is insufficient by

itself, solve the minimal repair ledger, and then prove the cubic perturbative Legendre-transform identity used to move cleanly between ordinary and Hamiltonian descriptions at 3PN.

Second, the middle stages verify the full comparable-mass reduction and fixed-chart uniqueness chain. They check the exact imported COM target and the resulting residual vector, the generic-frame projection and seed-repair story, the COM-null ideal and the smaller contact/gauge orbit, and finally the Hamiltonian-first lift that fixes the generic-frame representative uniquely in the ADM chart. In the language of the main text, this is the stage chain that justifies the exact sign-flip rule

$$\Delta H_3 = -\Delta L_3. \quad (94)$$

Third, the later stages verify the grouped real P_2 theorem chain. They check that the grouped target data are packaged correctly, that the minimal grouped local lift fails, that the richer grouped family succeeds, and that the exact middle-block compiler is nondegenerate with

$$\det M_{\text{mid}} = -\frac{4}{27}. \quad (95)$$

The suite therefore verifies the strongest positive conservative claim of the paper: the full nine-slot 3PN middle block is carried by the grouped real P_2 module and does not require a qualitatively new non- P_2 conservative carrier.

Fourth, the terminal stages verify the two apparent leftovers and the final three-lane theorem package. They check the sigma-collapse that removes the apparent static P_0/g ambiguity, the unique geometry-side completion, the pure-kinetic collapse to the universal free relativistic two-body COM Hamiltonian, and the exact final split

$$\Delta \hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}} + \Delta l_{15}^{(g)} U^4. \quad (96)$$

In other words, the suite verifies that the conservative 3PN bottleneck is no longer algebraic once the declared hierarchy and the fixed ADM chart are accepted.

Four reproducibility conventions are worth recording explicitly.

Named stage checks. Each stage carries a human-readable title and reduces its main claim to an explicit symbolic residual, determinant check, solved coefficient relation, or exact coefficient comparison. The reader is therefore not asked to trust a single final “zero” or “PASS” line without context.

Local assumptions and chart choices. The relevant hypotheses are attached to the stages that need them. COM reduction, Hamiltonian-first lifting, grouped isotropy bookkeeping, and the fixed-chart convention are all made explicit where they enter, so the script structure reflects the same claim taxonomy used in the paper.

Embedded integrity metadata. Because each stage source is shipped inside the master script together with its SHA256 digest, the archive records exactly which symbolic derivation text is being executed. This makes accidental stage drift easier to detect than in a loose notebook workflow.

Explicit bases and modular audit trail. Reproducibility comes from exposing the actual bases and invariant ledgers being used: the 15-slot COM basis, the generic-frame block decomposition, the COM-null ideal, the grouped real P_2 target pack, and the final three-lane split. The theorem is therefore audited in the same modular language in which it is proved.

12.5 What the suite does not verify

It is equally important to state the limits of the present archive.

First, the referee suite does not solve the fully dynamical moving-throat bulk PDE. The paper remains a full conservative 3PN derivation only within the same declared closure hierarchy already used at lower order. The scripts verify the algebra of that theorem attempt; they do not replace the missing microscopic completion.

Second, the suite does not by itself derive the future microscopic origin of the grouped real P_2 constitutive law or of the geometry-side completion. Those objects are algebraically fixed in the present paper, but their true moving-throat realization is still deferred to the later PDE-side program. Likewise, the suite does not close the passive/outgoing quadrupole normalization problem on the true branch. The 3PN result is a conservative theorem beneath that bridge, not a replacement for it.

Third, the suite is intentionally limited to the sectors addressed in the present paper. It does not claim to verify spin couplings, strong-field completion, dissipative leakage or radiation reaction, or higher-PN sectors beyond the present conservative 3PN gate.

Bottom line. The purpose of the archive is not to ask the reader to trust a computation instead of a derivation. It is to provide an auditable backstop showing that, under the stated hierarchy, the load-bearing symbolic statements close exactly:

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16}, \quad \Delta H_3 = -\Delta L_3, \quad \det M_{\text{mid}} = -\frac{4}{27}, \quad (97)$$

while the final theorem package itself takes the exact form

$$\Delta \hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}} + \Delta l_{15}^{(g)} U^4. \quad (98)$$

What remains open is not another algebraic coefficient search, but the microscopic moving-throat realization of the grouped real P_2 lane, the geometry-side completion, and the final passive/outgoing quadrupole normalization on the true branch.

13 Conclusions

This paper closes the conservative two-body 3PN problem in the same precise sense that the earlier full papers closed the conservative 1PN and 2PN sectors: not as an assumption-free theorem of a fully solved moving-throat PDE, but as an exact algebraic assembly within a declared closure hierarchy. The gain is larger than one more coefficient list. By 3PN the defect is forced to carry a genuinely nontrivial comparable-mass middle interaction block, and the present theorem shows that this burden is borne exactly by the grouped real P_2 lane, with the remaining static scalar burden collapsing to a unique geometry-side completion and the apparent pure-kinetic remainder collapsing to the ordinary-chart image of the universal free relativistic two-body Hamiltonian. In that sharpened sense, the conservative 3PN bottleneck is no longer an unresolved coefficient search.

13.1 What has been established

The proof line established in the paper may be stated compactly.

1. The strict one-body Schwarzschild gate through c^{-6} is fixed exactly, and its minimal repair ledger is not one-dimensional but

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16}. \quad (99)$$

So the carried 2PN self-sector package does not extend to 3PN by a single denominator coefficient alone.

2. The exact cubic ordinary/Hamiltonian compiler is rigid, and once the lower-order ledger is frozen the correct comparable-mass lift is Hamiltonian-first. In the fixed ADM chart this becomes the exact sign-flip identity

$$\Delta H_3 = -\Delta L_3, \quad (100)$$

so the generic-frame ordinary representative is unique after the Hamiltonian target is imposed.

3. The richer local grouped real P_2 family closes the full nine-slot middle interaction block exactly. Its compiler is nondegenerate,

$$\det M_{\text{mid}} = -\frac{4}{27}, \quad (101)$$

and the exact GR coefficient triples $(\lambda_A^{\text{GR}}, \sigma_A^{\text{GR}}, \tau_A^{\text{GR}})$ reproduce the entire solved middle block with no residual left behind.

4. The apparent static P_0/g ambiguity is not a surviving freedom. Sigma-collapse removes the one-parameter family and leaves one unique geometry-side completion,

$$\Delta l_{15}^{(g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}. \quad (102)$$

5. The apparent center-of-mass pure-kinetic remainder is not a new throat-response coefficient. It is fixed automatically by the exact free two-body relativistic Hamiltonian and therefore appears only as the ordinary-Lagrangian compiler image

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}. \quad (103)$$

When these facts are assembled, the full comparable-mass c^{-6} coefficient is assigned by three and only three lanes,

$$\Delta \hat{L}_3^{\text{GR}} = \underbrace{\Delta l_1 v^8}_{\text{free-Hamiltonian compiler image}} + \underbrace{L_{P_2}^{\text{mid}}}_{\text{exact grouped real } P_2 \text{ closure}} + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}. \quad (104)$$

There is no fourth independent conservative carrier hiding behind the COM bookkeeping. Within the declared hierarchy, the final Hamiltonian statement is therefore exact:

$$H_3^{\text{derived}} = H_3^{\text{ADM}}, \quad \Delta H_3 \equiv H_3^{\text{derived}} - H_3^{\text{ADM}} = 0. \quad (105)$$

Conceptually, the paper does three things at once. First, it upgrades the series from the full conservative 2PN ADM match to the first genuinely nontrivial higher conservative grouped- P_2 gate. Second, it proves that the middle interaction burden is already carried by the grouped real quadrupole lane, rather than by some qualitatively different hidden conservative carrier. Third, it sharpens the bookkeeping of what is fixed, what is closure-level, and what must still remain dynamical. The accompanying referee archive supports this reading in an auditable way: the 18-stage SymPy master suite checks the one-body gate, the cubic compiler, the imported target extraction, the Hamiltonian-first lift, the grouped- P_2 closure, the sigma-collapse, the pure-kinetic collapse, and the final theorem ledger inside the same declared hierarchy.

13.2 What remains open

The negative statement is equally important.

First, the paper does not solve the fully dynamical moving-throat PDE and then recover the 3PN theorem as an automatic bulk consequence. The present result remains a full conservative derivation only within the declared reduction and closure hierarchy already used by the lower-order papers. That status is not a weakness of the theorem; it is the correct statement of its scope.

Second, the paper does not yet derive the grouped real P_2 constitutive law or the geometry-side completion microscopically from the completed moving-throat branch. It proves that the grouped real P_2 module is the correct conservative carrier of the nine-slot middle block and that the residual static slot is uniquely geometry-side, but it does not yet compute those lanes from the underlying distributed throat dynamics.

Third, the dynamic pole/compliance observables carried forward from the earlier papers remain genuinely open objects of the unsolved moving-throat response problem. Representative scales are

$$\Omega_{1\perp}, \quad \Omega_{10}, \quad \Omega_0, \quad \Omega_{20}, \quad \Omega_{21}, \quad \Omega_{22}, \quad \Omega_g. \quad (106)$$

These are not leftover algebraic residues in the present 3PN theorem. They are dynamical response data of the microscopic mouth/support/geometry system.

Fourth, the present paper does not close the already narrowed 2.5PN outgoing quadrupole normalization problem. The remaining universal target may be stated in the invariant form

$$\boxed{\hat{m}_0^2 \Gamma_5 = \frac{2G}{5c^5}.} \quad (107)$$

The 3PN theorem removes the conservative algebraic uncertainty *beneath* this bridge, but it does not by itself determine the final passive/outgoing normalization on the physical moving-throat branch.

Finally, several larger sectors remain outside the honest scope of the present paper: dissipative leakage and radiation reaction beyond the already narrowed quadrupole route, spin couplings, strong-field completion, and higher-PN work beyond the solved conservative 3PN scope. What remains open is therefore no longer coefficient matching. It is the deeper dynamical throat explanation of data that are already algebraically assigned.

13.3 Interface to the next calculation

This gives the project a much cleaner handoff than before. The next job is no longer to reopen the 3PN comparable-mass algebra, to search for another hidden contact freedom, or to rerun a vague whole-theory higher-PN scan. That work is substantially finished in the present chart. The next job is the moving-throat grouped-quadrupole calculation itself.

On the conservative side, the physical branch must first determine the actual low-frequency grouped real P_2 data and decide whether the minimal isotropic verdict holds,

$$\boxed{a_2 = 0, \quad b_2 = 0.} \quad (108)$$

If that first gate fails, then the minimal isotropic quadrupole branch is already dead before outgoing normalization is attempted. If it passes, then the next conservative task is the $O(\omega^4)$ grouped moment test and the associated single-pole bookkeeping,

$$\boxed{\bar{u}_4 = 4\bar{u}_2^2, \quad \Omega_Q^2 = \frac{1}{4\bar{u}_2}, \quad \bar{K}_0^{\text{target}} = \frac{2G}{45c^5\bar{u}_2^{5/2}.} \quad (109)$$

Equivalently, once the physical grouped branch is known through its first even moments, the later outgoing Burke–Thorne coefficient is no longer an independent datum.

So the correct next calculation is sharply defined. One must derive the grouped real P_2 constitutive data and the geometry-side completion from the physical moving-throat branch, determine whether the isotropic passive/outgoing branch is actually realized there, and then evaluate the single invariant normalization product in (107). That is the only place where the present theorem still needs to be sharpened from a closure-level conservative result into a fully microscopic one.

The strongest honest endpoint of the paper is therefore also the strongest honest starting point for the next one: within the declared hierarchy, the conservative 3PN bottleneck is gone, the grouped real P_2 and geometry lanes are the correct carriers, and the only serious universal bridge question left open is the passive/outgoing normalization of that already identified quadrupole branch.

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A Carry-forward formulas from the earlier papers

This appendix collects the specific formulas and status statements imported from the parent 4D paper and the full conservative 1PN, 2PN, and 2.5PN packages and used without rederivation in the present 3PN work. Its role is technical rather than historical. We do not repeat the full parent action, the full lower-order proofs, or the longer appendix-side derivations that fixed the wake, breathing, support, and quadrupole-response data. Instead, we record only the load-bearing identities, reduced equations, and frozen coefficients that the present 3PN theorem chain actually uses.

The claim-status boundary is the same one used throughout the earlier papers. Exact projection identities from the parent 4D theory are kept visibly exact. The near-zone Poisson hook, the zero-mode Maxwell law, and the worldtube point-particle reduction are carried only as controlled reductions. The conservative Newtonian+1PN and 2PN ledgers are treated as frozen outputs of the earlier program within its declared closure hierarchy, not as live parameters to be retuned in the present paper. The 2.5PN package is used in the same way: not as a finished moving-throat PDE theorem, but as the theorem audit that narrows the surviving higher-order universal point-particle route to the orbital/worldtube STF quadrupole branch and repackages its conservative data as the grouped real $P_{20} \oplus P_{21} \oplus P_{22}$ bundle.

A.1 Newtonian and 1PN carried data

Before recording the Newtonian and 1PN coefficient ledger itself, it is useful to state the minimal parent 4D reduction dictionary through which that ledger is carried into the post-Newtonian sector. The parent theory lives on a $(4 + 1)$ -dimensional spacetime with coordinates

$$X^M = (t, x, y, z, w), \quad \mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \quad \mathbf{x} = (x, y, z) \in \mathbb{R}^3.$$

The bulk variables carried into the present paper are the matter field $\psi(\mathbf{X}, t)$, its density

$$\rho(\mathbf{X}, t) = |\psi(\mathbf{X}, t)|^2,$$

the optional localized gauge field $A_M(\mathbf{X}, t)$ with localization profile $Z(w)$, and the effective throat geometry variables

$$a(t), \quad L(t).$$

When electric charge is named explicitly in this inherited downstream notation, it is always the corrected branch quantity

$$q_* = \eta_Q e_*, \quad q_{\text{eff}} = \frac{q_*}{\sqrt{Z_{\text{int}}}}, \quad Z_{\text{int}} = \int_{-\infty}^{\infty} Z(w) dw,$$

not the gravity-side scalar coefficient κ_ρ used in the post-Newtonian ledger below.

Brane observables are defined by a fixed normalized projection kernel $W(w)$,

$$\int_{-\infty}^{\infty} W(w) dw = 1, \quad \mathcal{P}_W[Q](t, \mathbf{x}) \equiv \int_{-\infty}^{\infty} W(w) Q(t, \mathbf{x}, w) dw.$$

In particular,

$$\rho_{\text{br}} = \mathcal{P}_W[\rho], \quad \mathbf{J}_{\text{br}} = \mathcal{P}_W[(j^x, j^y, j^z)], \quad \mathbf{v}_{\text{br}} = \frac{\mathbf{J}_{\text{br}}}{\rho_{\text{br}}} \quad (\rho_{\text{br}} > 0).$$

Projection is not the same thing as reduction. Projection defines what a brane observer measures; reduction refers to the further elimination of the extra coordinate under an explicit ansatz or scaling regime.

From exact bulk continuity,

$$\partial_t \rho + \partial_i j^i = 0, \quad i \in \{x, y, z, w\},$$

one obtains the exact open-system brane continuity law

$$\partial_t \rho_{\text{br}} + \nabla_3 \cdot \mathbf{J}_{\text{br}} = S_{\text{leak}},$$

with leakage source

$$S_{\text{leak}} = -[W(w)j^w]_{-\infty}^{+\infty} + \int_{-\infty}^{\infty} W'(w)j^w dw.$$

After Helmholtz decomposition of the brane velocity,

$$\mathbf{v}_{\text{br}} = \nabla_3 \phi_{\text{br}} + \mathbf{v}_T, \quad \nabla_3 \cdot \mathbf{v}_T = 0,$$

projection continuity becomes the exact longitudinal identity

$$\rho_{\text{br}} \nabla_3^2 \phi_{\text{br}} = S_{\text{leak}} - \partial_t \rho_{\text{br}} - (\nabla_3 \rho_{\text{br}}) \cdot (\nabla_3 \phi_{\text{br}} + \mathbf{v}_T).$$

Under the usual quasi-static near-zone simplifications this becomes the carried Poisson hook

$$\nabla_3^2 \phi_{\text{br}} \simeq \frac{S_{\text{eff}}}{\rho_0}, \quad \Phi_N = \lambda_\Phi \phi_{\text{br}},$$

which is the origin of the Newtonian scalar sector used in the lower-order particle reduction. The corresponding coherent-defect / small-body worldtube limit gives the point-particle force law

$$M_A \ddot{\mathbf{X}}_A = -M_A \nabla \Phi_{\text{ext}}(\mathbf{X}_A) + O\left(\frac{a^2}{r^2} \frac{\Phi_{\text{ext}}}{r}\right),$$

with the first finite-size correction quadrupolar and therefore suppressed in the controlled small-worldtube regime. For completeness, the same parent paper also carries a controlled zero-mode Maxwell reduction. Under the far-field brane assumptions

$$A_w \approx 0, \quad \partial_w A_\mu \approx 0, \quad J^w \approx 0, \quad F_{\mu w} \approx 0,$$

integration over w gives the effective brane Maxwell law

$$\partial_\mu F^{\mu\nu} = \mu_0^{\text{eff}} J_{\text{eff}}^\nu, \quad \mu_0^{\text{eff}} = \frac{\mu_0}{Z_{\text{int}}}.$$

This reduction suppresses the mixed channels only in the controlled far-field limit. It does not erase $(A_w, J^w, F_{\mu w})$ from the microscopic ontology.

The Newtonian point-particle block carried into every later paper is therefore

$$L_0 = \frac{1}{2}m_A v_A^2 + \frac{1}{2}m_B v_B^2 + \frac{Gm_A m_B}{r}.$$

At 1PN the scalar/optical one-body reduction fixes the lower-order viability conditions

$$\kappa_\rho = 1, \quad n = 5, \quad \kappa_{\text{add}} = \frac{1}{2}, \quad \kappa_{\text{PV}} = \frac{3}{2}, \quad \beta_{\text{1PN}} = 3.$$

The same response-side closure also freezes the breathing ledger

$$E_w : E_f : E_{\text{PV}} = 11 : 2 : 5, \quad \frac{d \ln a_*}{d \ln \rho} = -\frac{57}{64}.$$

In the parity-even wake sector, the constructive isotropic Fourier-space closure fixes

$$\alpha^2 = \frac{3}{4}, \quad a_H = 0, \quad K_{\text{vec}} = \frac{2}{\pi^2},$$

which in turn give the EIH cross coefficients

$$C_{\parallel} = -\frac{7}{2}, \quad C_L = -\frac{1}{2}.$$

With these values frozen, the full carried Newtonian+1PN two-body block is

$$L_1 = \frac{1}{8}(m_A v_A^4 + m_B v_B^4) + \frac{Gm_A m_B}{r} \left[\frac{3}{2}(v_A^2 + v_B^2) - \frac{7}{2}v_{AB} - \frac{1}{2}v_{An}v_{Bn} \right] - \frac{G^2 m_A m_B (m_A + m_B)}{2r^2}.$$

Nothing in the present 3PN paper is allowed to retune these lower-order values. They are carry-forward anchors rather than open coefficients.

A.2 2PN carried data and the $P_0 \oplus P_2$ support hierarchy

The full conservative 2PN paper closes the comparable-mass c^{-4} ledger against the standard generic-frame ADM target within the same declared closure hierarchy, but the present 3PN paper imports from it only the load-bearing formulas and the response-side structure that sharpened the ontology of the defect. At the strict one-body level, the exact isotropic Schwarzschild gate through 2PN is packaged by

$$D_{\text{target}}(u) = 1 - 4u + 8u^2 + O(u^3), \quad u \equiv \frac{U}{c^2},$$

and the pure-static 2PN gate fixes

$$\lambda_\rho = \frac{1}{2}, \quad L_{2,\text{static}} = \frac{G^3 m_A m_B (m_A^2 + m_B^2)}{4r^3}.$$

The natural 2PN self/static seed promoted from the exact one-body target is

$$L_{2,\text{self+static}} = \frac{m_A v_A^6 + m_B v_B^6}{16} + \frac{7Gm_A m_B}{8r}(v_A^4 + v_B^4) \\ + \frac{2G^2 m_A m_B}{r^2}(m_B v_A^2 + m_A v_B^2) \\ + \frac{G^3 m_A m_B(m_A^2 + m_B^2)}{4r^3}.$$

The full carried 2PN coefficient inserted into the Newtonian+1PN ledger is

$$L_{2,\text{full}} = \frac{m_A v_A^6 + m_B v_B^6}{16} \\ + \frac{Gm_A m_B}{r} \left[\frac{7}{8}(v_A^4 + v_B^4) - \frac{7}{4}v_{AB}(v_A^2 + v_B^2) \right. \\ \left. - \frac{1}{4}v_{An}v_{Bn}(v_A^2 + v_B^2) + \frac{11}{8}v_A^2 v_B^2 \right. \\ \left. + \frac{1}{4}v_{AB}^2 - \frac{5}{8}(v_A^2 v_{Bn}^2 + v_B^2 v_{An}^2) + \frac{3}{2}v_{AB}v_{An}v_{Bn} + \frac{3}{8}v_{An}^2 v_{Bn}^2 \right] \\ + \frac{G^2 m_A m_B}{r^2} \left[\left(2m_B + \frac{11}{8}m_A\right)v_A^2 + \left(2m_A + \frac{11}{8}m_B\right)v_B^2 \right. \\ \left. - \frac{15}{4}(m_A + m_B)v_{AB} + \frac{15}{8}(m_A v_{An}^2 + m_B v_{Bn}^2) \right] \\ + \frac{G^3 m_A m_B}{4r^3}(m_A^2 + 5m_A m_B + m_B^2).$$

So by the start of the present paper, the full conservative Newtonian+1PN+2PN ledger is already frozen in the declared hierarchy.

Conceptually, the decisive ontology change at 2PN is that the defect is no longer described adequately as a pure scalar monopole. The conservative cross sector already splits into three response lanes:

1. a carried odd dipole wake sector,
2. a genuine even $P_0 \oplus P_2$ mouth/support layer,
3. and a separate geometry-closure channel.

The appendix-side zero-frequency throat data make this explicit. The odd dipole residues are frozen to

$$R_{1\perp} = \frac{7}{2}, \quad R_{10} = 4,$$

while the canonical even support residues are

$$R_0 = R_{20} = R_{21} = R_{22} = 1.$$

The same solved static even operator fixes the source-weight vector

$$J = \left(\frac{4}{\sqrt{5}}, \frac{5}{4}, 0, 0, 0, 0 \right), \quad J \cdot J = \frac{381}{80},$$

and the pure-potential geometry-closure deficit

$$\Delta_{\text{geom}} = \frac{281}{80}.$$

The monopole wall add-on

$$\Delta K_{00} = \frac{109}{280}$$

is kept distinct from Δ_{geom} ; they are not the same object and should not be merged in later bookkeeping.

What remains open after the conservative 2PN theorem is not these zero-frequency residues but the fully dynamical completion. In particular, the pole or compliance scales

$$\Omega_{1\perp}, \Omega_{10}, \Omega_0, \Omega_{20}, \Omega_{21}, \Omega_{22}, \Omega_g$$

are not claimed as already derived from the completed moving-throat PDE. This status boundary is central for the present paper. The 3PN calculation is allowed to use the fixed 2PN residues and source data, but it is not allowed to treat the fully dynamical throat completion as already solved.

A.3 2.5PN narrowing to the grouped real P_2 target

The 2.5PN theorem audit is what makes the present 3PN problem sharply focused rather than diffuse. The decisive benchmark of that audit shows that the frozen conservative hierarchy by itself cannot generate a nonzero local 2.5PN reaction theorem, while a minimal retarded quadrupole odd term lands on the standard Burke–Thorne / Iyer–Will family member

$$\alpha = 4, \quad \beta = 5.$$

For the present conservative paper, however, the important carry-forward message is narrower: after the scalar, dipole, and quadrupole channel audits, the live higher-order universal point-particle route collapses to the orbital/worldtube STF quadrupole branch. In grouped real language, that means the right conservative front end is the grouped real

$$P_{20} \oplus P_{21} \oplus P_{22}$$

bundle, not an unfocused collection of generic higher-order coefficients.

Accordingly, the grouped conservative response is organized as

$$Y_{20}(\omega) = 1 + u_2^{(20)}\omega^2 + u_4^{(20)}\omega^4 + O(\omega^6),$$

$$Y_{21}(\omega) = 1 + u_2^{(21)}\omega^2 + u_4^{(21)}\omega^4 + O(\omega^6),$$

$$Y_{22}(\omega) = 1 + u_2^{(22)}\omega^2 + u_4^{(22)}\omega^4 + O(\omega^6).$$

It is convenient to repackage these grouped triples into trace/anomaly variables. At $O(\omega^2)$ one defines

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \quad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \quad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2},$$

and at $O(\omega^4)$

$$\bar{u}_4 = \frac{u_4^{(20)} + 2u_4^{(21)} + 2u_4^{(22)}}{5}, \quad a_4 = \frac{2u_4^{(20)} - u_4^{(21)} - u_4^{(22)}}{10}, \quad b_4 = \frac{u_4^{(21)} - u_4^{(22)}}{2}.$$

The exact inverse map is

$$u_2^{(20)} = \bar{u}_2 + 4a_2, \quad u_2^{(21)} = \bar{u}_2 - a_2 + b_2, \quad u_2^{(22)} = \bar{u}_2 - a_2 - b_2,$$

$$u_4^{(20)} = \bar{u}_4 + 4a_4, \quad u_4^{(21)} = \bar{u}_4 - a_4 + b_4, \quad u_4^{(22)} = \bar{u}_4 - a_4 - b_4.$$

So the first isotropy gate for the minimal branch is simply

$$a_2 = 0, \quad b_2 = 0,$$

and the later $O(\omega^4)$ branch test is

$$a_4 = 0, \quad b_4 = 0.$$

At the level relevant for the present 3PN paper, the important point is that the grouped real conservative front end is already completely specified by the three $O(\omega^2)$ coefficients

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)},$$

or, equivalently, by $(\bar{u}_2, a_2, b_2)$. That is why the 3PN paper can ask whether the conservative middle block closes in grouped- P_2 language without pretending to have finished the full outgoing normalization problem.

For later reference, the minimal isotropic single-pole branch of the same 2.5PN package implies the exact formulas

$$\Omega_Q^2 = \frac{1}{4u_2}, \quad \Gamma_5^{\text{norm}} = 9u_2^{5/2}, \quad \bar{K}_0^{\text{target}} = \frac{2G}{45c^5 u_2^{5/2}},$$

together with the $O(\omega^4)$ branch identity

$$u_4 = 4u_2^2.$$

These are not part of the present conservative 3PN theorem, but they explain why the 2.5PN narrowing makes the grouped real P_2 front end the natural higher-order conservative payload.

B Exact 3PN one-body Schwarzschild expansion

The strict one-body gate for the present 3PN program is the isotropic Schwarzschild worldline action. Its role is narrow but load-bearing. Before any one-body repair ledger or comparable-mass lift is attempted, the reduced hierarchy must know exactly what the test-mass sector is allowed to reduce to. The point of the present appendix is therefore simply to record the exact Schwarzschild expansion through total order c^{-6} and isolate the new 3PN block.

Use the standard one-body potential variable

$$u \equiv \frac{U}{c^2} = \frac{GM}{rc^2}. \quad (110)$$

The exact test-mass worldline Lagrangian in isotropic Schwarzschild coordinates is

$$\frac{L_{\text{Schw}}}{m} = -c^2 \sqrt{\left(\frac{1-u/2}{1+u/2}\right)^2 - (1+u/2)^4 \frac{v^2}{c^2}}. \quad (111)$$

Expanding (111) through total order c^{-6} gives the full one-body target carried into the present paper,

$$\begin{aligned} \frac{L_{\text{Schw}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left(\frac{1}{8}v^4 + \frac{3}{2}Uv^2 - \frac{1}{2}U^2 \right) \\ & + \frac{1}{c^4} \left(\frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 + \frac{1}{4}U^3 \right) \\ & + \frac{1}{c^6} \left(\frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4 \right) + O(c^{-8}). \end{aligned} \quad (112)$$

The coefficient of c^{-6} is therefore the exact 3PN one-body destination,

$$\boxed{\frac{L_{\text{3PN,1body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4.} \quad (113)$$

This is the strict test-mass gate that every later comparable-mass construction must reduce to. It is the exact analogue at 3PN of the Schwarzschild one-body gate already used successfully at 2PN. The logic of the present paper is therefore deliberately sequential: do not begin with the generic two-body lift; first freeze the exact one-body target in (113), then ask what additional data the carried denominator/self/static hierarchy needs in order to hit it exactly. That repair algebra is the subject of the next appendix.

C One-body repair algebra

Appendix B freezes the exact isotropic Schwarzschild one-body target through total order c^{-6} . The natural next question is whether the carried 2PN denominator/self/static packaging can simply be pushed one order higher, or whether the strict test-mass gate already forces genuinely new 3PN one-body data. The point of the present appendix is to answer that question algebraically and coefficient by coefficient.

The result is rigid. One cubic denominator datum is needed to repair the U^3v^2 slot, one cubic static datum is needed to repair the pure U^4 slot, and one additional self-sector invariant is needed because the U^2v^4 slot remains wrong even after the first two repairs are imposed. So the full one-body 3PN repair ledger is already three-parameter at the strict Schwarzschild gate.

C.1 Cubic denominator datum

The first test is the minimal cubic continuation of the carried denominator-side self sector,

$$L_{\text{red}} = -mc^2(1-u)\sqrt{1 - \frac{v^2/c^2}{D(u)}}, \quad u \equiv \frac{U}{c^2}, \quad D(u) = 1 - 4u + 8u^2 + d_3u^3 + O(u^4). \quad (114)$$

Expanding (114) through total order c^{-6} gives

$$\begin{aligned} \frac{L_{\text{red}}}{m} = & -c^2 + \frac{1}{2}v^2 + U + \frac{1}{c^2} \left(\frac{1}{8}v^4 + \frac{3}{2}Uv^2 \right) \\ & + \frac{1}{c^4} \left(\frac{1}{16}v^6 + \frac{7}{8}Uv^4 + 2U^2v^2 \right) \\ & + \frac{1}{c^6} \left(\frac{5}{128}v^8 + \frac{11}{16}Uv^6 + 3U^2v^4 + \left(-4 - \frac{d_3}{2} \right) U^3v^2 \right) + O(c^{-8}). \end{aligned} \quad (115)$$

Two features of (115) are immediate. First, the carried denominator logic already reproduces the universal 3PN coefficients of v^8 and Uv^6 automatically. Second, it leaves the U^3v^2 slot open through the new cubic coefficient d_3 .

Matching the exact Schwarzschild target (113) in the U^3v^2 channel gives

$$-4 - \frac{d_3}{2} = \frac{13}{8} \quad \implies \quad \boxed{d_3 = -\frac{45}{4}}. \quad (116)$$

So the cubic denominator datum is fixed exactly by the U^3v^2 slot.

It is worth recording that the unmodified 2PN invariant-denominator continuation would have predicted a completely different cubic coefficient. If one simply extends the carried 2PN denominator-side packaging without opening any genuinely new cubic invariant, the audit gives

$$d_3^{\text{carry}} = \frac{21783}{2432}, \quad (117)$$

which is nowhere near the exact value in (116). So already at the U^3v^2 gate, the strict one-body problem forces a genuine new cubic invariant correction rather than a trivial continuation of the old 2PN denominator game.

C.2 Static gate

The pure static slot gives an independent gate. Extend the carried local mass-scaling sector by one cubic static coefficient $\mu_{\rho,3}$,

$$\frac{L_{\text{static,1body}}}{m} = U - \frac{1}{2} \frac{U^2}{c^2} + \frac{1}{4} \frac{U^3}{c^4} - \frac{\mu_{\rho,3}}{2} \frac{U^4}{c^6} + O(c^{-8}). \quad (118)$$

The exact Schwarzschild target requires the static 3PN term

$$-\frac{1}{8} \frac{U^4}{c^6}. \quad (119)$$

So the cubic static gate is fixed immediately by

$$-\frac{\mu_{\rho,3}}{2} = -\frac{1}{8} \quad \Longrightarrow \quad \boxed{\mu_{\rho,3} = \frac{1}{4}}. \quad (120)$$

This is the direct 3PN analogue of the static gate that already appeared at 2PN. The important point is that the static channel remains a separate exact gate of its own; it is not a residual bookkeeping convenience.

C.3 The extra self datum and why one cubic denominator correction is not enough

Once the cubic denominator datum and the cubic static gate are admitted, there is still one leftover mismatch. The minimal repaired one-body candidate is

$$\frac{L_{\text{cand}}}{m} = \frac{L_{\text{red}}}{m} - \frac{1}{2} \frac{U^2}{c^2} + \frac{1}{4} \frac{U^3}{c^4} - \frac{\mu_{\rho,3}}{2} \frac{U^4}{c^6} + s_{24} \frac{U^2 v^4}{c^6}, \quad (121)$$

where the first three displayed corrections simply restore the carried lower static ledger and the final term is the minimal new explicit self-sector slot. Subtracting (121) from the exact target gives

$$\left(\frac{L_{\text{3PN,1body}}}{m} - \frac{L_{\text{cand}}}{m} \right)_{c^{-6}} = \left(\frac{45}{8} + \frac{d_3}{2} \right) U^3 v^2 + \left(\frac{47}{16} - 3 - s_{24} \right) U^2 v^4 + \left(-\frac{1}{8} + \frac{\mu_{\rho,3}}{2} \right) U^4. \quad (122)$$

So the one-body repair algebra separates cleanly by slot.

The first and third coefficients in (122) vanish once (116) and (120) are imposed. But even after those two exact repairs, the denominator-style self sector still predicts

$$3U^2 v^4 = \frac{48}{16} U^2 v^4, \quad (123)$$

whereas the exact Schwarzschild target demands

$$\frac{47}{16} U^2 v^4. \quad (124)$$

So there is an uncanceled one-body discrepancy of

$$-\frac{1}{16} U^2 v^4. \quad (125)$$

This cannot be absorbed by d_3 , because d_3 controls only the $U^3 v^2$ slot, and it cannot be absorbed by $\mu_{\rho,3}$, because $\mu_{\rho,3}$ controls only the pure static U^4 slot. The minimal repair is therefore to admit the explicit self invariant already written in (121), with coefficient fixed by

$$\frac{47}{16} - 3 - s_{24} = 0 \quad \Longrightarrow \quad \boxed{s_{24} = -\frac{1}{16}}. \quad (126)$$

Collecting everything, the exact one-body repair ledger is

$$\boxed{\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16}.} \quad (127)$$

This is the cleanest strict one-body lesson of the whole 3PN analysis. The 3PN gate does *not* say “continue the 2PN denominator idea by one more coefficient.” It says that three distinct data are already needed at the strict test-mass level:

1. a cubic static gate,
2. a cubic denominator datum,
3. and one genuinely new self-sector slot.

That is exactly why the full 3PN program had to begin with a one-body audit before the comparable-mass lift was even well posed. Once the exact target and its minimal repair ledger are frozen, the next step is to promote the strict one-body structure into the natural two-body self/static seed, which is the subject of the next appendix.

D Natural 3PN self/static seed derivation

Appendices B and C freeze both ingredients that have to exist before any honest comparable-mass 3PN lift can begin: the exact isotropic Schwarzschild one-body target, and the minimal one-body repair ledger needed to realize it inside the carried self/static packaging. Once those are fixed, the next object is not a generic two-body ansatz. It is the unique symmetric *bodywise* promotion of the exact one-body data into the two-body sector.

The purpose of the present appendix is to write that promotion explicitly and to explain why it is the correct starting seed. The logic is the same one that already worked at 2PN: keep the seed strictly bodywise and symmetric, so that (i) the strict test-mass reduction is automatic, and (ii) no genuinely comparable-mass interaction data are smuggled into the seed before the residual solve even begins.

D.1 Bodywise promotion of the exact one-body target

Start from the exact one-body 3PN target from Appendix B,

$$\frac{L_{3\text{PN},1\text{body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4. \quad (128)$$

For body A moving in the field of body B , the strict bodywise substitutions are

$$U_A = \frac{Gm_B}{r}, \quad m \rightarrow m_A, \quad v \rightarrow v_A. \quad (129)$$

For body B moving in the field of body A , they are

$$U_B = \frac{Gm_A}{r}, \quad m \rightarrow m_B, \quad v \rightarrow v_B. \quad (130)$$

So the admissible self/static seed is fixed by the rule

$$L_{3,\text{seed}} = m_A \left[\frac{L_{3\text{PN},1\text{body}}}{m} \right]_{U \rightarrow U_A, v \rightarrow v_A} + m_B \left[\frac{L_{3\text{PN},1\text{body}}}{m} \right]_{U \rightarrow U_B, v \rightarrow v_B}. \quad (131)$$

Evaluating (131) gives the natural symmetric self/static 3PN seed,

$$\boxed{L_{3,\text{seed}} = \frac{5}{128}(m_A v_A^8 + m_B v_B^8) + \frac{11Gm_A m_B}{16r}(v_A^6 + v_B^6)} \quad (132)$$

$$\boxed{+\frac{47G^2m_Am_B}{16r^2}(m_Bv_A^4+m_Av_B^4)+\frac{13G^3m_Am_B}{8r^3}(m_B^2v_A^2+m_A^2v_B^2)-\frac{G^4m_Am_B}{8r^4}(m_B^3+m_A^3)}, \quad (133)$$

with the whole block understood as the coefficient of c^{-6} in the full conservative expansion.

Written term by term, the map from the one-body monomials in (128) to their two-body bodywise images is completely rigid:

$$mv^8 \mapsto m_Av_A^8 + m_Bv_B^8, \quad (134)$$

$$mUv^6 \mapsto \frac{Gm_Am_B}{r}(v_A^6 + v_B^6), \quad (135)$$

$$mU^2v^4 \mapsto \frac{G^2m_Am_B}{r^2}(m_Bv_A^4 + m_Av_B^4), \quad (136)$$

$$mU^3v^2 \mapsto \frac{G^3m_Am_B}{r^3}(m_B^2v_A^2 + m_A^2v_B^2), \quad (137)$$

$$mU^4 \mapsto \frac{G^4m_Am_B}{r^4}(m_B^3 + m_A^3). \quad (138)$$

No extra freedom appears in this dictionary once the seed is required to remain bodywise and symmetric.

D.2 Exact test-mass reduction and uniqueness

The reason the bodywise rule is the right one is that it preserves the strict one-body gate automatically. Take the test-mass limit with body A light, body B heavy, and the heavy body unperturbed at leading order:

$$m_A \rightarrow \mu, \quad m_B \rightarrow M, \quad v_B \rightarrow 0, \quad U_A \rightarrow \frac{GM}{r}. \quad (139)$$

Then (132)–(133) reduce immediately to

$$L_{3,\text{seed}} \rightarrow \mu \left(\frac{5}{128}v_A^8 + \frac{11}{16}U_Av_A^6 + \frac{47}{16}U_A^2v_A^4 + \frac{13}{8}U_A^3v_A^2 - \frac{1}{8}U_A^4 \right), \quad (140)$$

which is exactly the strict Schwarzschild target (128) for the light body. The heavy-body copy collapses because $v_B \rightarrow 0$, so it does not contaminate the light-body dynamics.

This already gives the uniqueness statement needed later in the paper. Suppose one asks for a 3PN seed with the following three properties:

1. it is symmetric under $A \leftrightarrow B$,
2. it reproduces the exact one-body target in the strict test-mass limit,
3. it contains only bodywise/self-static images of the one-body monomials, and no genuinely comparable-mass cross-velocity structures.

Then each monomial in (128) has exactly one admissible symmetric image, namely the one listed in (134)–(138). So the seed (132)–(133) is not just natural; within that bodywise/self-static class it is unique.

D.3 Why this is the correct starting seed for the 3PN lift

The seed has three structural properties that make it the correct starting point for the full 3PN lift.

First, it isolates the universal bodywise content and nothing else. The seed is built only from the exact one-body data already forced by the strict Schwarzschild gate. It therefore contains

the universal free v^8 term, the bodywise Uv^6 , U^2v^4 , and U^3v^2 self sectors, and the static U^4 sector, but it contains no genuinely comparable-mass middle-block structures of the form

$$v_A^2 v_B^6, \quad v_A^4 v_B^4, \quad \text{or analogous mixed invariant combinations.}$$

Those belong to the comparable-mass residual, not to the seed.

Second, it makes the later residual split conceptually clean. The full 3PN problem can therefore be organized as

$$L_3 = L_{3,\text{seed}} + \Delta L_3, \quad (141)$$

where ΔL_3 is the pure comparable-mass interaction residual to be solved on the later 24/17/8/1 generic-frame scaffold. In other words, once the seed is frozen, every remaining 3PN interaction coefficient can be interpreted as “beyond the exact one-body/self-static core” rather than as a blur of bodywise and comparable-mass information.

Third, the seed is sparse when carried into the later COM chart. Before the lower-order compiler feedback is applied, only the self/static lanes are switched on directly. That sparseness is precisely what makes the imported COM residual finite and interpretable: the later exact 15-slot residual vector is not “the whole 3PN answer,” but the part of the 3PN answer that remains once the unique bodywise seed has already been removed.

That is why the object derived here is the right seed and not merely a plausible starting guess. It is the unique symmetric promotion of the exact one-body 3PN data into the same self/static lane that already organized the successful 2PN construction, and it is the last universal input that should be frozen before the genuine comparable-mass 3PN solve begins.

E Exact cubic Legendre-transform identity at 3PN

The 2PN assembly already relied on the perturbative Legendre transform through quadratic order. The first genuinely new compiler ingredient at 3PN is the *cubic-order* extension. The purpose of the present appendix is to freeze that identity in the compact matrix form actually used throughout the 3PN construction.

The role of this appendix is therefore very specific. It does not solve the full comparable-mass problem by itself. What it does is fix the exact ordinary Lagrangian $\rightarrow$ Hamiltonian compiler through order ε^3 with $\varepsilon = c^{-2}$. Once that identity is available, the later reduced COM and generic-frame lifts are controlled algebraically rather than guessed.

E.1 Perturbative setup

Write the conservative ordinary Lagrangian as

$$L(v) = L_0(v) + \varepsilon L_1(v) + \varepsilon^2 L_2(v) + \varepsilon^3 L_3(v), \quad \varepsilon = c^{-2}, \quad (142)$$

with quadratic zeroth-order block

$$L_0(v) = \frac{1}{2} v^T M v, \quad (143)$$

where the Newtonian mass matrix M is constant at the order relevant for the perturbative inversion. Throughout this appendix, gradients with respect to v are taken as column vectors, and Hessians are taken as matrices.

The canonical momentum is

$$p \equiv \frac{\partial L}{\partial v} = M v + \varepsilon \frac{\partial L_1}{\partial v} + \varepsilon^2 \frac{\partial L_2}{\partial v} + \varepsilon^3 \frac{\partial L_3}{\partial v}. \quad (144)$$

Define the Newtonian momentum-space velocity

$$v_0 = M^{-1} p, \quad (145)$$

and the lower-order derivative data evaluated at v_0 ,

$$A_0 \equiv \left. \frac{\partial L_1}{\partial v} \right|_{v_0}, \quad B_0 \equiv \left. \frac{\partial L_2}{\partial v} \right|_{v_0}, \quad C_0 \equiv \left. \frac{\partial^2 L_1}{\partial v^2} \right|_{v_0}. \quad (146)$$

The perturbative momentum inversion is written as

$$v = v_0 + \varepsilon v_1 + \varepsilon^2 v_2 + O(\varepsilon^3). \quad (147)$$

For the cubic Hamiltonian coefficient H_3 , only v_1 and v_2 are needed explicitly.

E.2 Order-by-order momentum inversion

Insert (147) into (144) and expand around v_0 . To the order needed for H_3 , the Taylor expansions are

$$\frac{\partial L_1}{\partial v}(v) = A_0 + \varepsilon C_0 v_1 + O(\varepsilon^2), \quad \frac{\partial L_2}{\partial v}(v) = B_0 + O(\varepsilon). \quad (148)$$

Using $Mv_0 = p$, the momentum equation becomes

$$0 = \varepsilon(Mv_1 + A_0) + \varepsilon^2(Mv_2 + B_0 + C_0 v_1) + O(\varepsilon^3). \quad (149)$$

So the coefficients are fixed order by order:

$$Mv_1 + A_0 = 0, \quad Mv_2 + B_0 + C_0 v_1 = 0. \quad (150)$$

Therefore

$$\boxed{v_1 = -M^{-1}A_0, \quad v_2 = M^{-1}C_0 M^{-1}A_0 - M^{-1}B_0.} \quad (151)$$

This is the exact perturbative inversion needed through cubic order.

E.3 Exact cubic Hamiltonian identity

Now substitute (147) and (151) into

$$H(p) = p \cdot v - L(v). \quad (152)$$

The Hamiltonian expansion takes the form

$$H = H_0 + \varepsilon H_1 + \varepsilon^2 H_2 + \varepsilon^3 H_3 + O(\varepsilon^4), \quad (153)$$

with zeroth-order piece

$$H_0 = \frac{1}{2} p^T M^{-1} p. \quad (154)$$

Carrying the cubic expansion through exactly gives

$$H_1 = -L_1(v_0), \quad (155)$$

$$H_2 = -L_2(v_0) + \frac{1}{2} A_0^T M^{-1} A_0, \quad (156)$$

and the key 3PN compiler identity

$$\boxed{H_3 = -L_3(v_0) + A_0^T M^{-1} B_0 - \frac{1}{2} A_0^T M^{-1} C_0 M^{-1} A_0.} \quad (157)$$

Equation (157) is the exact cubic analogue of the quadratic compiler identity already used at 2PN. It is the formula actually used to carry the ordinary 3PN data into the Hamiltonian chart.

It is also useful to record the reduced COM specialization. In the normalized COM variables used later, the Newtonian quadratic block is the identity, so $M = I$ and therefore

$$v_0 = p, \quad H_3 = -L_3(p) + A_0 \cdot B_0 - \frac{1}{2} A_0^T C_0 A_0. \quad (158)$$

This is the compact form that the referee audit checks explicitly before the later slot-by-slot COM compiler is built.

E.4 Practical meaning for the 3PN lift

The practical meaning of (157) is the same one that made the 2PN paper manageable, but now one order higher. Once the lower-order 1PN/2PN ledger is frozen, the feedback tensors A_0 , B_0 , and C_0 are already known. So the genuinely new 3PN Hamiltonian content enters only through the explicit $-L_3(v_0)$ term, while the remaining cubic feedback terms are completely fixed by carried lower-order data.

That observation is what turns the comparable-mass 3PN problem into a finite algebraic residual rather than a blind full derivation campaign. In the later reduced COM basis the map generated by (157) diagonalizes, and in the fixed generic-frame Hamiltonian chart the residual compiler becomes rigid enough that the remaining 3PN bookkeeping can be solved slot by slot.

F COM basis and diagonal compiler derivation

Appendix E fixes the exact cubic ordinary-Lagrangian $\rightarrow$ Hamiltonian compiler through order c^{-6} . The purpose of the present appendix is to specialize that compiler to the isotropic reduced center-of-mass chart actually used in the 3PN construction and to show that, in the natural 15-slot basis, the map reduces *diagonally*.

That diagonalization is one of the main algebraic simplifications of the whole 3PN program. Once the lower-order reduced COM blocks are frozen, the target Hamiltonian coefficients h_i determine the ordinary-Lagrangian coefficients l_i slot by slot, with no further nontrivial inversion. This is the precise sense in which the COM stage of the 3PN lift is finite rather than open-ended.

F.1 Reduced COM ordinary-Lagrangian basis

Write the reduced conservative COM ordinary Lagrangian in the standard isotropic chart as

$$\hat{L} = \hat{L}_0 + c^{-2}\hat{L}_1 + c^{-4}\hat{L}_2 + c^{-6}\hat{L}_3 + O(c^{-8}), \quad (159)$$

with lower-order blocks already frozen by the carried Newtonian, 1PN, and 2PN papers. In the reduced COM variables used throughout the 3PN session, those blocks are

$$\hat{L}_0 = \frac{1}{2}v^2 + \frac{1}{r}, \quad (160)$$

$$\hat{L}_1 = \frac{1-3\nu}{8}v^4 + \frac{1}{r} \left[\frac{3+\nu}{2}v^2 + \frac{\nu}{2}\dot{r}^2 \right] - \frac{1}{2r^2}, \quad (161)$$

$$\begin{aligned} \hat{L}_2 = & \frac{1-5\nu+5\nu^2}{16}v^6 + \frac{1}{r} \left[\left(\frac{7}{8} - \frac{7}{4}\nu - \frac{1}{8}\nu^2 \right) v^4 + \left(\frac{1}{4}\nu - \frac{1}{4}\nu^2 \right) v^2\dot{r}^2 + \frac{3}{8}\nu^2\dot{r}^4 \right] \\ & + \frac{1}{r^2} \left[\left(2 - \frac{7}{8}\nu \right) v^2 + \frac{15}{8}\nu\dot{r}^2 \right] + \frac{1}{r^3} \left(\frac{1}{4} + \frac{3}{4}\nu \right). \end{aligned} \quad (162)$$

Here

$$v^2 = \dot{r}^2 + v_\perp^2, \quad \nu \equiv \frac{m_A m_B}{(m_A + m_B)^2}, \quad (163)$$

and the reduced COM normalization is the same one used throughout the main text.

The new 3PN coefficient is then expanded in the natural 15-slot ordinary basis

$$\begin{aligned} \hat{L}_3 = & l_1 v^8 + l_2 v^6 \dot{r}^2 + l_3 v^4 \dot{r}^4 + l_4 v^2 \dot{r}^6 + l_5 \dot{r}^8 \\ & + \frac{1}{r} (l_6 v^6 + l_7 v^4 \dot{r}^2 + l_8 v^2 \dot{r}^4 + l_9 \dot{r}^6) \\ & + \frac{1}{r^2} (l_{10} v^4 + l_{11} v^2 \dot{r}^2 + l_{12} \dot{r}^4) + \frac{1}{r^3} (l_{13} v^2 + l_{14} \dot{r}^2) + \frac{l_{15}}{r^4}. \end{aligned} \quad (164)$$

This is the reduced COM ordinary-Lagrangian ledger used everywhere in the 3PN solve. Later exact comparable-mass results are always represented as corrections to these 15 slots beyond the frozen one-body/self-static seed.

F.2 Reduced COM Hamiltonian basis

The matching isotropic reduced COM Hamiltonian expansion is

$$\hat{H} = \hat{H}_0 + c^{-2}\hat{H}_1 + c^{-4}\hat{H}_2 + c^{-6}\hat{H}_3 + O(c^{-8}), \quad (165)$$

and the 3PN coefficient is written in the parallel 15-slot Hamiltonian basis

$$\begin{aligned} \hat{H}_3 = & h_1 p^8 + h_2 p^6 p_r^2 + h_3 p^4 p_r^4 + h_4 p^2 p_r^6 + h_5 p_r^8 \\ & + \frac{1}{r} (h_6 p^6 + h_7 p^4 p_r^2 + h_8 p^2 p_r^4 + h_9 p_r^6) \\ & + \frac{1}{r^2} (h_{10} p^4 + h_{11} p^2 p_r^2 + h_{12} p_r^4) + \frac{1}{r^3} (h_{13} p^2 + h_{14} p_r^2) + \frac{h_{15}}{r^4}. \end{aligned} \quad (166)$$

Here

$$p^2 = p_r^2 + p_\perp^2 \quad (167)$$

is the reduced isotropic COM momentum invariant. The choice of the monomials in (166) is not arbitrary. It is precisely the choice for which the cubic compiler from Appendix E closes on the same 15-slot monomial ladder and becomes diagonal.

The reduced COM specialization of the exact cubic compiler is

$$\hat{H}_3 = -\hat{L}_3(v_0) + A_0 \cdot B_0 - \frac{1}{2} A_0^T C_0 A_0, \quad v_0 = p, \quad (168)$$

with carried lower-order derivative data

$$A_0 \equiv \left. \frac{\partial \hat{L}_1}{\partial v} \right|_{v=p}, \quad B_0 \equiv \left. \frac{\partial \hat{L}_2}{\partial v} \right|_{v=p}, \quad C_0 \equiv \left. \frac{\partial^2 \hat{L}_1}{\partial v^2} \right|_{v=p}. \quad (169)$$

Because the carried lower-order blocks are isotropic polynomials in $(v^2, \dot{r}^2, 1/r)$, the feedback term $A_0 \cdot B_0 - \frac{1}{2} A_0^T C_0 A_0$ is an even polynomial in $(p_r, p_\perp)$. Re-expanding it in the basis $(p^2, p_r^2, 1/r)$ lands exactly on the same 15 monomials as (166). That is the structural reason the final map diagonalizes.

F.3 Exact diagonal map and its inverse

Applying (168) to the basis (164), and then re-expanding the result in the Hamiltonian basis (166), gives the exact slotwise map

$$\begin{aligned} h_1 &= -l_1 + \frac{3}{16}\nu - \frac{21}{16}\nu^2 + \frac{9}{4}\nu^3, & h_2 &= -l_2, \\ h_3 &= -l_3, & h_4 &= -l_4, \\ h_5 &= -l_5. \end{aligned} \quad (170)$$

$$\begin{aligned} h_6 &= -l_6 + \frac{1}{4} + \frac{7}{8}\nu - \frac{35}{8}\nu^2 - \frac{21}{4}\nu^3, & h_7 &= -l_7 + \frac{11}{8}\nu^2 - \frac{9}{2}\nu^3, \\ h_8 &= -l_8 + \frac{3}{4}\nu^2 - \frac{9}{4}\nu^3, & h_9 &= -l_9. \end{aligned} \quad (171)$$

$$\begin{aligned} h_{10} &= -l_{10} + \frac{5}{4} + \frac{15}{8}\nu + \frac{123}{8}\nu^2 + \frac{13}{4}\nu^3, \\ h_{11} &= -l_{11} + \frac{7}{8}\nu + \frac{41}{8}\nu^2 + \frac{31}{4}\nu^3, \\ h_{12} &= -l_{12} + \frac{9}{2}\nu^2 + 4\nu^3. \end{aligned} \quad (172)$$

$$\begin{aligned}
h_{13} &= -l_{13} - \frac{3}{2} - \frac{59}{4}\nu - \frac{25}{4}\nu^2 - \frac{1}{2}\nu^3, \\
h_{14} &= -l_{14} + \frac{7}{4}\nu - \frac{31}{4}\nu^2 - \frac{7}{2}\nu^3, \\
h_{15} &= -l_{15}.
\end{aligned} \tag{173}$$

So the chosen ordinary basis diagonalizes the reduced COM cubic compiler exactly. Equivalently,

$$\frac{\partial h_i}{\partial l_j} = -\delta_{ij} \tag{174}$$

throughout the full 15-slot basis.

The inverse map is therefore immediate:

$$\begin{aligned}
l_1 &= \frac{3}{16}\nu - \frac{21}{16}\nu^2 + \frac{9}{4}\nu^3 - h_1, & l_2 &= -h_2, \\
l_3 &= -h_3, & l_4 &= -h_4, \\
l_5 &= -h_5.
\end{aligned} \tag{175}$$

$$\begin{aligned}
l_6 &= \frac{1}{4} + \frac{7}{8}\nu - \frac{35}{8}\nu^2 - \frac{21}{4}\nu^3 - h_6, & l_7 &= \frac{11}{8}\nu^2 - \frac{9}{2}\nu^3 - h_7, \\
l_8 &= \frac{3}{4}\nu^2 - \frac{9}{4}\nu^3 - h_8, & l_9 &= -h_9.
\end{aligned} \tag{176}$$

$$\begin{aligned}
l_{10} &= \frac{5}{4} + \frac{15}{8}\nu + \frac{123}{8}\nu^2 + \frac{13}{4}\nu^3 - h_{10}, \\
l_{11} &= \frac{7}{8}\nu + \frac{41}{8}\nu^2 + \frac{31}{4}\nu^3 - h_{11}, \\
l_{12} &= \frac{9}{2}\nu^2 + 4\nu^3 - h_{12}.
\end{aligned} \tag{177}$$

$$\begin{aligned}
l_{13} &= -\frac{3}{2} - \frac{59}{4}\nu - \frac{25}{4}\nu^2 - \frac{1}{2}\nu^3 - h_{13}, \\
l_{14} &= \frac{7}{4}\nu - \frac{31}{4}\nu^2 - \frac{7}{2}\nu^3 - h_{14}, \\
l_{15} &= -h_{15}.
\end{aligned} \tag{178}$$

There is no hidden COM-only kernel here. Once the target Hamiltonian coefficients are supplied, the ordinary coefficients follow immediately and uniquely.

That is the exact algebraic reason the later 3PN reduced COM target import is so efficient. Appendix G only has to provide the target list h_i ; after that, the corresponding ordinary coefficients l_i are already fixed by (175)–(178). In this sense the diagonal COM compiler is not just a calculational convenience. It is the device that turns the reduced 3PN lift from a nonlinear inversion problem into a slotwise algebraic translation.

G Imported GR COM target coefficients

This appendix records the exact reduced center-of-mass 3PN target in the same isotropic 15-slot Hamiltonian basis used in Appendix F, translates that Hamiltonian target back into the ordinary-Lagrangian chart by the exact inverse compiler, and then isolates the residual beyond the frozen one-body/self-static seed. This is the point at which the 3PN comparable-mass problem becomes a finite algebraic target rather than an open-ended generic two-body search.

G.1 15-slot target list

Using the standard reduced COM ADM Hamiltonian through 3PN in the same isotropic basis as Appendix F, the exact GR target coefficients are

$$h_1 = \frac{-5 + 35\nu - 70\nu^2 + 35\nu^3}{128}, \quad h_2 = h_3 = h_4 = h_5 = 0. \quad (179)$$

$$h_6 = \frac{-7 + 42\nu - 53\nu^2 - 5\nu^3}{16}, \quad h_7 = \frac{(2 - 3\nu)\nu^2}{16}, \quad (180)$$

$$h_8 = \frac{3(1 - \nu)\nu^2}{16}, \quad h_9 = -\frac{5\nu^3}{16}.$$

$$h_{10} = \frac{-27 + 136\nu + 109\nu^2}{16}, \quad h_{11} = \frac{(17 + 30\nu)\nu}{16}, \quad (181)$$

$$h_{12} = \frac{(5 + 43\nu)\nu}{12}.$$

$$h_{13} = \frac{-600 + (3\pi^2 - 1340)\nu - 552\nu^2}{192}, \quad h_{14} = -\frac{(340 + 3\pi^2 + 112\nu)\nu}{64}. \quad (182)$$

$$h_{15} = \frac{12 + (872 - 63\pi^2)\nu}{96}. \quad (183)$$

The strict test-mass limit reproduces the exact one-body Schwarzschild slots,

$$h_1 \rightarrow -\frac{5}{128}, \quad h_6 \rightarrow -\frac{7}{16}, \quad h_{10} \rightarrow -\frac{27}{16}, \quad h_{13} \rightarrow -\frac{25}{8}, \quad h_{15} \rightarrow \frac{1}{8}, \quad (184)$$

so the imported COM Hamiltonian target is compatible with the exact one-body 3PN gate from the start.

G.2 Exact ordinary-Lagrangian solution

Applying the inverse diagonal map from Appendix F slot by slot gives the exact COM ordinary-Lagrangian coefficients

$$l_1 = \frac{5 - 11\nu - 98\nu^2 + 253\nu^3}{128}, \quad l_2 = l_3 = l_4 = l_5 = 0. \quad (185)$$

$$l_6 = \frac{11 - 28\nu - 17\nu^2 - 79\nu^3}{16}, \quad l_7 = \frac{20\nu^2 - 69\nu^3}{16}, \quad (186)$$

$$l_8 = \frac{9\nu^2 - 33\nu^3}{16}, \quad l_9 = \frac{5\nu^3}{16}.$$

$$l_{10} = \frac{47 - 106\nu + 137\nu^2 + 52\nu^3}{16}, \quad l_{11} = \frac{-3\nu + 52\nu^2 + 124\nu^3}{16}, \quad (187)$$

$$l_{12} = \frac{-5\nu + 11\nu^2 + 48\nu^3}{12}.$$

$$l_{13} = \frac{13}{8} - \left(\frac{373}{48} + \frac{\pi^2}{64} \right) \nu - \frac{27}{8} \nu^2 - \frac{1}{2} \nu^3. \quad (188)$$

$$l_{14} = \frac{452 + 3\pi^2}{64} \nu - 6\nu^2 - \frac{7}{2} \nu^3, \quad l_{15} = -\frac{1}{8} - \frac{872 - 63\pi^2}{96} \nu. \quad (189)$$

Three structural facts are worth keeping explicit.

First, the pure quartic-radial slots remain absent,

$$l_2 = l_3 = l_4 = l_5 = 0. \quad (190)$$

Second, the strict one-body limit reproduces the carried self/static seed,

$$l_1 \rightarrow \frac{5}{128}, \quad l_6 \rightarrow \frac{11}{16}, \quad l_{10} \rightarrow \frac{47}{16}, \quad l_{13} \rightarrow \frac{13}{8}, \quad l_{15} \rightarrow -\frac{1}{8}. \quad (191)$$

Third, the π^2 dependence first enters only in the deep-potential r^{-3} and r^{-4} slots, exactly where the genuinely new static 3PN structure sits.

G.3 Residual beyond the self/static seed

Let the frozen self/static seed occupy the nonzero COM ordinary slots

$$l_1^{\text{seed}} = \frac{5}{128} - \frac{35}{128}\nu + \frac{35}{64}\nu^2 - \frac{35}{128}\nu^3, \quad (192)$$

$$l_6^{\text{seed}} = \frac{11}{16} - \frac{33}{8}\nu + \frac{99}{16}\nu^2 - \frac{11}{8}\nu^3, \quad (193)$$

$$l_{10}^{\text{seed}} = \frac{47}{16} - \frac{235}{16}\nu + \frac{235}{16}\nu^2, \quad (194)$$

$$l_{13}^{\text{seed}} = \frac{13}{8} - \frac{13}{2}\nu + \frac{13}{4}\nu^2, \quad l_{15}^{\text{seed}} = -\frac{1}{8} + \frac{3}{8}\nu, \quad (195)$$

with all other seed slots zero. Define the exact comparable-mass residual beyond that seed by

$$\Delta l_i \equiv l_i^{\text{GR}} - l_i^{\text{seed}}. \quad (196)$$

Then

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}, \quad \Delta l_2 = \Delta l_3 = \Delta l_4 = \Delta l_5 = 0. \quad (197)$$

$$\Delta l_6 = \frac{\nu(38 - 116\nu - 57\nu^2)}{16}, \quad \Delta l_7 = \frac{\nu^2(20 - 69\nu)}{16}, \quad (198)$$

$$\Delta l_8 = \frac{3\nu^2(3 - 11\nu)}{16}, \quad \Delta l_9 = \frac{5\nu^3}{16}.$$

$$\Delta l_{10} = \frac{\nu(129 - 98\nu + 52\nu^2)}{16}, \quad \Delta l_{11} = \frac{\nu(-3 + 52\nu + 124\nu^2)}{16}, \quad (199)$$

$$\Delta l_{12} = \frac{\nu(-5 + 11\nu + 48\nu^2)}{12}.$$

$$\Delta l_{13} = -\frac{\nu(244 + 3\pi^2 + 1272\nu + 96\nu^2)}{192}. \quad (200)$$

$$\Delta l_{14} = \frac{\nu(452 + 3\pi^2 - 384\nu - 224\nu^2)}{64}, \quad \Delta l_{15} = \frac{\nu(-908 + 63\pi^2)}{96}. \quad (201)$$

Because the COM compiler is diagonal with the same lower-order feedback for both seed and target, the residual obeys the exact slotwise identity

$$\Delta l_i = -\Delta h_i. \quad (202)$$

This is the compact way to remember the entire COM target-import step: the ordinary center-of-mass problem is already completely solved, and the remaining content is a purely comparable-mass 15-slot data vector. In the later appendices this vector is what gets projected into the compact generic-frame 24/17/8/1 scaffold and, after that, into the grouped real P_2 inverse problem.

H Generic-frame residual scaffold

Once the exact one-body 3PN gate, the natural bodywise self/static seed, and the reduced COM target are frozen, the next task is to state the honest generic-frame comparable-mass interaction scaffold before any contact/gauge quotient is imposed. The guiding choice is deliberately conservative. We do not try to guess the smallest basis too early. Instead, we start from an exchange-symmetric *overcomplete* residual scaffold that is guaranteed to preserve the exact one-body gate and then postpone every reduction of that scaffold until after the target Hamiltonian chart has been imported.

Write

$$\mathbf{r} \equiv \mathbf{x}_A - \mathbf{x}_B, \quad r \equiv |\mathbf{r}|, \quad \mathbf{n} \equiv \frac{\mathbf{r}}{r}, \quad (203)$$

and introduce the standard generic-frame invariants

$$a = v_A^2, \quad b = v_B^2, \quad c = \mathbf{v}_A \cdot \mathbf{v}_B, \quad d = v_{An} = \mathbf{v}_A \cdot \mathbf{n}, \quad e = v_{Bn} = \mathbf{v}_B \cdot \mathbf{n}, \quad (204)$$

together with the mass labels

$$p \equiv m_A, \quad q \equiv m_B. \quad (205)$$

The scaffold is symmetrized under body exchange,

$$A \leftrightarrow B : \quad (a, b, d, e, p, q) \mapsto (b, a, e, d, q, p), \quad c \mapsto c, \quad (206)$$

and every retained basis element is required to vanish in the strict test-mass branch

$$p \rightarrow 0, \quad q \rightarrow 1, \quad b \rightarrow 0, \quad c \rightarrow 0, \quad e \rightarrow 0. \quad (207)$$

That last condition is what keeps the exact one-body gate cleanly separated from the genuine comparable-mass lift.

The resulting interaction scaffold has block sizes

$$24, \quad 17, \quad 8, \quad 1, \quad (208)$$

for the G/r , G^2/r^2 , G^3/r^3 , and G^4/r^4 blocks, respectively, so the full generic-frame 3PN interaction ledger contains 50 directions before any later quotient by COM blindness or ordinary contact freedom.

H.1 G/r sextic block

At order G/r the new comparable-mass residual is an exchange-symmetric sextic polynomial in the invariants (204). The overcomplete sextic basis has 24 elements,

$$Q_{01} = a^2b + ab^2, \quad Q_{02} = a^2c + b^2c, \quad Q_{03} = a^2de + b^2de, \quad Q_{04} = a^2e^2 + b^2d^2, \quad (209)$$

$$Q_{05} = abc, \quad Q_{06} = abd^2 + abe^2, \quad Q_{07} = abde, \quad Q_{08} = ac^2 + bc^2, \quad (210)$$

$$Q_{09} = acd^2 + bce^2, \quad Q_{10} = acde + bcde, \quad Q_{11} = ace^2 + bcd^2, \quad Q_{12} = ad^2e^2 + bd^2e^2, \quad (211)$$

$$Q_{13} = ad^3e + bde^3, \quad Q_{14} = ade^3 + bd^3e, \quad Q_{15} = ae^4 + bd^4, \quad Q_{16} = c^2d^2 + c^2e^2, \quad (212)$$

$$Q_{17} = c^2de, \quad Q_{18} = c^3, \quad Q_{19} = cd^2e^2, \quad Q_{20} = cd^3e + cde^3, \quad (213)$$

$$Q_{21} = cd^4 + ce^4, \quad Q_{22} = d^3e^3, \quad Q_{23} = d^4e^2 + d^2e^4, \quad Q_{24} = d^5e + de^5. \quad (214)$$

So the corresponding residual contribution is

$$\frac{Gm_A m_B}{r} \sum_{i=1}^{24} q_i Q_i. \quad (215)$$

This is already broad enough to span the later reduced COM image of the velocity-six interaction block; later appendices only remove redundancy, not missing structure.

H.2 G^2/r^2 quartic block

At order G^2/r^2 one extra mass power must be carried explicitly, so the comparable-mass residual is organized in an exchange-symmetric quartic mass-weighted basis with 17 elements,

$$T_{01} = a^2p + b^2q, \quad T_{02} = abp + abq, \quad T_{03} = acp + bcq, \quad T_{04} = acq + bcp, \quad (216)$$

$$T_{05} = ad^2p + be^2q, \quad T_{06} = adep + bdeq, \quad T_{07} = adeq + bdep, \quad T_{08} = ae^2p + bd^2q, \quad (217)$$

$$T_{09} = ae^2q + bd^2p, \quad T_{10} = c^2p + c^2q, \quad T_{11} = cd^2p + ce^2q, \quad T_{12} = cd^2q + ce^2p, \quad (218)$$

$$T_{13} = cdep + cdeq, \quad T_{14} = d^2e^2p + d^2e^2q, \quad T_{15} = d^3ep + de^3q, \quad (219)$$

$$T_{16} = d^3eq + de^3p, \quad T_{17} = d^4p + e^4q. \quad (220)$$

The quartic residual block is therefore

$$\frac{G^2 m_A m_B}{r^2} \sum_{j=1}^{17} t_j T_j. \quad (221)$$

This is the first block in which the later seed-obstruction analysis becomes nontrivial. The basis is intentionally broader than the final solved representative because the purpose of the present appendix is to preserve the honest algebraic search space before the middle-block repair is known.

H.3 G^3/r^3 quadratic block

At order G^3/r^3 the residual is quadratic in the velocity invariants and carries two extra explicit mass powers. The exchange-symmetric basis has eight elements,

$$S_{01} = ap^2 + bq^2, \quad S_{02} = apq + bpq, \quad S_{03} = cp^2 + cq^2, \quad S_{04} = cpq, \quad (222)$$

$$S_{05} = d^2p^2 + e^2q^2, \quad S_{06} = d^2pq + e^2pq, \quad S_{07} = dep^2 + deq^2, \quad S_{08} = depq. \quad (223)$$

So the quadratic residual block is

$$\frac{G^3 m_A m_B}{r^3} \sum_{k=1}^8 s_k S_k. \quad (224)$$

This is the other block in which the later COM projection reveals a genuine middle-block obstruction relative to the naive seed placement. Again, that obstruction is a bookkeeping issue, not an indication that the present block is too small.

H.4 G^4/r^4 static block

At order G^4/r^4 there is only one exchange-symmetric static cross polynomial, namely

$$U_{01} = p^2q + pq^2. \quad (225)$$

So the static residual block is simply

$$\frac{G^4 m_A m_B}{r^4} u_1 U_{01}. \quad (226)$$

Combining all four interaction blocks, the honest generic-frame comparable-mass scaffold is

$$L_{3,\text{full}} = L_{3,\text{seed}} + L_{3,\text{res}}, \quad (227)$$

with

$$L_{3,\text{res}} = \frac{Gm_A m_B}{r} \sum_{i=1}^{24} q_i Q_i + \frac{G^2 m_A m_B}{r^2} \sum_{j=1}^{17} t_j T_j + \frac{G^3 m_A m_B}{r^3} \sum_{k=1}^8 s_k S_k + \frac{G^4 m_A m_B}{r^4} u_1 U_{01}. \quad (228)$$

The point of (228) is not that the final 3PN answer needs all 50 coefficients. It does not. The point is that the full generic-frame lift should begin from a scaffold that is manifestly exchange-symmetric, manifestly compatible with the exact one-body gate, and manifestly finite before any later quotient by seed repair, COM blindness, or contact/gauge reduction is imposed. Later appendices will show that once the fixed ADM Hamiltonian chart is used, the remaining algebraic freedom collapses dramatically; but the right place to impose that collapse is later, not here.

I COM projection matrices, seed obstruction, and seed repair

The reduced COM target of the previous appendices is already exact, and the exchange-symmetric generic-frame scaffold from the comparable-mass setup is also already finite. The next honest question is therefore not whether a lift exists in principle, but how much of that lift is already fixed by center-of-mass projection and where the first genuine mismatch occurs. This appendix records that first exact answer.

The result is both cleaner and smaller than one might have expected. COM projection fixes a large part of the generic-frame interaction lift immediately. The first mismatch is not in the one-body gate, not in the G/r sextic block, and not in the static G^4/r^4 slot. It sits entirely in small ν^3 tails in the middle G^2/r^2 and G^3/r^3 blocks, and it is removed exactly by a tiny ν -dressed seed correction that vanishes in the strict one-body limit. So the first generic-frame 3PN obstruction is a placement/bookkeeping issue, not a failure of the full scaffold.

I.1 COM reduction maps and the blockwise projection matrices

Use the generic-frame invariants

$$a = v_A^2, \quad b = v_B^2, \quad c = \mathbf{v}_A \cdot \mathbf{v}_B, \quad d = v_{An}, \quad e = v_{Bn}, \quad (229)$$

with masses

$$p = m_A, \quad q = m_B, \quad M = p + q, \quad \nu \equiv \frac{pq}{M^2}. \quad (230)$$

In the COM frame,

$$\mathbf{v}_A = X_B \mathbf{v}, \quad \mathbf{v}_B = -X_A \mathbf{v}, \quad X_A + X_B = 1, \quad \nu = X_A X_B, \quad (231)$$

so

$$a = X_B^2 v^2, \quad b = X_A^2 v^2, \quad c = -\nu v^2, \quad (232)$$

$$d = X_B \dot{r}, \quad e = -X_A \dot{r}. \quad (233)$$

Therefore each exchange-symmetric generic-frame basis monomial has a definite image in the reduced COM slot basis.

Write the interaction scaffold, after subtraction of the carried one-body/self-static seed, as

$$L_{3,\text{res}} = \frac{Gm_A m_B}{r} \sum_{i=1}^{24} q_i Q_i + \frac{G^2 m_A m_B}{r^2} \sum_{j=1}^{17} t_j T_j + \frac{G^3 m_A m_B}{r^3} \sum_{k=1}^8 s_k S_k + \frac{G^4 m_A m_B}{r^4} u_1 U_{01}, \quad (234)$$

where $\{Q_i\}$, $\{T_j\}$, $\{S_k\}$, and U_{01} are the compact exchange-symmetric basis elements introduced in the generic-frame scaffold appendix. Let

$$\mathbf{q} = (q_1, \dots, q_{24})^T, \quad \mathbf{t} = (t_1, \dots, t_{17})^T, \quad \mathbf{s} = (s_1, \dots, s_8)^T. \quad (235)$$

Then the COM reduction induces exact coefficient-extraction maps

$$\Pi_Q(\mathbf{q}) = M_Q \mathbf{q}, \quad M_Q \in \text{Mat}_{12 \times 24}(\mathbb{Q}), \quad (236)$$

$$\Pi_T(\mathbf{t}) = M_T \mathbf{t}, \quad M_T \in \text{Mat}_{6 \times 17}(\mathbb{Q}), \quad (237)$$

$$\Pi_S(\mathbf{s}) = M_S \mathbf{s}, \quad M_S \in \text{Mat}_{4 \times 8}(\mathbb{Q}), \quad (238)$$

$$\Pi_U(u_1) = M_U u_1, \quad M_U \in \text{Mat}_{1 \times 1}(\mathbb{Q}[\pi^2]). \quad (239)$$

Here the entries of the projection matrices are defined by coefficient extraction in the reduced COM slots:

$$\Pi_Q(\mathbf{q}) \equiv ([u]\delta l_6, [\nu^2]\delta l_6, [\nu^3]\delta l_6, \dots, [\nu^3]\delta l_9)^T, \quad (240)$$

$$\Pi_T(\mathbf{t}) \equiv ([u]\delta l_{10}, [\nu^2]\delta l_{10}, [\nu]\delta l_{11}, [\nu^2]\delta l_{11}, [\nu]\delta l_{12}, [\nu^2]\delta l_{12})^T, \quad (241)$$

$$\Pi_S(\mathbf{s}) \equiv ([u]\delta l_{13}, [\nu^2]\delta l_{13}, [\nu]\delta l_{14}, [\nu^2]\delta l_{14})^T, \quad (242)$$

$$\Pi_U(u_1) \equiv ([u]\delta l_{15}). \quad (243)$$

So the phrase ‘‘COM projection matrix’’ means the exact linear map that sends a block of generic-frame coefficients to the polynomial coefficients of the reduced COM slots.

The audit establishes the exact ranks

$$\text{rank } M_Q = 12, \quad \text{rank } M_T = 6, \quad \text{rank } M_S = 4, \quad \text{rank } M_U = 1, \quad (244)$$

with nullities

$$\dim \ker M_Q = 12, \quad \dim \ker M_T = 11, \quad \dim \ker M_S = 4, \quad \dim \ker M_U = 0. \quad (245)$$

Equivalently, the total COM-blind interaction freedom is

$$\boxed{12 + 11 + 4 + 0 = 27.} \quad (246)$$

In the compact basis ordering used by the audit, the pivot columns may be taken as

$$Q : \{1, 2, 3, 4, 5, 7, 12, 13, 14, 22, 23, 24\}, \quad (247)$$

$$T : \{1, 2, 5, 6, 14, 16\}, \quad S : \{1, 2, 5, 6\}, \quad (248)$$

while the U block is of course already unique. These pivot data are not used later as physical objects; they are simply a compact way to say that the COM image is already completely explicit before any gauge/contact quotient is taken.

The structural interpretation of (244) is immediate. The Q block can support ν , ν^2 , and ν^3 in each of its four COM slots; the T block supports only ν and ν^2 in each of its three slots; the S block likewise supports only ν and ν^2 in each of its two slots, with coefficients allowed in $\mathbb{Q}(\pi^2)$; and the U block reduces only to a coefficient proportional to ν . This is exactly the mass-degree fact that produces the first obstruction.

I.2 The exact middle-block obstruction relative to the natural seed

Subtract the natural 3PN one-body/self-static seed from the exact COM target. The G/r sextic block and the G^4/r^4 static slot are already compatible with the images of M_Q and M_U . The only incompatibility sits in the middle two blocks. Explicitly, the exact COM residuals are

$$\Delta l_{10} = \nu \left(\frac{129}{16} - \frac{98}{16} \nu \right) + \boxed{\frac{13}{4} \nu^3}, \quad (249)$$

$$\Delta l_{11} = \nu \left(-\frac{3}{16} + \frac{52}{16}\nu \right) + \boxed{\frac{31}{4}\nu^3}, \quad (250)$$

$$\Delta l_{12} = \nu \left(-\frac{5}{12} + \frac{11}{12}\nu \right) + \boxed{4\nu^3}, \quad (251)$$

for the G^2/r^2 block, while for the G^3/r^3 block one finds

$$\Delta l_{13} = \nu \left[-\frac{244 + 3\pi^2}{192} - \frac{1272}{192}\nu \right] + \boxed{-\frac{1}{2}\nu^3}, \quad (252)$$

$$\Delta l_{14} = \nu \left[\frac{452 + 3\pi^2}{64} - 6\nu \right] + \boxed{-\frac{7}{2}\nu^3}. \quad (253)$$

Because the T and S projection matrices carry only ν and ν^2 support, none of the boxed ν^3 pieces in (249)–(253) can arise from the present constant-coefficient generic-frame middle blocks. The exact obstruction vector is therefore

$$\boxed{\delta l_{10}^{\text{obs}} = \frac{13}{4}\nu^3, \quad \delta l_{11}^{\text{obs}} = \frac{31}{4}\nu^3, \quad \delta l_{12}^{\text{obs}} = 4\nu^3}, \quad (254)$$

$$\boxed{\delta l_{13}^{\text{obs}} = -\frac{1}{2}\nu^3, \quad \delta l_{14}^{\text{obs}} = -\frac{7}{2}\nu^3}. \quad (255)$$

This is the first exact generic-frame lift obstruction. The natural seed is already exact in the strict one-body sector, but it is not yet aligned with the mass-degree structure of the middle interaction blocks.

I.3 Minimal ν -dressed seed repair

The obstruction is repaired by a remarkably small correction sector. Define

$$\boxed{\Delta T_\nu = \frac{pq}{4(p+q)} (13ab + 31cde + 16d^2e^2)}, \quad (256)$$

$$\boxed{\Delta S_\nu = \frac{p^2q^2}{2(p+q)^2} (c + 7de)}. \quad (257)$$

These terms have exactly the two properties one wants.

First, they vanish in the strict one-body limit. On the branch

$$m_A \rightarrow 0, \quad \mathbf{v}_B \rightarrow 0, \quad p \rightarrow 0, \quad b, c, e \rightarrow 0, \quad (258)$$

one has

$$\boxed{\Delta T_\nu \rightarrow 0, \quad \Delta S_\nu \rightarrow 0}. \quad (259)$$

So the seed repair never touches the exact one-body gate.

Second, under COM reduction these terms reproduce precisely the missing obstruction tails. In slot language,

$$\Delta T_\nu \xrightarrow{\text{COM}} \left(\frac{13}{4}\nu^3, \frac{31}{4}\nu^3, 4\nu^3 \right) \quad \text{in } (l_{10}, l_{11}, l_{12}), \quad (260)$$

$$\Delta S_\nu \xrightarrow{\text{COM}} \left(-\frac{1}{2}\nu^3, -\frac{7}{2}\nu^3 \right) \quad \text{in } (l_{13}, l_{14}). \quad (261)$$

So the refined seed is simply

$$\boxed{L_{3,\text{seed}}^{\text{ref}} = L_{3,\text{seed}} + \frac{G^2pq}{r^2} \Delta T_\nu + \frac{G^3pq}{r^3} \Delta S_\nu}. \quad (262)$$

This is the exact ν -dressed seed repair advertised in the main text. It is minimal in the literal sense: it touches only the obstructed middle blocks, it vanishes in the strict one-body limit, and it removes all of the COM mismatch.

I.4 One exact repaired representative in each interaction block

After the refined seed is frozen, each block admits an exact generic-frame representative with the correct COM image. The representative is not unique — that is the subject of the next appendix — but existence is now explicit. Using the compact basis monomials from the generic-frame scaffold appendix, one may take

$$L_{Q,\text{part}} = \frac{Gm_A m_B}{r} \left[\frac{9}{4} Q_1 - \frac{19}{8} Q_2 + \frac{5}{4} Q_4 + \frac{61}{16} Q_5 + \frac{29}{16} Q_7 + \frac{9}{16} Q_{12} + \frac{15}{32} Q_{14} - \frac{5}{16} Q_{22} \right], \quad (263)$$

$$L_{T,\text{part}} = \frac{G^2 m_A m_B}{r^2} \left[\frac{129}{16} (a^2 p + b^2 q) + \frac{289}{16} (abp + abq) - \frac{3}{16} (ad^2 p + be^2 q) \right] \\ + \frac{G^2 m_A m_B}{r^2} \left[-\frac{43}{16} (ade p + bde q) - \frac{1}{3} (d^2 e^2 p + d^2 e^2 q) + \frac{5}{12} (d^3 e q + de^3 p) \right], \quad (264)$$

$$L_{S,\text{part}} = \frac{G^3 m_A m_B}{r^3} \left[-\frac{880 + 3\pi^2}{192} (ap^2 + bq^2) - \frac{244 + 3\pi^2}{192} (apq + bpq) \right] \\ + \frac{G^3 m_A m_B}{r^3} \left[\frac{260 + 3\pi^2}{64} (d^2 p^2 + e^2 q^2) + \frac{452 + 3\pi^2}{64} (d^2 pq + e^2 pq) \right], \quad (265)$$

and the unique static block

$$L_{U,\text{part}} = \frac{G^4 m_A m_B}{r^4} \frac{63\pi^2 - 908}{96} (p^2 q + pq^2). \quad (266)$$

By construction, the COM reductions of (263)–(266) reproduce the exact refined targets in every block. The important point is not that these formulas are especially canonical — they are not — but that, after the seed repair, blockwise COM compatibility is completely explicit.

I.5 What the projection stage has fixed — and what it has not

The full content of the present appendix can now be summarized in one line:

COM projection plus the minimal ν -dressed seed repair removes every blockwise compatibility obstruction, but it still leaves a 27-dimensional COM-blind family.

(267)

Equivalently,

$$Q : 24 \rightarrow 12, \quad T : 17 \rightarrow 6, \quad S : 8 \rightarrow 4, \quad U : 1 \rightarrow 1, \quad (268)$$

so after repair the unfixed interaction nullities remain

$$12, \quad 11, \quad 4, \quad 0, \quad (269)$$

for a total of

$$\boxed{27} \quad (270)$$

COM-blind directions.

So the present appendix does not finish the full generic-frame theorem. What it finishes is the exact COM-constraint layer. It shows that

1. the Q block was already compatible,
2. the U block is already unique once the π^2 coefficient is admitted,
3. the only true mismatch sat in five tiny ν^3 tails in the middle blocks,

4. those tails are absorbed exactly by the refined seed (262), and
5. after that repair one explicit COM-compatible representative exists in every interaction block.

What remains is therefore no longer a compatibility problem. It is the genuine COM-null/gauge/contact quotient problem taken up in the next appendix.

J The COM-null ideal and canonical quotient slice

After the exact ν -dressed seed repair, the generic-frame 3PN lift is no longer obstructed by any middle-block mismatch. What remains is a quotient problem. The repaired 24/17/8/1 interaction scaffold still contains nontrivial generic-frame directions that vanish identically after COM reduction, and the next honest task is to identify those directions exactly and choose a convenient representative of their quotient class.

This appendix records that algebraic structure. The main conclusion is simple. The remaining unfixed family is not unexplained physics. It is precisely the COM-blind family generated by the standard COM identities. Once that family is recognized as an ideal, it becomes natural to choose a canonical sparse representative by quotienting it away. That choice is exact algebraically and already sufficient to write down a clean generic-frame representative of the full solved COM target. What it is *not* yet is the final ordinary gauge/contact theorem; that sharper statement is postponed to the next appendix.

Continue to write the repaired generic-frame interaction scaffold as

$$L_{3,\text{res}} = \frac{Gpq}{r} \sum_{i=1}^{24} q_i Q_i + \frac{G^2 pq}{r^2} \sum_{j=1}^{17} t_j T_j + \frac{G^3 pq}{r^3} \sum_{k=1}^8 s_k S_k + \frac{G^4 pq}{r^4} u_1 U_{01}, \quad (271)$$

where the compact exchange-symmetric basis elements Q_i , T_j , S_k , and U_{01} were fixed in the generic-frame scaffold appendix, and where the coefficient vectors are

$$\mathbf{q} = (q_1, \dots, q_{24})^T, \quad \mathbf{t} = (t_1, \dots, t_{17})^T, \quad \mathbf{s} = (s_1, \dots, s_8)^T. \quad (272)$$

The previous appendix showed that after seed repair the COM image is completely fixed block by block. The present appendix identifies exactly what survives in the kernel of that projection.

J.1 The COM-null ideal

In the COM frame the body velocities satisfy three linear-momentum identities,

$$\boxed{\mathcal{C}_1 \equiv pa + qc = 0, \quad \mathcal{C}_2 \equiv pc + qb = 0, \quad \mathcal{C}_3 \equiv pd + qe = 0,} \quad (273)$$

and three collinearity identities,

$$\boxed{\mathcal{C}_4 \equiv ab - c^2 = 0, \quad \mathcal{C}_5 \equiv ae - cd = 0, \quad \mathcal{C}_6 \equiv bd - ce = 0.} \quad (274)$$

These generate the COM-null ideal

$$\boxed{\mathcal{I}_{\text{COM}} = \langle \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3, \mathcal{C}_4, \mathcal{C}_5, \mathcal{C}_6 \rangle.} \quad (275)$$

It is also convenient to name the smaller linear-momentum ideal

$$\mathcal{I}_{\text{lin}} = \langle \mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3 \rangle. \quad (276)$$

Then the full 27-dimensional COM-blind family is exactly the generic-frame freedom invisible in the quotient algebra

$$\mathbb{Q}[a, b, c, d, e, p, q] / \mathcal{I}_{\text{COM}}. \quad (277)$$

A useful refinement already visible in the referee audit is that the T - and S -block null directions lie in the smaller ideal $\mathfrak{J}_{\text{lin}}$, whereas the Q -block null directions require the full ideal $\mathfrak{J}_{\text{COM}}$.

The sparse null generators returned by the exact COM-projection audit may be chosen as follows.

T-block null generators. A sparse basis of $\ker M_T$ is

$$\begin{aligned} N_{T1} &= abp + abq + acp + bcq, & N_{T2} &= a^2p + acq + b^2q + bcp, \\ N_{T3} &= ad^2p + adeq + bdep + be^2q, & N_{T4} &= adep + ae^2p + bd^2q + bdeq, \\ N_{T5} &= adep + ae^2q + bd^2p + bdeq, & N_{T6} &= -abp - abq + c^2p + c^2q, \\ N_{T7} &= -adep - bdeq + cd^2p + ce^2q, & N_{T8} &= ad^2p + be^2q + cd^2q + ce^2p, \\ N_{T9} &= adep + bdeq + cdep + cdeq, & N_{T10} &= d^3ep + d^2e^2p + d^2e^2q + de^3q, \\ N_{T11} &= d^4p + d^3eq + de^3p + e^4q. \end{aligned}$$

The exact audit verifies

$$N_{Ti} \in \mathfrak{J}_{\text{lin}} \quad (i = 1, \dots, 11). \quad (278)$$

S-block null generators. A sparse basis of $\ker M_S$ is

$$\begin{aligned} N_{S1} &= apq + bpq + cp^2 + cq^2, & N_{S2} &= \frac{1}{2}ap^2 + \frac{1}{2}bq^2 + cpq, \\ N_{S3} &= d^2pq + dep^2 + deq^2 + e^2pq, & N_{S4} &= \frac{1}{2}d^2p^2 + depq + \frac{1}{2}e^2q^2. \end{aligned}$$

Again the audit gives

$$N_{Si} \in \mathfrak{J}_{\text{lin}} \quad (i = 1, \dots, 4). \quad (279)$$

Q-block null generators. A sparse basis of $\ker M_Q$ is

$$\begin{aligned} N_{Q1} &= -a^2e^2 + abd^2 + abe^2 - b^2d^2, & N_{Q2} &= -a^2b - ab^2 + ac^2 + bc^2, \\ N_{Q3} &= -a^2de + acd^2 - b^2de + bce^2, & N_{Q4} &= -a^2e^2 + acde - b^2d^2 + bcde, \\ N_{Q5} &= -2abde + ace^2 + bcd^2, & N_{Q6} &= -ad^2e^2 + ae^4 + bd^4 - bd^2e^2, \\ N_{Q7} &= -a^2e^2 - b^2d^2 + c^2d^2 + c^2e^2, & N_{Q8} &= -abde + c^2de, \\ N_{Q9} &= -abc + c^3, & N_{Q10} &= -\frac{1}{2}ade^3 - \frac{1}{2}bd^3e + cd^2e^2, \\ N_{Q11} &= -ad^2e^2 - bd^2e^2 + cd^3e + cde^3, & N_{Q12} &= -ad^3e - bde^3 + cd^4 + ce^4. \end{aligned}$$

For these generators the exact statement is only

$$N_{Qi} \in \mathfrak{J}_{\text{COM}} \quad (i = 1, \dots, 12), \quad (280)$$

not membership in the smaller linear ideal.

The conceptual consequence of (278)–(280) is important. The entire COM-blind family is generated by exact algebraic COM identities, but only the T and S kernels are already linear-momentum-null. The Q block genuinely needs the collinearity relations as well.

J.2 What COM fixes and what it does not fix

After the minimal ν -dressed repair, COM matching determines one exact image in each interaction block, but it does not fix the full generic-frame lift. Block by block, the exact nullities are

$$\boxed{\dim \ker M_Q = 12, \quad \dim \ker M_T = 11, \quad \dim \ker M_S = 4, \quad \dim \ker M_U = 0.} \quad (281)$$

Equivalently, the total COM-blind interaction freedom is

$$\boxed{12 + 11 + 4 + 0 = 27.} \quad (282)$$

So COM reduction fixes the image layer but leaves a 27-dimensional quotient family unfixed. In that precise sense, COM data alone determine the target only modulo the ideal $\mathfrak{I}_{\text{COM}}$.

This also clarifies what the previous appendix actually achieved. The ν -dressed repair removed the only genuine image obstruction. Once that is done, the leftover freedom is entirely quotient freedom. There is no remaining middle-block compatibility failure hiding inside the 27 dimensions of (282); what remains is the choice of a representative in a fixed quotient class.

J.3 Canonical quotient slice

The natural sparse algebraic choice is therefore to quotient by the COM-null ideal and set those 27 null coefficients to zero. Equivalently, one chooses a complement to the nullspace of the COM projection matrices and keeps only that complement. This defines a canonical quotient slice: a distinguished algebraic representative of the full COM equivalence class.

The usefulness of this choice is immediate. It turns the abstract statement “the generic-frame lift is determined only up to COM-blind directions” into an explicit sparse answer. Write that canonical representative as

$$L_{3,\text{can}} = L_{3,\text{seed}} + \frac{Gpq}{r}Q_{\text{can}} + \frac{G^2pq}{r^2}T_{\text{can}} + \frac{G^3pq}{r^3}S_{\text{can}} + \frac{G^4}{r^4}U_{\text{can}}. \quad (283)$$

By construction,

$$\Pi_Q(Q_{\text{can}}) = (\Delta l_6, \Delta l_7, \Delta l_8, \Delta l_9), \quad \Pi_T(T_{\text{can}}) = (\Delta l_{10}, \Delta l_{11}, \Delta l_{12}), \quad (284)$$

$$\Pi_S(S_{\text{can}}) = (\Delta l_{13}, \Delta l_{14}), \quad \Pi_U(U_{\text{can}}) = (\Delta l_{15}). \quad (285)$$

The next four subsections simply record the corresponding block formulas.

J.4 G/r sextic block

The canonical sextic representative may be taken as

$$\begin{aligned} Q_{\text{can}} = & \frac{9}{4}(a^2b + ab^2) - \frac{19}{8}(a^2c + b^2c) + \frac{5}{4}(a^2e^2 + b^2d^2) + \frac{61}{16}abc + \frac{29}{16}abde \\ & + \frac{9}{16}(ad^2e^2 + bd^2e^2) + \frac{15}{32}(ade^3 + bd^3e) - \frac{5}{16}d^3e^3. \end{aligned} \quad (286)$$

This is the unique sparse representative of the sextic block in the chosen quotient slice.

J.5 G^2/r^2 quartic block

The canonical quartic representative may be taken as

$$\begin{aligned} T_{\text{can}} = & \frac{129}{16}(a^2p + b^2q) + \frac{289}{16}(abp + abq) - \frac{3}{16}(ad^2p + be^2q) - \frac{43}{16}(adep + bdeq) \\ & - \frac{1}{3}(d^2e^2p + d^2e^2q) + \frac{5}{12}(d^3eq + de^3p) + \frac{pq}{4(p+q)}(13ab + 31cde + 16d^2e^2). \end{aligned} \quad (287)$$

The final term is precisely the compact way in which the ν -dressed repair is incorporated into the canonical quotient representative.

J.6 G^3/r^3 quadratic block

The canonical quadratic representative may be taken as

$$\begin{aligned} S_{\text{can}} = & -\frac{880 + 3\pi^2}{192}(ap^2 + bq^2) - \frac{244 + 3\pi^2}{192}(apq + bpq) + \frac{260 + 3\pi^2}{64}(d^2p^2 + e^2q^2) \\ & + \frac{452 + 3\pi^2}{64}(d^2pq + e^2pq) + \frac{p^2q^2}{2(p+q)^2}(c + 7de). \end{aligned} \quad (288)$$

Again the last term is the compact generic-frame image of the repaired middle-block placement.

J.7 G^4/r^4 static block

The static block is already unique once the COM target is fixed. Its canonical representative is therefore simply

$$U_{\text{can}} = \frac{63\pi^2 - 908}{96}(p^2q + pq^2). \quad (289)$$

Substituting (286)–(289) into (283) gives one explicit sparse generic-frame answer whose COM reduction reproduces the exact solved 3PN target in every slot.

J.8 Status of the canonical slice

The canonical quotient slice is an exact algebraic representative, and it is extremely useful because it turns the whole COM-blind family into a sparse explicit formula. But it is still only a quotient choice. Nothing in the construction above says that every COM-blind direction is an ordinary contact transformation, or even that the full 27-dimensional quotient family is physically gauge.

That distinction is the real point of this appendix. The canonical slice gives the right algebraic reference chart, but not yet the final physical uniqueness theorem. The next appendix sharpens the result by computing the actual scaffold-preserving ordinary contact/gauge orbit and showing that it is only an 11-dimensional subspace of the larger 27-dimensional COM-blind family. Equivalently,

COM blindness and ordinary contact/gauge freedom are not identical.

So Appendix K is not a reworking of the same material. It is the additional step that turns the algebraic quotient picture into the true ordinary contact/gauge statement.

K Contact generator and gauge-orbit derivation

The previous appendix identified the full 27-dimensional COM-blind family and used it to define a canonical sparse quotient representative. That algebraic slice is exact and extremely useful, but by itself it does not yet answer the sharper physical bookkeeping question: how much of that COM-blind freedom is actually generated by ordinary 3PN contact transformations? The purpose of this appendix is to answer that question exactly.

The result is stricter than the naive quotient picture. Once the generic-frame 3PN interaction lift is required both to stay inside the compact 24/17/8/1 scaffold and to preserve the already solved COM target, the true ordinary contact/gauge family is only 11-dimensional. So COM blindness and ordinary contact freedom are not identical. The earlier canonical quotient slice is therefore the right sparse algebraic representative, but it is larger than the actual ordinary gauge orbit.

Continue to write the repaired generic-frame residual scaffold as

$$L_{3,\text{res}} = \frac{Gpq}{r} \sum_{i=1}^{24} q_i Q_i + \frac{G^2pq}{r^2} \sum_{j=1}^{17} t_j T_j + \frac{G^3pq}{r^3} \sum_{k=1}^8 s_k S_k + \frac{G^4pq}{r^4} u_1 U_{01}, \quad (290)$$

with the compact exchange-symmetric scaffold basis fixed in the earlier appendices. In the invariant shorthand used throughout the generic-frame lift,

$$a = v_A^2, \quad b = v_B^2, \quad c = \mathbf{v}_A \cdot \mathbf{v}_B, \quad d = v_{An}, \quad e = v_{Bn}, \quad (291)$$

while

$$p = m_A, \quad q = m_B, \quad r = |\mathbf{x}_A - \mathbf{x}_B|. \quad (292)$$

The scaffold block sizes are

$$\boxed{\dim Q = 24, \quad \dim T = 17, \quad \dim S = 8, \quad \dim U = 1.} \quad (293)$$

The present appendix reconstructs the ordinary total-derivative orbit inside that fixed scaffold.

K.1 Raw odd-generator families

At 3PN an ordinary contact transformation is generated by a total derivative of an odd scalar. The crucial exchange-parity rule differs from the even-scalar rule used for the interaction scaffold, because $\mathbf{n}$ changes sign under label interchange. The correct odd rule is

$$A \leftrightarrow B, \quad c \mapsto c, \quad d \mapsto -e, \quad e \mapsto -d, \quad p \leftrightarrow q. \quad (294)$$

The referee audit shows that the raw odd-scalar generator families relevant at formal 3PN are

$$r \times (\text{degree-7 odd scalars}), \quad G \times (\text{degree-5 odd scalars}), \quad (295)$$

$$\frac{G^2}{r} \times (\text{degree-3 odd scalars}), \quad \frac{G^3}{r^2} \times (\text{degree-1 odd scalars}). \quad (296)$$

Counting exchange-symmetric odd monomials with the strict test-mass-vanishing rule gives the exact raw dimensions

$$\boxed{31, \quad 12, \quad 8, \quad 2,} \quad (297)$$

for a total raw odd family of

$$\boxed{31 + 12 + 8 + 2 = 53.} \quad (298)$$

So before any scaffold or COM condition is imposed there are fifty-three candidate odd generators whose Newtonian-order-reduced total derivatives could, in principle, shift the ordinary 3PN representative.

K.2 Scaffold-preserving and COM-preserving reduction

Not every total derivative generated from the raw odd family stays inside the compact 24/17/8/1 scaffold. Let F be any raw odd scalar and define the induced ordinary shift after Newtonian order reduction by

$$\delta L_{\text{cont}}[F] \equiv \left. \frac{dF}{dt} \right|_{\text{Newt red.}}. \quad (299)$$

Demanding that $\delta L_{\text{cont}}[F]$ project entirely onto the compact $Q/T/S/U$ scaffold imposes a linear constraint system of rank 31 on the 53-dimensional raw family. The resulting scaffold-preserving kernel therefore has dimension

$$\boxed{53 - 31 = 22.} \quad (300)$$

The second condition is that the contact shift must leave the solved COM target unchanged. Imposing zero COM image on that 22-dimensional scaffold-preserving family produces an exact COM image rank of 11, so the actual ordinary contact/gauge orbit has dimension

$$\boxed{22 - 11 = 11.} \quad (301)$$

Equivalently, the exact reduction chain is

$$\boxed{53 \longrightarrow 22 \longrightarrow 11.} \quad (302)$$

This is the sharp bookkeeping statement that the previous appendix did not yet make explicit: the full COM-blind quotient is larger than the true ordinary contact family.

K.3 Sparse basis for the actual 11-generator orbit

It is convenient to isolate the part of the COM-null ideal that actually enters the ordinary contact family. In addition to the earlier linear generators

$$\mathcal{C}_1 \equiv pa + qc, \quad \mathcal{C}_2 \equiv pc + qb, \quad (303)$$

the four generators that appear in the sparse contact basis are

$$\boxed{\mathcal{C}_3 \equiv pd + qe, \quad \mathcal{C}_4 \equiv ab - c^2, \quad \mathcal{C}_5 \equiv ae - cd, \quad \mathcal{C}_6 \equiv bd - ce.} \quad (304)$$

A sparse basis for the actual scaffold-preserving and COM-preserving ordinary contact orbit may then be chosen as

$$\Gamma_1 = G(-a\mathcal{C}_5 + b\mathcal{C}_6), \quad \Gamma_2 = G(b\mathcal{C}_5 - a\mathcal{C}_6), \quad (305)$$

$$\Gamma_3 = G(d + e)(e\mathcal{C}_5 - d\mathcal{C}_6), \quad \Gamma_4 = -G(d - e)\mathcal{C}_4, \quad (306)$$

$$\Gamma_5 = -Gde(\mathcal{C}_5 - \mathcal{C}_6), \quad \Gamma_6 = G(-d^2\mathcal{C}_5 + e^2\mathcal{C}_6), \quad (307)$$

$$\Gamma_7 = \frac{G^2}{r}(q\mathcal{C}_5 - p\mathcal{C}_6), \quad \Gamma_8 = \frac{G^2}{r}(a - b)\mathcal{C}_3, \quad (308)$$

$$\Gamma_9 = \frac{G^2}{r}(-p\mathcal{C}_5 + q\mathcal{C}_6), \quad \Gamma_{10} = \frac{G^2}{r}(d^2 - e^2)\mathcal{C}_3, \quad (309)$$

$$\Gamma_{11} = -\frac{G^3}{r^2}(p - q)\mathcal{C}_3. \quad (310)$$

The factorized form in (305)–(310) is more informative than the fully expanded version printed by the referee audit, because it makes three structural facts immediate.

First, every surviving generator factors through

$$\Gamma_i \in \langle \mathcal{C}_3, \mathcal{C}_4, \mathcal{C}_5, \mathcal{C}_6 \rangle, \quad i = 1, \dots, 11, \quad (311)$$

so every generator already vanishes in the COM quotient. Second, no generator uses $\mathcal{C}_1$ or $\mathcal{C}_2$, so the actual ordinary contact orbit lives only in the smaller subideal

$$\boxed{\mathcal{I}_{\text{contact}} = \langle \mathcal{C}_3, \mathcal{C}_4, \mathcal{C}_5, \mathcal{C}_6 \rangle \subset \mathcal{I}_{\text{COM}}.} \quad (312)$$

Third, because none of the generators carries a purely static G^4/r^4 piece, the simple ordinary contact family leaves the static block untouched.

K.4 Image inside the COM-null family

After Newtonian order reduction and projection back onto the compact scaffold, the image of the eleven generators above has exact rank

$$\boxed{\text{rank Im}(\Gamma_1, \dots, \Gamma_{11}) = 11.} \quad (313)$$

By contrast, the full COM-null family identified in the previous appendix has rank

$$\boxed{\text{rank } \mathcal{I}_{\text{COM}} = 27.} \quad (314)$$

So the residual algebraic quotient left after dividing out the actual ordinary contact orbit has dimension

$$\boxed{27 - 11 = 16.} \quad (315)$$

Equivalently, if M_{full} denotes the full COM projection matrix on the compact scaffold, then the exact referee statement is

$$M_{\text{full}} \text{Im}(\Gamma_1, \dots, \Gamma_{11}) = 0, \quad (316)$$

but the converse fails because the COM-null kernel is larger than the contact image.

The blockwise contact-image ranks are

$$\boxed{\text{rank}_Q = 6, \quad \text{rank}_T = 9, \quad \text{rank}_S = 4, \quad \text{rank}_U = 0.} \quad (317)$$

The vanishing U -block rank is the exact statement that the simple ordinary contact family does not move the static G^4/r^4 slot at all. More broadly, (317) explains why the canonical quotient slice from the previous appendix is larger than the actual ordinary contact orbit: the contact image only sweeps out one 11-dimensional suborbit inside the full 27-dimensional COM-blind family.

This is the precise conclusion needed before the generic-frame uniqueness theorem can be stated without overclaiming. The canonical quotient slice is still the right sparse algebraic representative, but it is not exhausted by ordinary gauge alone. That is why the final uniqueness claim must be deferred to the Hamiltonian-first fixed-chart argument of the next appendix rather than being read off from ordinary contact freedom at this stage.

L Generic-frame target import and Hamiltonian-first lift

This appendix records the last conceptual clarification needed before the full fixed-chart generic-frame uniqueness theorem can be stated. The ordinary generic-frame target itself is not the problem. After the one-body gate, the natural self/static seed, the reduced COM target, the COM-null ideal, and the ordinary contact family have all been separated cleanly, the remaining question is only how the imported generic-frame representative should be compared with COM data. The answer is now exact: the apparent inconsistency comes entirely from doing the reduction in the wrong order. Naive ordinary-Lagrangian COM substitution is not the right quotient principle; Hamiltonian-first lift is.

L.1 Ordinary generic-frame target import

Use again the exchange-symmetric generic-frame invariants

$$a = v_A^2, \quad b = v_B^2, \quad c = \mathbf{v}_A \cdot \mathbf{v}_B, \quad d = v_{An} = \mathbf{v}_A \cdot \mathbf{n}, \quad e = v_{Bn} = \mathbf{v}_B \cdot \mathbf{n}, \quad (318)$$

with

$$p \equiv m_A, \quad q \equiv m_B, \quad \mathbf{r} \equiv \mathbf{x}_A - \mathbf{x}_B, \quad r \equiv |\mathbf{r}|, \quad \mathbf{n} \equiv \frac{\mathbf{r}}{r}. \quad (319)$$

The full generic-frame ordinary 3PN target used in the paper is imported from the standard literature with the ambiguity fixed by

$$\lambda = -\frac{3}{11}\omega_s - \frac{1987}{3080}, \quad \omega_s = 0, \quad \implies \quad \lambda = -\frac{1987}{3080}. \quad (320)$$

It is convenient to split the imported target into the frozen natural one-body/self-static seed and the genuinely comparable-mass residual. Write

$$L_{3,\text{gen}}^{\text{imp}} = L_{3,\text{seed}} + \frac{Gm_A m_B}{r} Q_{\text{res}} + \frac{G^2 m_A m_B}{r^2} T_{\text{res}} + \frac{G^3 m_A m_B}{r^3} S_{\text{res}} + \frac{G^4 m_A m_B}{r^4} U_{\text{res}}, \quad (321)$$

where the seed polynomials are

$$\begin{aligned} Q_{\text{seed}} &= \frac{11}{16}(a^3 + b^3), & T_{\text{seed}} &= \frac{47}{16}(qa^2 + pb^2), \\ S_{\text{seed}} &= \frac{13}{8}(q^2 a + p^2 b), & U_{\text{seed}} &= -\frac{1}{8}(q^3 + p^3). \end{aligned} \quad (322)$$

The exact import audit shows that after subtraction of (322) the residual lands entirely inside the compact generic-frame interaction scaffold,

$$Q_{\text{res}} \in \text{span}\{Q_i\}_{i=1}^{24}, \quad T_{\text{res}} \in \text{span}\{T_j\}_{j=1}^{17}, \quad S_{\text{res}} \in \text{span}\{S_k\}_{k=1}^8, \quad U_{\text{res}} \in \text{span}\{U_{01}\}, \quad (323)$$

so the imported target may be written exactly as

$$\begin{aligned} L_{3,\text{gen}}^{\text{imp}} = & L_{3,\text{seed}} + \frac{Gm_A m_B}{r} \sum_{i=1}^{24} q_i^{\text{imp}} Q_i + \frac{G^2 m_A m_B}{r^2} \sum_{j=1}^{17} t_j^{\text{imp}} T_j \\ & + \frac{G^3 m_A m_B}{r^3} \sum_{k=1}^8 s_k^{\text{imp}} S_k + \frac{G^4 m_A m_B}{r^4} u_1^{\text{imp}} U_{01}. \end{aligned} \quad (324)$$

No extra generic-frame direction outside the 24/17/8/1 scaffold is required. The import also preserves the strict one-body gate, because on the test-mass branch

$$p \rightarrow 0, \quad q \rightarrow 1, \quad b \rightarrow 0, \quad c \rightarrow 0, \quad e \rightarrow 0, \quad (325)$$

one has

$$Q_{\text{res}} \rightarrow 0, \quad T_{\text{res}} \rightarrow 0, \quad S_{\text{res}} \rightarrow 0, \quad U_{\text{res}} \rightarrow 0. \quad (326)$$

So the imported ordinary generic-frame target is exact, compatible with the one-body gate, and already expressible in the finite interaction scaffold.

L.2 Naive COM reduction mismatch

The decisive subtlety appears only after one tries to compare the imported ordinary generic-frame target directly with the previously solved COM ordinary ledger. In the COM frame,

$$\mathbf{v}_A = X_B \mathbf{v}, \quad \mathbf{v}_B = -X_A \mathbf{v}, \quad X_A + X_B = 1, \quad \nu = X_A X_B, \quad (327)$$

so one might try to obtain the reduced COM ordinary target by simple substitution into (324). That procedure fails.

If one performs this naive ordinary reduction, the resulting COM coefficients do not coincide with the exact COM ordinary target of the earlier appendices. The exact mismatches are

$$\delta l_6 = \frac{\nu(80\nu^2 + 48\nu - 17)}{16}, \quad \delta l_7 = \frac{3\nu(24\nu^2 - 10\nu + 1)}{16}, \quad \delta l_8 = \frac{3\nu^2(4\nu - 1)}{8}, \quad \delta l_9 = 0, \quad (328)$$

$$\delta l_{10} = \frac{\nu(-52\nu^2 - 123\nu + 34)}{16}, \quad \delta l_{11} = \frac{\nu(-124\nu^2 - 97\nu + 32)}{16}, \quad \delta l_{12} = \nu^2(1 - 4\nu), \quad (329)$$

$$\delta l_{13} = \frac{\nu(4\nu^2 + 27\nu - 7)}{8}, \quad \delta l_{14} = \frac{\nu(28\nu^2 + 69\nu - 19)}{8}, \quad \delta l_{15} = 0. \quad (330)$$

So naive ordinary COM reduction agrees only in the l_9 and l_{15} slots. This is not a contradiction and it is not a defect of the imported target. Rather, it is the familiar 3PN reduction-ordering subtlety: the reduced relative ordinary Lagrangian does not follow from the general-frame ordinary representative by naive substitution alone. Therefore naive ordinary COM reduction cannot be used as the final generic-to-COM quotient principle.

L.3 Hamiltonian-first recovery theorem

The correct cure is to apply the exact cubic Legendre transform in the generic frame first, and only then reduce the resulting Hamiltonian to COM. Carry forward the exact cubic compiler from Appendix E,

$$H_3 = -L_3(v_0) + A_0^T M^{-1} B_0 - \frac{1}{2} A_0^T M^{-1} C_0 M^{-1} A_0, \quad (331)$$

with

$$v_0 = M^{-1}P, \quad A_0 = \left. \frac{\partial L_1}{\partial v} \right|_{v_0}, \quad B_0 = \left. \frac{\partial L_2}{\partial v} \right|_{v_0}, \quad C_0 = \left. \frac{\partial^2 L_1}{\partial v^2} \right|_{v_0}. \quad (332)$$

Because the lower-order 1PN/2PN ledger is already frozen, the feedback terms in (331) are completely determined before the new generic-frame 3PN residual is inserted. One therefore lifts (324) to the generic-frame Hamiltonian first and imposes the COM relations only afterward.

That procedure gives the exact theorem

$$\boxed{H_{3,\text{COM}}^{(\text{lift})} = H_{3,\text{COM}}^{\text{target}}} \quad (333)$$

Equivalently,

$$\boxed{\text{generic-frame Legendre lift first} \implies H_{3,\text{COM}}^{\text{lift}} = H_{3,\text{COM}}^{\text{GR}}} \quad (334)$$

So the earlier mismatch is diagnosed completely:

$$\text{naive COM reduction of ordinary } L_3 \neq \text{correct COM ordinary target}, \quad (335)$$

but the imported generic-frame ordinary target itself is exact and perfectly compatible with the one-body gate. The error was the order of operations, not any defect in the target.

L.4 What Appendix L fixes for the later lift

Appendix L settles the last conceptual issue in the generic-frame import stage. Its output can be summarized in four statements.

First, the imported generic-frame ordinary target is exact, lives entirely inside the compact 24/17/8/1 interaction scaffold, and continues to vanish on the strict test-mass branch. So the generic-frame target itself is fully consistent with the lower-order backbone.

Second, naive ordinary-Lagrangian COM reduction is not the right quotient principle. The exact nonzero mismatches (328)–(330) show that direct ordinary COM substitution loses information at precisely the level where the 3PN chart subtlety matters.

Third, Hamiltonian-first lift cures the problem exactly. Once the imported ordinary target is lifted through the cubic compiler before COM reduction, the standard COM 3PN Hamiltonian is recovered without residue. This vindicates both the imported target and the lower-order compiler chain.

Fourth, after this appendix the remaining ambiguity question is no longer “is the target right?” but rather “which ordinary representative/canonical slice survives in the fixed Hamiltonian chart?” That is precisely the point at which the next appendix takes over. From there onward the lift is no longer a COM-quotient problem; it becomes the fixed-chart generic-frame Hamiltonian compiler and uniqueness theorem.

M Generic-frame Hamiltonian compiler and uniqueness

This appendix records the point at which the generic-frame 3PN lift stops being a COM-quotient problem and becomes an exact compiler theorem. The previous appendix already showed that the imported ordinary generic-frame target is exact and that the correct comparison with COM data is Hamiltonian-first rather than naive ordinary substitution. What remains is therefore sharper: does the previously identified 27-dimensional COM-blind family represent a genuine 3PN ambiguity, or only a blindness of COM reduction? In the fixed ADM Hamiltonian chart the answer is now exact: it is only a blindness.

The earlier appendices had already done four things:

1. frozen the exact one-body 3PN gate and the natural one-body/self-static seed;

2. solved the reduced COM 15-slot Hamiltonian target and its ordinary-side image;
3. built the 50-slot generic-frame interaction scaffold with block sizes

$$24, \quad 17, \quad 8, \quad 1; \quad (336)$$

4. and identified the 27-dimensional COM-blind family together with the smaller 11-generator ordinary contact orbit.

The present appendix proves that once the lower-order ledger and the natural 3PN seed are frozen, the full generic-frame Hamiltonian compiler has no kernel and therefore removes the whole COM-null family at once.

M.1 Fixed generic-frame momentum basis

Use the same formal invariant symbols as in the ordinary generic-frame scaffold, but reinterpret them on the Hamiltonian side as Newtonian-order momentum invariants:

$$a = \frac{P_A^2}{m_A^2}, \quad b = \frac{P_B^2}{m_B^2}, \quad c = \frac{P_A \cdot P_B}{m_A m_B}, \quad d = \frac{\mathbf{n} \cdot P_A}{m_A}, \quad e = \frac{\mathbf{n} \cdot P_B}{m_B}. \quad (337)$$

Write again

$$p \equiv m_A, \quad q \equiv m_B, \quad \mathbf{r} \equiv \mathbf{x}_A - \mathbf{x}_B, \quad r \equiv |\mathbf{r}|, \quad \mathbf{n} \equiv \frac{\mathbf{r}}{r}. \quad (338)$$

In these variables the generic-frame Hamiltonian interaction scaffold has the same formal block structure as the ordinary one:

- a 24-slot sextic block at order G/r ,
- a 17-slot quartic block at order G^2/r^2 ,
- an 8-slot quadratic block at order G^3/r^3 ,
- and a 1-slot static block at order G^4/r^4 .

So the full generic-frame 3PN interaction ledger is still a 50-slot basis. Nothing new is being hidden in a different chart; only the kinematic interpretation of the invariants changes.

Write the ordinary interaction residual beyond the already frozen one-body/self- static seed as

$$\Delta L_3 = \frac{Gpq}{r} \sum_{i=1}^{24} q_i Q_i + \frac{G^2 pq}{r^2} \sum_{j=1}^{17} t_j T_j + \frac{G^3 pq}{r^3} \sum_{k=1}^8 s_k S_k + \frac{G^4 pq}{r^4} u_1 U_{01}, \quad (339)$$

and write the Hamiltonian residual in the same formal invariant chart as

$$\Delta H_3 = \frac{Gpq}{r} \sum_{i=1}^{24} \hat{q}_i Q_i + \frac{G^2 pq}{r^2} \sum_{j=1}^{17} \hat{t}_j T_j + \frac{G^3 pq}{r^3} \sum_{k=1}^8 \hat{s}_k S_k + \frac{G^4 pq}{r^4} \hat{u}_1 U_{01}. \quad (340)$$

The problem of uniqueness is therefore exactly the problem of how the 50 ordinary coefficients in (339) are mapped into the 50 Hamiltonian coefficients in (340) once all lower-order data are frozen.

M.2 Exact compiler theorem

Carry forward the exact cubic Legendre identity proved earlier:

$$H_3 = -L_3(v_0) + A_0^T M^{-1} B_0 - \frac{1}{2} A_0^T M^{-1} C_0 M^{-1} A_0, \quad (341)$$

with

$$v_0 = M^{-1} P, \quad A_0 = \left. \frac{\partial L_1}{\partial v} \right|_{v_0}, \quad B_0 = \left. \frac{\partial L_2}{\partial v} \right|_{v_0}, \quad C_0 = \left. \frac{\partial^2 L_1}{\partial v^2} \right|_{v_0}. \quad (342)$$

Now split the full ordinary 3PN block as

$$L_3 = L_{3,\text{seed}} + \Delta L_3, \quad (343)$$

where $L_{3,\text{seed}}$ is the natural one-body/self-static seed already frozen in the previous appendices. The decisive point is that the second and third terms in (341) depend only on the already frozen lower-order ledger (L_1, L_2) . They do not depend on the new 3PN interaction residual ΔL_3 . Therefore the full generic-frame residual compiler is exact and trivial:

$$\boxed{\Delta H_3 = -\Delta L_3(v_0)}. \quad (344)$$

Comparing (339) and (340), this is literally the slotwise sign flip

$$\boxed{\hat{q}_i = -q_i, \quad \hat{t}_j = -t_j, \quad \hat{s}_k = -s_k, \quad \hat{u}_1 = -u_1}. \quad (345)$$

Equivalently, if the 50 ordinary residual coordinates are assembled into one column vector Δl and the 50 Hamiltonian residual coordinates into Δh , then

$$\boxed{\Delta h = C_{\text{gen}} \Delta l, \quad C_{\text{gen}} = -I_{50}}. \quad (346)$$

It is useful to display the blockwise form explicitly:

$$C_Q = -I_{24}, \quad C_T = -I_{17}, \quad C_S = -I_8, \quad C_U = -I_1, \quad C_{\text{gen}} = \text{diag}(C_Q, C_T, C_S, C_U). \quad (347)$$

The referee audit therefore gives the exact linear-algebraic status

$$\text{rank } C_{\text{gen}} = 50, \quad \ker C_{\text{gen}} = \{0\}. \quad (348)$$

So the fixed-chart generic-frame Hamiltonian compiler has full rank and no kernel. Once the lower-order blocks and the natural 3PN seed are frozen, the generic-frame lift is no longer a nonlinear inversion problem. The Hamiltonian residual is simply the negative of the ordinary residual in the same invariant chart.

M.3 Direct recovery of the ordinary representative

Let the exact imported ordinary target from the previous appendix be written on the compact interaction scaffold as the coefficient vectors

$$\mathbf{q}^{\text{imp}} \in \mathbb{R}^{24}, \quad \mathbf{t}^{\text{imp}} \in \mathbb{R}^{17}, \quad \mathbf{s}^{\text{imp}} \in \mathbb{R}^8, \quad u_1^{\text{imp}} \in \mathbb{R}, \quad (349)$$

so that

$$\Delta L_3^{\text{imp}} = \frac{Gpq}{r} \sum_{i=1}^{24} q_i^{\text{imp}} Q_i + \frac{G^2 pq}{r^2} \sum_{j=1}^{17} t_j^{\text{imp}} T_j + \frac{G^3 pq}{r^3} \sum_{k=1}^8 s_k^{\text{imp}} S_k + \frac{G^4 pq}{r^4} u_1^{\text{imp}} U_{01}. \quad (350)$$

Then the fixed-chart Hamiltonian target is obtained immediately by the sign flip of (345):

$$\mathbf{q}_H^{\text{target}} = -\mathbf{q}_H^{\text{imp}}, \quad \mathbf{t}_H^{\text{target}} = -\mathbf{t}_H^{\text{imp}}, \quad \mathbf{s}_H^{\text{target}} = -\mathbf{s}_H^{\text{imp}}, \quad u_{1,H}^{\text{target}} = -u_{1,H}^{\text{imp}}. \quad (351)$$

Because the inverse compiler is the same matrix,

$$C_{\text{gen}}^{-1} = C_{\text{gen}} = -I_{50}, \quad (352)$$

one recovers the ordinary representative directly from the fixed-chart Hamiltonian target:

$$\mathbf{q}^{\text{imp}} = -\mathbf{q}_H^{\text{target}}, \quad \mathbf{t}^{\text{imp}} = -\mathbf{t}_H^{\text{target}}, \quad \mathbf{s}^{\text{imp}} = -\mathbf{s}_H^{\text{target}}, \quad u_1^{\text{imp}} = -u_{1,H}^{\text{target}}. \quad (353)$$

So once the fixed ADM Hamiltonian chart is chosen, there is no further algebraic quotienting step. The exact ordinary representative is read off immediately by sign reversal of the Hamiltonian residual coordinates.

M.4 Fixed-chart uniqueness and the fate of the COM-null family

This theorem resolves the earlier 27-dimensional COM-blind family exactly. Before the full Hamiltonian chart was imposed, the ordinary generic-frame lift had

- a 27-dimensional family invisible to COM reduction,
- inside it an 11-generator ordinary contact/gauge orbit,
- and therefore a residual 16-dimensional quotient after the true contact family was removed.

That was a real COM-side obstruction, but it was only a COM-side obstruction. Because the full generic-frame Hamiltonian compiler has zero kernel, none of those 27 directions is Hamiltonian-blind in the fixed ADM chart. So the full generic-frame Hamiltonian target removes the whole COM-null family at once. Equivalently,

$$\boxed{\text{COM blindness is not Hamiltonian blindness.}} \quad (354)$$

The 11-generator ordinary contact family is not an exception to this. Inside the fixed ADM Hamiltonian chart, those directions do not lie in the compiler kernel; they correspond instead to moving to a different canonical chart. Once that chart is frozen, the Hamiltonian target selects a single ordinary representative.

The uniqueness theorem may therefore be written in its sharpest form as follows. Suppose ΔL_3 and $\Delta L'_3$ are two generic-frame ordinary residuals that lead to the same fixed-chart Hamiltonian target. Then by (344),

$$0 = \Delta H_3 - \Delta H'_3 = -(\Delta L_3 - \Delta L'_3), \quad (355)$$

so

$$\boxed{\Delta L'_3 = \Delta L_3.} \quad (356)$$

In other words, once the fixed ADM Hamiltonian chart is chosen, the generic-frame ordinary representative is unique.

At that point the 3PN generic-frame lift is no longer the bottleneck. The live conservative work moves entirely into the physical interpretation of the solved residual: the grouped real P_2 middle block, the unique static geometry completion, and the pure-kinetic compiler image.

M.5 What this appendix fixes for the later split

This appendix settles the last generic-frame algebra question in four steps.

First, it fixes the Hamiltonian-side invariant chart in exactly the same 24/17/8/1 interaction scaffold used on the ordinary side, so no hidden degree count is being changed when one passes to momentum variables.

Second, it proves that the exact cubic Legendre transform acts on the full 50-slot interaction residual by pure sign flip, $C_{\text{gen}} = -I_{50}$, once the lower-order ledger and the natural 3PN seed are frozen.

Third, it shows that the compiler has zero kernel. The previously found 27-dimensional COM-blind family, and the smaller 11-generator ordinary contact orbit sitting inside it, therefore do not survive as true algebraic freedoms in the fixed ADM chart.

Fourth, it gives the fixed-chart uniqueness theorem (356). From this point onward, the live conservative questions are no longer about which generic-frame representative is allowed. They are only about how the unique residual decomposes physically into its grouped real P_2 , geometry, and pure-kinetic lanes.

N Grouped real P_2 target pack and co-rotating front-end scaffold

This appendix freezes the exact conservative grouped- P_2 inverse problem that stands between the solved comparable-mass 3PN target and any future moving-throat constitutive interpretation. Its job is twofold. First, it states the exact 3PN data vector that any grouped real $P_{20} \oplus P_{21} \oplus P_{22}$ conservative module must face once the frozen one-body/self-static seed is subtracted. Second, it writes the first exact co-rotating time-local $O(\omega^2)$ front-end scaffold in grouped language, before any throat-side closure is guessed.

The point of the appendix is therefore not to complete the 3PN grouped module—that comes only after the richer local lift is admitted—but to make clear exactly what information the grouped branch carries, how sharply it is constrained, and what the smallest kinematic front end looks like before any constitutive dictionary is imposed.

N.1 Exact 3PN data vector seen by the grouped front end

From Appendix G, the exact comparable-mass residual beyond the frozen one-body/self-static seed is

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}, \quad \Delta l_2 = \Delta l_3 = \Delta l_4 = \Delta l_5 = 0. \quad (357)$$

$$\Delta l_6 = \frac{\nu(38 - 116\nu - 57\nu^2)}{16}, \quad \Delta l_7 = \frac{\nu^2(20 - 69\nu)}{16}, \quad (358)$$

$$\Delta l_8 = \frac{3\nu^2(3 - 11\nu)}{16}, \quad \Delta l_9 = \frac{5\nu^3}{16}.$$

$$\Delta l_{10} = \frac{\nu(129 - 98\nu + 52\nu^2)}{16}, \quad \Delta l_{11} = \frac{\nu(-3 + 52\nu + 124\nu^2)}{16}, \quad (359)$$

$$\Delta l_{12} = \frac{\nu(-5 + 11\nu + 48\nu^2)}{12}.$$

$$\Delta l_{13} = -\frac{\nu(244 + 3\pi^2 + 1272\nu + 96\nu^2)}{192}. \quad (360)$$

$$\Delta l_{14} = \frac{\nu(452 + 3\pi^2 - 384\nu - 224\nu^2)}{64}, \quad \Delta l_{15} = \frac{\nu(-908 + 63\pi^2)}{96}. \quad (361)$$

The grouped conservative front end can only hope to carry the middle block $(\Delta l_6, \dots, \Delta l_{14})$. The pure-kinetic slot Δl_1 and the static scalar/geometry slot Δl_{15} therefore sit outside the first minimal grouped inverse problem from the beginning.

At 3PN the grouped conservative payload through order $O(\omega^2)$ contains only three real data,

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)}. \quad (362)$$

So the grouped front end faces an exact 11-slot residual vector while carrying only three real conservative unknowns. Before any isotropy assumption is made, this is already an eight-fold overdetermined test. On the minimal isotropic branch it becomes a ten-fold overdetermined test. That sharp mismatch of data counts is the real reason the grouped- P_2 route is such a useful pre-PDE conservative gate: once a constitutive dictionary is written, the test is brutally exact rather than qualitative.

N.2 Grouped trace, anisotropy variables, and the isotropy gate

It is often more transparent to rewrite the three grouped coefficients in terms of one grouped trace and two anisotropy defects. Define

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \quad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \quad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2}. \quad (363)$$

The inverse map is exact:

$$u_2^{(20)} = \bar{u}_2 + 4a_2, \quad u_2^{(21)} = \bar{u}_2 - a_2 + b_2, \quad u_2^{(22)} = \bar{u}_2 - a_2 - b_2. \quad (364)$$

Equivalently, the grouped weighted-sum identity is automatic,

$$(u_2^{(20)} - \bar{u}_2) + 2(u_2^{(21)} - \bar{u}_2) + 2(u_2^{(22)} - \bar{u}_2) = 0. \quad (365)$$

The natural grouped anisotropy norm is therefore

$$\mathcal{A}_2^2 = 4a_2^2 + \frac{4}{5}b_2^2. \quad (366)$$

So the exact isotropy gate for the minimal branch is simply

$$\boxed{a_2 = 0, \quad b_2 = 0}, \quad (367)$$

or equivalently $\mathcal{A}_2^2 = 0$. If that gate passes, then all three $O(\omega^2)$ grouped coefficients coincide and one may write

$$u_2^{(20)} = u_2^{(21)} = u_2^{(22)} \equiv u_2. \quad (368)$$

This is the same isotropy test that later feeds directly into the narrowed 2.5PN quadrupole bridge.

N.3 Co-rotating real- P_2 source coefficients

Work in the instantaneous co-rotating pair frame and decompose the relative velocity as

$$\mathbf{v} = \mathbf{u} + d\mathbf{n}, \quad \mathbf{u} \cdot \mathbf{n} = 0, \quad v^2 = u^2 + d^2, \quad U = \frac{GM}{r}. \quad (369)$$

In this frame the grouped real source coefficients of the normalized STF orbital quadrupole $\ddot{I}_{ij}^{\text{orb}}/(2\mu)$ are

$$C_{20} = \sqrt{\frac{2}{3}} \left(d^2 - \frac{u^2}{2} - U \right), \quad (370)$$

$$C_{21c} = \sqrt{2} d u_x, \quad C_{21s} = \sqrt{2} d u_y, \quad (371)$$

$$C_{22c} = \frac{u_x^2 - u_y^2}{\sqrt{2}}, \quad C_{22s} = \sqrt{2} u_x u_y. \quad (372)$$

Using the exact co-rotating pair-frame kinematics

$$\dot{d} = \frac{u^2 - U}{r}, \quad \dot{u}_x = -\frac{d}{r}u_x, \quad \dot{u}_y = -\frac{d}{r}u_y, \quad \dot{U} = -\frac{d}{r}U, \quad (373)$$

one obtains the exact grouped derivative norms

$$\dot{C}_{20}^2 = \frac{2d^2}{3r^2}(3u^2 - U)^2, \quad (374)$$

$$\dot{C}_{21c}^2 + \dot{C}_{21s}^2 = \frac{2u^2}{r^2}(u^2 - d^2 - U)^2, \quad (375)$$

$$\dot{C}_{22c}^2 + \dot{C}_{22s}^2 = \frac{2d^2u^4}{r^2}. \quad (376)$$

These formulas are exact kinematic identities of the grouped source map itself. No throat-side constitutive closure has yet been used.

N.4 First exact time-local grouped- P_2 front-end scaffold

The smallest time-local grouped conservative ansatz built from the exact source norms is

$$L_{P_2, \omega^2}^{\text{front}} = \frac{1}{2}c_2^{(20)}\dot{C}_{20}^2 + \frac{1}{2}c_2^{(21)}(\dot{C}_{21c}^2 + \dot{C}_{21s}^2) + \frac{1}{2}c_2^{(22)}(\dot{C}_{22c}^2 + \dot{C}_{22s}^2), \quad (377)$$

with three real grouped coefficients $c_2^{(20)}, c_2^{(21)}, c_2^{(22)}$. Substituting (374)–(376) gives the exact co-rotating front-end scaffold

$$L_{P_2, \omega^2}^{\text{front}} = \frac{c_2^{(20)}}{3r^2}d^2(3u^2 - U)^2 + \frac{c_2^{(21)}}{r^2}u^2(u^2 - d^2 - U)^2 + \frac{c_2^{(22)}}{r^2}d^2u^4. \quad (378)$$

This is still only a kinematic front end. But it turns the conservative grouped- P_2 problem into a concrete inverse problem: exact middle-block target data on one side, and three grouped $O(\omega^2)$ coefficients on the other.

On the isotropic branch,

$$a_2 = b_2 = 0, \quad c_2^{(20)} = c_2^{(21)} = c_2^{(22)} \equiv \bar{c}_2, \quad (379)$$

and the front-end scaffold collapses exactly to

$$L_{P_2, \omega^2}^{\text{iso}} = \frac{\bar{c}_2}{3r^2} \left(3v^6 - 3d^2v^4 - 6Uv^4 + 12Ud^2v^2 - 6Ud^4 + 3U^2v^2 - 2U^2d^2 \right). \quad (380)$$

This isotropic collapse is useful later because it makes the degree of overdetermination completely explicit: once isotropy is imposed, one conservative trace datum is being asked to account for the full 3PN middle block.

That is why Appendix N is the right place to stop before any throat-side constitutive guess is made. The exact 3PN grouped data, its isotropy gate, the co-rotating source coefficients, and the first exact front-end scaffold are now fully explicit. The next question—taken up in the following appendix—is whether the smallest local constitutive lift of this front end is already enough. It is not, and that precise negative result is what forces the richer grouped local family used later in the paper.

O Failure of the minimal local grouped- P_2 demotion

This appendix records the first exact negative result of the grouped real P_2 program at 3PN. The point is not that the grouped- P_2 ontology fails. The point is narrower and more useful: the *smallest* local constitutive lift of the grouped front end is too rigid to reproduce the exact GR middle block. This no-go is what forces the richer grouped compiler used in the main text.

O.1 Exact grouped target pack and the overdetermined 3PN gate

At 3PN, the grouped conservative front end is controlled by exactly three real $O(\omega^2)$ data,

$$u_2^{(20)}, \quad u_2^{(21)}, \quad u_2^{(22)}, \quad (381)$$

or equivalently by the grouped trace and anisotropy defects

$$\bar{u}_2 = \frac{u_2^{(20)} + 2u_2^{(21)} + 2u_2^{(22)}}{5}, \quad a_2 = \frac{2u_2^{(20)} - u_2^{(21)} - u_2^{(22)}}{10}, \quad b_2 = \frac{u_2^{(21)} - u_2^{(22)}}{2}. \quad (382)$$

So the exact isotropy gate for the minimal branch is

$$a_2 = 0, \quad b_2 = 0. \quad (383)$$

The structural reason this is already a hard theorem gate is that the exact 3PN comparable-mass COM target presents an 11-slot residual vector in the interaction sector, whereas the grouped front end offers only three conservative unknowns. On the isotropic branch that shrinks to one parameter. In that sense, 3PN is an overdetermined algebraic test of the grouped real P_2 idea rather than a loose plausibility exercise.

The exact co-rotating time-local front end built from the real STF orbital source coefficients takes the form

$$L_{P_2, \omega^2}^{\text{front}} = \frac{c_2^{(20)}}{3r^2} \dot{r}^2 (3u_\perp^2 - U)^2 + \frac{c_2^{(21)}}{r^2} u_\perp^2 (u_\perp^2 - \dot{r}^2 - U)^2 + \frac{c_2^{(22)}}{r^2} \dot{r}^2 u_\perp^4, \quad (384)$$

where $u_\perp^2 = v^2 - \dot{r}^2$. This is the first exact grouped kinematic scaffold. The question addressed here is whether its *minimal local demotion* can already match the exact GR middle block.

O.2 Why the direct grouped front end is too rigid

The direct grouped front end has uniform virial weight five, so by itself it is too high to be a formal 3PN ordinary-Lagrangian block. The unique local isotropic one-step demotion that lowers the weight by exactly one unit is multiplication by one power of $U^{-1} \sim r$. After that demotion, the image lands only in the nine interaction slots

$$(l_6, \dots, l_{14}), \quad (385)$$

with no support at all for the pure-kinetic slots $(l_1, \dots, l_5)$ and no support for the static slot l_{15} .

This is already informative: even before comparison with the exact GR target, the minimally demoted grouped lift can at most hope to carry the middle block. It cannot possibly account for the pure-kinetic or static sectors. The real obstruction, however, is stronger than that. Because the demoted image depends on only three grouped coefficients, its nine-slot image has rank three and therefore obeys six exact linear relations.

O.3 The isotropic image and the six exact slot relations

On the isotropic branch $a_2 = b_2 = 0$, the demoted grouped image is especially rigid. In the standard isotropic COM basis it gives

$$l_6 = \bar{u}_2, \quad l_7 = -\bar{u}_2, \quad l_8 = l_9 = 0, \quad (386)$$

$$l_{10} = -2\bar{u}_2, \quad l_{11} = 4\bar{u}_2, \quad l_{12} = -2\bar{u}_2, \quad l_{13} = \bar{u}_2, \quad l_{14} = -\frac{2}{3}\bar{u}_2. \quad (387)$$

So even on the most favorable branch, the minimally demoted image is far too constrained to be a generic middle-block carrier.

More generally, because the nine slots depend on only three grouped coefficients, the minimal image obeys six exact relations. A convenient basis is

$$2l_6 + 2l_7 + l_8 = 0, \quad (388)$$

$$l_9 = l_6 + l_7, \quad (389)$$

$$l_{10} = -2l_6, \quad (390)$$

$$l_{11} + l_{12} = 2l_6, \quad (391)$$

$$l_{13} = l_6, \quad (392)$$

$$l_{14} = -\frac{1}{6}l_{11}. \quad (393)$$

Any 3PN target produced by the minimally demoted local grouped- P_2 scaffold must satisfy all six simultaneously.

O.4 The exact GR target violates all six relations

Substituting the exact GR residual slots $(\Delta l_6, \dots, \Delta l_{14})$ into these relations gives the obstruction polynomials

$$2\Delta l_6 + 2\Delta l_7 + \Delta l_8 = \frac{\nu(-285\nu^2 - 183\nu + 76)}{16}, \quad (394)$$

$$-\Delta l_6 - \Delta l_7 + \Delta l_9 = \frac{\nu(131\nu^2 + 96\nu - 38)}{16}, \quad (395)$$

$$\Delta l_{10} + 2\Delta l_6 = \frac{\nu(-62\nu^2 - 330\nu + 205)}{16}, \quad (396)$$

$$\Delta l_{11} + \Delta l_{12} - 2\Delta l_6 = \frac{\nu(906\nu^2 + 896\nu - 257)}{48}, \quad (397)$$

$$\Delta l_{13} - \Delta l_6 = \frac{\nu(588\nu^2 + 120\nu - 700 - 3\pi^2)}{192}, \quad (398)$$

$$\Delta l_{14} + \frac{1}{6}\Delta l_{11} = \frac{\nu(-424\nu^2 - 1048\nu + 9\pi^2 + 1350)}{192}. \quad (399)$$

All six are nonzero for generic ν . In particular, the obstruction does not disappear at equal mass. The point is not one exceptional coefficient mismatch; it is a structural rank failure of the minimally demoted grouped map.

So the exact conclusion is

the minimally demoted local grouped- P_2 image
 does not contain the exact GR 3PN middle block

(400)

This is the first exact grouped- P_2 no-go of the 3PN program.

O.5 Interpretation and consequence for the next step

This no-go does *not* kill the grouped ontology. It kills only the first, overly rigid constitutive guess. The grouped real P_2 bundle is still the right front end; what fails is the minimal local demotion of the direct channel norms. The practical lesson is therefore very sharp:

1. the direct grouped $O(\omega^2)$ scaffold is the correct kinematic starting point,
2. the unique one-step local demotion by one inverse orbital weight is too small,
3. the true 3PN grouped constitutive lift must introduce additional middle-block mixing,

4. and separate mechanisms are still needed for the pure-kinetic and static sectors.

This is exactly why the next appendix enlarges the grouped local family from the three demoted T_A directions to the full nine-basis (T_A, S_A, V_A) compiler. The failure recorded here is therefore a useful theorem gate: it prevents the grouped program from being pursued in an underfit form and directly motivates the richer exact middle-block closure used in the main theorem chain.

P Richer grouped- P_2 compiler and inverse map

This appendix records the exact positive grouped- P_2 result that replaces the no-go of the minimal local lift. The point is not that the grouped ontology was too small. The point is that the first local constitutive choice was too rigid. Once the natural local grouped family is enlarged in the unique minimal way described below, the entire nine-slot 3PN middle block becomes exactly reachable, with an explicit inverse compiler from the solved GR target to the grouped coefficients.

Throughout this appendix we use the reduced COM notation

$$d \equiv \dot{r}, \quad u_{\perp}^2 \equiv v^2 - d^2, \quad U \equiv \frac{GM}{r},$$

and the standard middle-block slot labels $(l_6, \dots, l_{14})$.

P.1 Why the minimal grouped lift had to be enlarged

The minimal local grouped- P_2 lift built only from the demoted dynamic family T_A lands in the correct nine interaction slots, but its image has rank three and obeys six exact slot relations. The solved GR middle block violates all six, so the minimal lift is too small. The correct interpretation is not that the grouped branch fails, but that the constitutive family is underfit.

The first demoted grouped family is

$$T_A \equiv U^{-1} \dot{C}_A^2, \quad A \in \{20, 21, 22\},$$

where the grouped source-square norms are

$$C_{20}^2 = \frac{1}{6}(3d^2 - v^2 - 2U)^2, \quad C_{21}^2 = 2d^2\nu_{\perp}^2, \quad C_{22}^2 = \frac{1}{2}\nu_{\perp}^4.$$

In explicit local form this gives

$$T_{20} = \frac{1}{3}Ud^2(3\nu_{\perp}^2 - U)^2, \quad T_{21} = U\nu_{\perp}^2(\nu_{\perp}^2 - d^2 - U)^2, \quad T_{22} = Ud^2\nu_{\perp}^4.$$

The exact positive result is that the grouped ontology itself is already large enough; only the constitutive family must be enlarged by the two natural local grouped companions

$$S_A \equiv U^2 C_A^2, \quad V_A \equiv \frac{v^2}{U} S_A = Uv^2 C_A^2, \quad A \in \{20, 21, 22\}.$$

So the richer grouped real P_2 family is

$$(T_{20}, T_{21}, T_{22}, S_{20}, S_{21}, S_{22}, V_{20}, V_{21}, V_{22}).$$

P.2 Exact middle-block compiler

Write the richer grouped lift as

$$L_{P_2}^{\text{mid}} = \sum_{A \in \{20, 21, 22\}} (\lambda_A T_A + \sigma_A S_A + \tau_A V_A),$$

with grouped coefficient triples

$$(\lambda_{20}, \lambda_{21}, \lambda_{22}), \quad (\sigma_{20}, \sigma_{21}, \sigma_{22}), \quad (\tau_{20}, \tau_{21}, \tau_{22}).$$

Expanding the nine basis elements in the COM monomial basis for the middle block,

$$(Uv^6, Uv^4d^2, Uv^2d^4, Ud^6, U^2v^4, U^2v^2d^2, U^2d^4, U^3v^2, U^3d^2),$$

produces an exact 9×9 linear map

$$M_{\text{mid}} : (\lambda_{20}, \lambda_{21}, \lambda_{22}, \sigma_{20}, \sigma_{21}, \sigma_{22}, \tau_{20}, \tau_{21}, \tau_{22}) \mapsto (l_6, \dots, l_{14}).$$

The decisive fact is that this compiler is exactly invertible:

$$\det M_{\text{mid}} = -\frac{4}{27}.$$

So the richer local grouped family closes the full nine-slot middle block exactly.

P.3 Exact inverse map

Because M_{mid} is invertible, the exact inverse compiler from the middle-block target

$$(L_6, \dots, L_{14})$$

to the grouped coefficients is algebraic and unique. In the notation above one finds

$$\lambda_{20} = 6(L_{10} + L_{11} + L_{12}) + 3(L_{13} + L_{14}) + 9(L_6 + L_7 + L_8 + L_9),$$

$$\lambda_{21} = -(L_{10} + L_{11} + L_{12}) + L_{13} - 3(L_6 + L_7 + L_8 + L_9),$$

$$\lambda_{22} = -22(L_{10} + L_{11} + L_{12}) - 5L_{13} - 9L_{14} - 39(L_6 + L_7 + L_8) - 38L_9,$$

$$\sigma_{20} = \frac{3}{2}(L_{10} + L_{11} + L_{12}) + 3(L_6 + L_7 + L_8 + L_9),$$

$$\sigma_{21} = \frac{17}{2}L_{10} + 8L_{11} + \frac{15}{2}L_{12} + 2L_{13} + 3L_{14} + \frac{27}{2}(L_6 + L_7 + L_8 + L_9),$$

$$\sigma_{22} = -\frac{5}{2}L_{10} - \frac{9}{2}L_{11} - \frac{9}{2}L_{12} + 4L_{13} - 15(L_6 + L_7 + L_8 + L_9),$$

$$\tau_{20} = \frac{3}{2}(L_6 + L_7 + L_8 + L_9),$$

$$\tau_{21} = \frac{1}{2}(L_{10} + L_{11} + L_{12} - L_{13}) + 3L_6 + \frac{5}{2}L_7 + 2L_8 + \frac{3}{2}L_9,$$

$$\tau_{22} = 2(L_{10} + L_{11} + L_{12} - L_{13}) + \frac{15}{2}L_6 + \frac{11}{2}(L_7 + L_8 + L_9).$$

This is the exact grouped- P_2 inverse map on the 3PN middle block.

P.4 Exact fit to the solved GR middle block

Now substitute the solved GR middle-block residual

$$(L_6, \dots, L_{14}) = (\Delta l_6, \dots, \Delta l_{14}).$$

The inverse map then gives one unique grouped coefficient set:

$$\begin{aligned}\lambda_{20}^{\text{GR}} &= \frac{\nu(2672 - 3154\nu - 276\nu^2 + 3\pi^2)}{32}, \\ \lambda_{21}^{\text{GR}} &= \frac{\nu(-3044 + 2236\nu + 2568\nu^2 - 3\pi^2)}{192}, \\ \lambda_{22}^{\text{GR}} &= \frac{\nu(-30136 + 32858\nu + 7650\nu^2 - 33\pi^2)}{96}, \\ \sigma_{20}^{\text{GR}} &= \frac{\nu(293 - 308\nu - 102\nu^2)}{16}, \\ \sigma_{21}^{\text{GR}} &= \frac{\nu(22006 - 23778\nu - 4188\nu^2 + 21\pi^2)}{192}, \\ \sigma_{22}^{\text{GR}} &= \frac{\nu(-2791 + 2478\nu + 3906\nu^2 - 3\pi^2)}{48}, \\ \tau_{20}^{\text{GR}} &= \frac{3\nu(38 - 87\nu - 154\nu^2)}{32}, \\ \tau_{21}^{\text{GR}} &= \frac{\nu(4412 - 5824\nu - 6672\nu^2 + 3\pi^2)}{384}, \\ \tau_{22}^{\text{GR}} &= \frac{\nu(3386 - 3367\nu - 2790\nu^2 + 3\pi^2)}{96}.\end{aligned}$$

With exactly these coefficients, the richer grouped lift reproduces the full solved GR middle block with no residual left:

$$(l_6, \dots, l_{14}) = (\Delta l_6, \dots, \Delta l_{14}).$$

P.5 Structural corollaries

Two corollaries matter for the carry-forward theorem ledger.

First, every basis element in (T_A, S_A, V_A) carries at least one power of U . Therefore the richer grouped compiler still forces

$$\boxed{l_1 = l_2 = l_3 = l_4 = l_5 = 0}$$

identically. So the pure-kinetic residual remains outside the grouped- P_2 module.

Second, the richer grouped compiler predicts a specific static companion rather than leaving the static slot arbitrary:

$$\boxed{l_{15}^{(P_2 \text{ pred})} = l_{10} + l_{11} + l_{12} + 2(l_6 + l_7 + l_8 + l_9).}$$

Evaluated on the solved GR middle block, this gives

$$\boxed{l_{15}^{(P_2 \text{ pred})} = \frac{\nu(293 - 308\nu - 102\nu^2)}{24}.$$

This is already close to the exact GR static residual, but it is not equal to it. So after exact grouped middle-block closure, the only remaining grouped-side obstruction is a one-dimensional static remainder, which the main text shows is uniquely geometry-side.

The correct carry-forward reading is therefore:

the richer grouped real P_2 family closes the full nine-slot 3PN middle block exactly,

but the static completion and the outgoing/isotropy questions remain separate.

Q Static P_0 /geometry counterterm and sigma-collapse

This appendix records the exact scalar-side conclusion that remains after the richer grouped real P_2 compiler has closed the full nine-slot 3PN middle block. The positive grouped result leaves only a one-dimensional static remainder, but the right carrier of that remainder is not obvious if one looks only at the COM chart. At first sight one finds an apparent one-parameter split between a body-local scalar P_0 family and a pair/geometry family. The point of this appendix is that the apparent freedom is algebraically fake. Once the exact generic-frame mass identity is used, the whole family collapses to a unique geometry-side completion.

Throughout this appendix we use the generic-frame mass labels

$$p \equiv m_A, \quad q \equiv m_B, \quad \nu \equiv \frac{pq}{(p+q)^2}, \quad U \equiv \frac{G(p+q)}{r}.$$

Q.1 Residual static slot after grouped closure

After exact grouped middle-block closure, the solved COM residual takes the form

$$\Delta \hat{L}_3^{\text{GR}} = \Delta l_1 v^8 + L_{P_2}^{\text{mid}}[\lambda_A^{\text{GR}}, \sigma_A^{\text{GR}}, \tau_A^{\text{GR}}] + \Delta l_{15}^{(0/g)} U^4.$$

So the grouped real P_2 module has done all it can do. The only remaining conservative scalar job is the one-dimensional static remainder

$$\Delta l_{15}^{(0/g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}.$$

In the fixed ADM chart this is the full scalar gap left after the grouped- P_2 middle block is closed.

Q.2 Why the old constant-coefficient U -block is too small

Inside the old constant-coefficient generic-frame interaction scaffold, the only static basis polynomial is

$$U_1 = p^2 q + pq^2$$

inside the common prefactor $G^4 pq/r^4$. In the COM chart this always reduces to a coefficient proportional to

$$\nu U^4,$$

so any such constant-coefficient static slot is linear in ν after the overall ν is factored out.

But the exact 3PN static remainder above contains ν^2 and ν^3 pieces. Therefore the old constant-coefficient U -block is too small to carry the actual scalar gap. This is the exact no-go behind the later counterterm interpretation: the remaining scalar completion cannot be represented by just “one more constant static basis coordinate” in the original generic-frame scaffold.

Q.3 Apparent body-local P_0 versus pair/geometry family

Once one allows a body-local scalar carrier as well as the pair/geometry carrier, the static completion initially looks like a genuine one-parameter family. The two natural mass polynomials are

$$U_0 = p^3 + q^3, \quad U_g = p^2 q + pq^2.$$

The body-local family behaves like a scalar P_0 wall/support term, while the pair family is the natural geometry-side carrier because it is intrinsically pair-coupled and vanishes in the strict test-mass branch.

At the generic-frame level one can therefore write the regular one-parameter family

$$\Delta L_{3,\text{ct}}^{(0)} = \frac{G^4 pq}{r^4} \sigma \nu (p^3 + q^3),$$

$$\Delta L_{3,\text{ct}}^{(g)} = \frac{G^4 pq}{r^4} \left[\frac{408\nu^2 + 1232\nu - 2080 + 63\pi^2}{96} - \sigma(1 - 3\nu) \right] (p^2 q + pq^2).$$

Their COM images combine to reproduce the exact scalar gap for every regular value of the split parameter σ . So if one stops at the COM bookkeeping stage, the scalar completion appears to admit a free P_0 -versus-geometry admixture.

A useful warning sign is obtained by asking for a “pure- P_0 ” slice. Solving for a split that removes the pair/geometry piece gives

$$\sigma_{\text{pure } P_0}(\nu) = \frac{-408\nu^2 - 1232\nu - 63\pi^2 + 2080}{96(3\nu - 1)},$$

which is singular at $\nu = 1/3$. So even before the exact collapse is imposed, the apparent pure- P_0 option is already unnatural.

Q.4 Exact sigma-collapse and unique geometry completion

The apparent σ -freedom is not physical. The exact mass identity

$$\boxed{\nu(p^3 + q^3) = (1 - 3\nu)(p^2 q + pq^2)}$$

collapses the whole family identically. Indeed,

$$\sigma \nu (p^3 + q^3) = \sigma(1 - 3\nu)(p^2 q + pq^2),$$

so the σ -dependence cancels between the body-local and pair pieces. The full static counterterm therefore reduces to the unique geometry-side expression

$$\boxed{\Delta L_{3,\text{ct}}^{\text{scalar}} = \frac{G^4 pq}{96r^4} (408\nu^2 + 1232\nu - 2080 + 63\pi^2)(p^2 q + pq^2)}.$$

There is no independent regular body-local P_0 completion left in the fixed ADM chart.

The same conclusion can be read directly on the target coefficients. The imported generic-frame static target coefficient is

$$c_U^{\text{target}} = -\frac{227}{24} + \frac{21\pi^2}{32},$$

while the richer grouped- P_2 middle-block fit already forces the static companion

$$c_U^{(P_2 \text{ pred})} = \frac{293 - 308\nu - 102\nu^2}{24}.$$

Their exact difference is the unique geometry coefficient

$$\boxed{c_U^{(g)} = \frac{408\nu^2 + 1232\nu - 2080 + 63\pi^2}{96}}.$$

On the COM side this reduces exactly to

$$\boxed{\Delta l_{15}^{(g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96}}.$$

So the sigma-collapse is not a matter of taste or chart choice. It is the exact statement that, after the grouped- P_2 middle block is closed, the remaining 3PN static scalar piece lands uniquely in the pair/geometry lane.

Q.5 Push through the generic-frame Hamiltonian compiler

Once the lower-order ledger is frozen, the generic-frame 3PN Hamiltonian compiler remains the exact sign flip

$$\Delta H_3 = -\Delta L_3.$$

So the unique scalar counterterm lifts immediately to

$$\Delta H_{3,\text{ct}}^{\text{scalar}} = -\Delta L_{3,\text{ct}}^{\text{scalar}}.$$

There is no further algebraic freedom to resolve. The carry-forward interpretation is therefore sharp:

after the grouped- P_2 middle block is closed,
the remaining static 3PN scalar piece is uniquely geometry-side

This is the scalar-side analog of the middle-block grouped- P_2 result. The algebraic bottleneck is gone; what remains is the physical question of what moving-throat wall/geometry dynamics microscopically realizes this geometry-side scalar completion.

R Pure-kinetic collapse derivation

This appendix records the final kinetic audit behind the conservative 3PN theorem. After the richer grouped real P_2 compiler has closed the full nine-slot middle block and the static scalar remainder has collapsed uniquely to the geometry side, one apparently independent center-of-mass kinetic slot is still left over. At first sight that leftover looks like one more independent 3PN throat-response datum. The exact free-body audit shows that this reading is wrong. The residual is not a new comparable-mass interaction family at all. It is only the ordinary-Lagrangian counterimage of the universal free relativistic two-body center-of-mass Hamiltonian.

The point of this appendix is therefore very narrow: to show explicitly that the apparent pure-kinetic remainder

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}$$

is generated automatically by the exact COM compiler map, and that no additional generic-frame pure-kinetic interaction module should be introduced in the fixed ADM chart.

R.1 Apparent residual after grouped and static closure

After exact grouped middle-block closure and unique static completion, the remaining COM kinetic slots are

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}, \quad \Delta l_2 = \Delta l_3 = \Delta l_4 = \Delta l_5 = 0.$$

In the isotropic 15-slot COM basis, this is the whole apparent pure-kinetic residue. Taken by itself, it looks as though one more independent 3PN kinetic throat coefficient must still be supplied.

That interpretation would only be correct if a genuinely new kinetic datum first appeared as a new generic-frame interaction family. The exact free-body audit shows that no such family exists. The pure-kinetic issue is therefore not a new interaction-sector problem. It is a compiler problem.

R.2 Exact free generic-frame block and its naive COM reduction

At generic-frame level, the exact free relativistic two-body ordinary Lagrangian is simply the sum of the two separate free-body square roots,

$$L_{\text{free}}^{\text{gen}} = -m_A c^2 \sqrt{1 - \frac{v_A^2}{c^2}} - m_B c^2 \sqrt{1 - \frac{v_B^2}{c^2}}.$$

Expanding through 3PN gives

$$\begin{aligned} L_{\text{free}}^{\text{gen}} = & -(m_A + m_B)c^2 + \frac{1}{2}(m_A v_A^2 + m_B v_B^2) + \frac{1}{8c^2}(m_A v_A^4 + m_B v_B^4) \\ & + \frac{1}{16c^4}(m_A v_A^6 + m_B v_B^6) + \frac{5}{128c^6}(m_A v_A^8 + m_B v_B^8) + O(c^{-8}). \end{aligned}$$

So the exact generic-frame 3PN pure-kinetic block is

$$L_{3,\text{free}}^{\text{gen}} = \frac{5}{128}(m_A v_A^8 + m_B v_B^8).$$

There is no mixed comparable-mass pure-kinetic interaction monomial here. In particular, the free generic-frame block contains no independent $m_A m_B$ -type octic interaction that could serve as a new 3PN constitutive datum.

Now impose the naive COM substitution

$$\mathbf{v}_A = \eta_B \mathbf{v}, \quad \mathbf{v}_B = -\eta_A \mathbf{v}, \quad \eta_A \equiv \frac{m_A}{M}, \quad \eta_B \equiv \frac{m_B}{M}, \quad M \equiv m_A + m_B,$$

with reduced mass

$$\mu \equiv \frac{m_A m_B}{M}$$

and symmetric mass ratio

$$\nu \equiv \frac{\mu}{M} = \frac{m_A m_B}{M^2}.$$

Then the carried COM seed coefficient is

$$\frac{1}{\mu} \frac{5}{128} (m_A \eta_B^8 + m_B \eta_A^8) v^8 = l_1^{\text{seed}} v^8,$$

with

$$l_1^{\text{seed}} = \frac{5}{128} - \frac{35}{128}\nu + \frac{35}{64}\nu^2 - \frac{35}{128}\nu^3.$$

So the earlier seed coefficient was nothing more than the naive COM reduction of the universal free generic-frame relativistic Lagrangian.

R.3 Exact free COM Hamiltonian target

The correct free reference object is instead the exact relativistic two-body COM Hamiltonian,

$$H_{\text{free}}^{\text{COM}} = \sqrt{m_A^2 c^4 + P^2 c^2} + \sqrt{m_B^2 c^4 + P^2 c^2},$$

with reduced momentum variable defined by

$$P = \mu p.$$

Subtracting the rest energy and dividing by μ gives the reduced expansion

$$\frac{H_{\text{free}}^{\text{COM}} - M c^2}{\mu} = \frac{1}{2} p^2 - \frac{1 - 3\nu}{8c^2} p^4 + \frac{1 - 5\nu + 5\nu^2}{16c^4} p^6 + \frac{h_1^{\text{free}}}{c^6} p^8 + O(c^{-8}),$$

where the exact 3PN free coefficient is

$$h_1^{\text{free}} = \frac{-5 + 35\nu - 70\nu^2 + 35\nu^3}{128}, \quad h_2 = h_3 = h_4 = h_5 = 0.$$

This is not a new observable of the throat. It is just the universal free relativistic two-body Hamiltonian written in reduced COM variables. So the relevant comparison is not between the ordinary seed and a new kinetic throat datum. It is between the ordinary seed and the already-fixed free COM Hamiltonian target.

R.4 Exact COM compiler map and collapse of the apparent residual

The exact COM compiler acts on the first pure-kinetic slot by

$$l_1 = F_1(\nu) - h_1, \quad F_1(\nu) = \frac{3}{16}\nu - \frac{21}{16}\nu^2 + \frac{9}{4}\nu^3.$$

Applying this map to the naive free ordinary seed gives its Hamiltonian image,

$$h_1^{\text{seed}} = F_1(\nu) - l_1^{\text{seed}} = -\frac{5}{128} + \frac{59}{128}\nu - \frac{119}{64}\nu^2 + \frac{323}{128}\nu^3.$$

So the shift between the naive seed and the true free COM target is

$$\begin{aligned} \Delta l_1 &= l_1^{\text{full}} - l_1^{\text{seed}} = h_1^{\text{seed}} - h_1^{\text{free}} \\ &= \frac{3}{16}\nu - \frac{21}{16}\nu^2 + \frac{9}{4}\nu^3 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}. \end{aligned}$$

Equivalently, the full ordinary coefficient required by the exact free COM Hamiltonian is

$$l_1^{\text{full}} = F_1(\nu) - h_1^{\text{free}} = \frac{5}{128} - \frac{11}{128}\nu - \frac{49}{64}\nu^2 + \frac{253}{128}\nu^3.$$

Thus the whole apparent leftover is exactly the ordinary-Lagrangian compensation required to reproduce the already-fixed free relativistic COM Hamiltonian.

The essential consequence is immediate:

$$\boxed{\text{the apparent COM pure-kinetic remainder is not a new dynamical throat datum.}}$$

It is only the ordinary-side image of the universal free Hamiltonian under the exact COM compiler.

R.5 Fixed-chart interpretation

The theorem-level reading is therefore sharp.

First, the exact generic-frame free-body ordinary block has no mixed pure-kinetic interaction monomial:

$$L_{3,\text{free}}^{\text{gen}} = \frac{5}{128}(m_A v_A^8 + m_B v_B^8).$$

Second, naive COM reduction of that block gives the carried seed coefficient l_1^{seed} and nothing more. Third, the exact free relativistic two-body COM Hamiltonian fixes the true target coefficient h_1^{free} with

$$h_2 = h_3 = h_4 = h_5 = 0.$$

So once the exact COM compiler is imposed, the residual

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}$$

is generated automatically, and no extra generic-frame pure-kinetic interaction module should be added in the fixed ADM chart.

This is the final kinetic-side analogue of the grouped- P_2 and sigma-collapse results. The algebraic bottleneck is gone. After grouped middle-block closure, unique static geometry completion, and pure-kinetic collapse, the conservative 3PN split really does stop with three lanes only: the compiler image of free kinematics, the exact grouped real P_2 module, and the unique geometry-side static completion.

S Consolidated theorem ledger

This appendix is the shortest single place to recover the final conservative 3PN status after the earlier appendices are assembled. Its purpose is not to re-derive the chain once more, but to freeze the exact theorem package in one compact location so that later work does not reopen already settled bookkeeping.

The right reading is the same as in the earlier full 1PN and 2PN papers: this is an exact conservative theorem package *within the declared closure hierarchy*, not an assumption-free theorem of the fully solved moving-throat PDE. Within that hierarchy, however, the algebraic 3PN bottleneck is gone.

S.1 Exact constants and exact compiler data

The strict one-body Schwarzschild gate fixes the minimal 3PN repair ledger to

$$\mu_{\rho,3} = \frac{1}{4}, \quad d_3 = -\frac{45}{4}, \quad s_{24} = -\frac{1}{16}.$$

These are the exact carried constants that replace the earlier possibility of a one-parameter continuation of the 2PN self sector.

Once the lower-order ledger is frozen and the fixed ADM Hamiltonian chart is chosen, the full generic-frame 3PN interaction compiler becomes

$$\Delta H_3 = -\Delta L_3, \quad C_{\text{gen}} = -I_{50}.$$

So the earlier COM-blind and contact/gauge bookkeeping freedoms do not survive as true algebraic ambiguity in the chosen chart.

For the grouped real P_2 middle block, the richer local nine-family compiler is exactly invertible, with

$$\det M_{\text{mid}} = -\frac{4}{27}.$$

This is the decisive exact statement behind the successful middle-block closure: the grouped conservative fit is neither underdetermined nor merely gauge-selected.

After the grouped closure is imposed, the remaining static scalar piece is uniquely geometry-side. In COM form the exact coefficient is

$$\Delta l_{15}^{(g)} = \frac{\nu(408\nu^2 + 1232\nu - 2080 + 63\pi^2)}{96},$$

and in generic-frame coefficient language

$$c_U^{(g)} = \frac{408\nu^2 + 1232\nu - 2080 + 63\pi^2}{96}.$$

Likewise, the apparent pure-kinetic remainder is

$$\Delta l_1 = \frac{3\nu(3\nu - 1)(4\nu - 1)}{16}.$$

The pure-kinetic collapse appendix shows that this is not a new throat datum, but the ordinary-Lagrangian counterimage of the universal free relativistic two-body COM Hamiltonian.

S.2 Exact split formulas

The exact isotropic one-body 3PN target remains

$$\frac{L_{3\text{PN},1\text{body}}}{m} = \frac{5}{128}v^8 + \frac{11}{16}Uv^6 + \frac{47}{16}U^2v^4 + \frac{13}{8}U^3v^2 - \frac{1}{8}U^4.$$

This is the immutable test-mass gate beneath the full comparable-mass lift.

Beyond the frozen one-body/self-static seed, the exact comparable-mass COM residual is fully assigned by the three-lane split

$$\Delta \hat{L}_3^{\text{GR}} = \underbrace{\Delta l_1 v^8}_{\text{compiler image of the free COM Hamiltonian}} + \underbrace{L_{P_2}^{\text{mid}}}_{\text{exact grouped real } P_2 \text{ middle-block closure}} + \underbrace{\Delta l_{15}^{(g)} U^4}_{\text{unique geometry completion}}.$$

So after the earlier appendices are assembled, the conservative 3PN residual really does stop with exactly three lanes and no fourth algebraic family.

The middle block itself is carried by the richer local grouped family

$$(T_{20}, T_{21}, T_{22}, S_{20}, S_{21}, S_{22}, V_{20}, V_{21}, V_{22}), \quad A \in \{20, 21, 22\},$$

with the useful identifications

$$S_A = U^2 C_A^2, \quad V_A = \frac{v^2}{U} S_A,$$

and the exact inverse map guaranteed by the richer grouped- P_2 compiler appendix. The exact grouped fit fixes the nine middle slots and leaves only the static scalar slot to be completed.

Writing

$$p \equiv m_A, \quad q \equiv m_B,$$

the full static completion is uniquely

$$\Delta L_{3,\text{ct}}^{\text{scalar}} = \frac{G^4 pq}{96r^4} (408\nu^2 + 1232\nu - 2080 + 63\pi^2) (p^2 q + p q^2),$$

with fixed-chart Hamiltonian image

$$\Delta H_{3,\text{ct}}^{\text{scalar}} = -\Delta L_{3,\text{ct}}^{\text{scalar}}.$$

This is the exact sigma-collapsed pair/geometry remainder from the static P_0 /geometry counterterm and sigma-collapse appendix.

The pure-kinetic sector likewise needs no new interaction module. The exact free generic-frame block is

$$L_{3,\text{free}}^{\text{gen}} = \frac{5}{128}(m_A v_A^8 + m_B v_B^8),$$

while the exact free COM Hamiltonian target has

$$h_1^{\text{free}} = \frac{-5 + 35\nu - 70\nu^2 + 35\nu^3}{128}, \quad h_2 = h_3 = h_4 = h_5 = 0.$$

The exact COM compiler then produces Δl_1 automatically, so no independent generic-frame pure-kinetic interaction family should be added in the fixed ADM chart.

S.3 Fixed versus open data

Within the declared conservative hierarchy, the following should now be treated as fixed.

1. The one-body 3PN repair ledger

$$(\mu_{\rho,3}, d_3, s_{24}) = \left(\frac{1}{4}, -\frac{45}{4}, -\frac{1}{16}\right).$$

2. The exact generic-frame uniqueness theorem in the fixed ADM chart,

$$\Delta H_3 = -\Delta L_3, \quad C_{\text{gen}} = -I_{50}.$$

3. The exact grouped real P_2 middle-block closure, with invertible richer compiler and

$$\det M_{\text{mid}} = -\frac{4}{27}.$$

4. The unique geometry-side completion of the remaining static slot.
5. The pure-kinetic collapse to the universal free relativistic COM Hamiltonian.

In that precise sense, the conservative 3PN two-body ledger is now fully assigned in the fixed ADM chart. What remains open is no longer algebraic coefficient matching.

The open problems are now dynamical and constitutive. First, the richer grouped real P_2 constitutive map should be derived directly from the moving-throat problem rather than by inverse matching. Second, the unique geometry completion should be derived directly from the throat-side geometry/compressibility dynamics. Third, those two conservative lanes must be carried into the already narrowed 2.5PN bridge, where the live universal route has already been reduced to the grouped real quadrupole bundle and its passive/outgoing normalization problem.

The final carry-forward statement is therefore

the conservative 3PN algebra is closed;
the remaining bottleneck is the moving-throat origin of the grouped- P_2 and geometry lanes.